

Structure-based design and evaluation of reversible KRAS G13D inhibitors

Supporting Information

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1. General procedures – Synthetic chemistry

General methods and techniques. All intermediates and final analogs were synthesized using common synthetic organic chemistry techniques. No unexpected or unusually high safety hazards were encountered.

Reagents and solvents. All reagents and solvents were purchased from commercial sources and used without further purification unless otherwise noted.

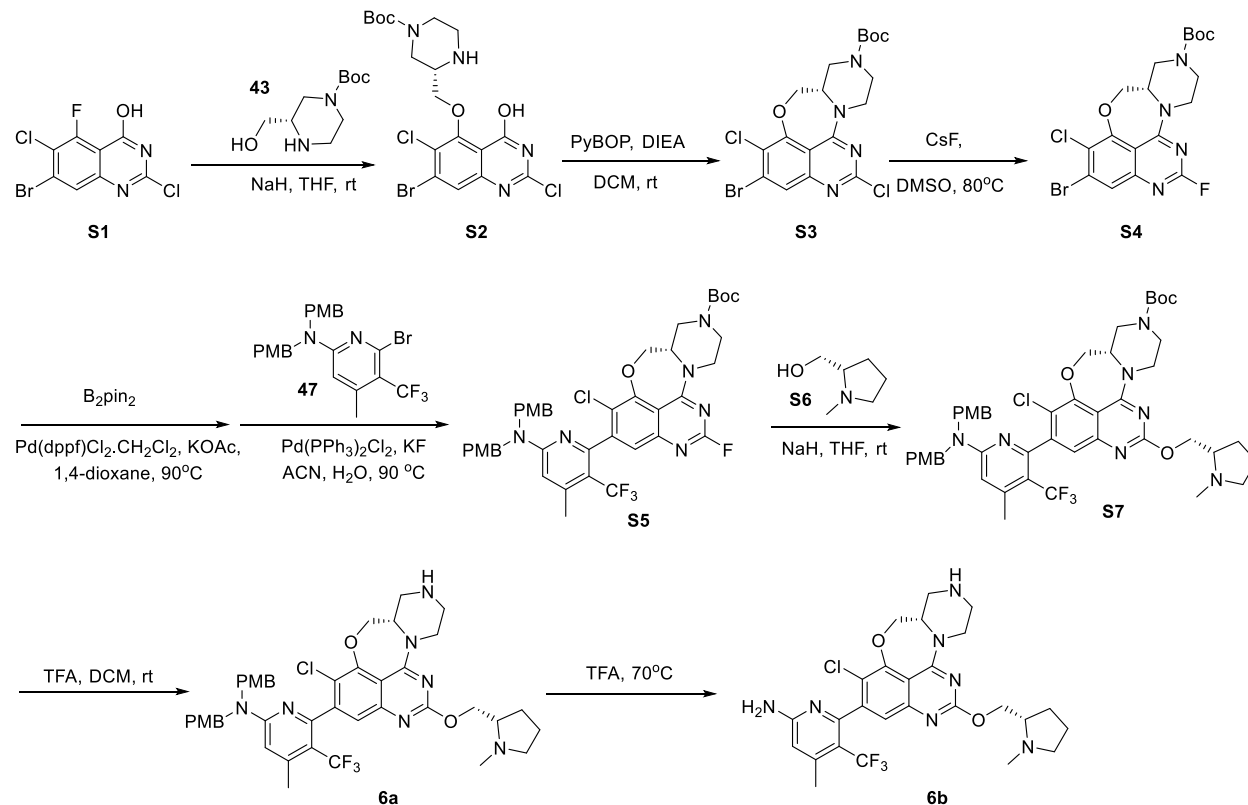
Column chromatography. Flash chromatographic purifications were performed on a Biotage IsoleraTM Prime 3.3.0 or a Teledyne Isco Combiflash Rf.

LCMS. The purity of compounds was assessed via LCMS using either a Shimadzu LCMS-2020 instrument (mobile phase A:water/0.05%TFA; mobile phase B:ACN/0.05%TFA; gradient: 5-95% or 5-100% B within 1.10 min-2.3 min, total flow 1.2 mL/min-1.5 mL/min; Column: HALO C18 - Length: 30 mm - Internal Diameter: 3.0 mm - Description: 2.0 μ m particles) or on an Agilent 1290 UHPLC instrument coupled with Agilent MSD (6140) mass spectrometer using ESI as ionization source (LC separation was done on a Phenomenex XB-C18, 1.7 μ m, 50 $\times$ 2.1 mm column at a flow rate of 0.4 ml / minute; mobile phase A was water with 0.1% FA and mobile phase B was acetonitrile with 0.1% FA; the gradient started at 2% MPB and ended at 98% MPB over 7 min and held at 98% MPB for 1.5 min following an equilibration for 1.5 min; LC column temperature was 40 $^{\circ}$ C; UV absorbance was collected at 220nm and 254nm and mass spec full scan was applied to all experiments.)

NMR. NMR spectra were acquired on Bruker AVANCE III HD 300 MHz/400 MHz spectrometer or a Bruker 400 MHz BBI probe with an Avance NEO console or a Bruker 400 MHz prodigy BBO cold probe with an Avance III HD console.

2. Representative experimental procedures for analog synthesis

Synthesis of compounds 6a and 6b



Step 1: To a mixture of *tert*-butyl (S)-3-(hydroxymethyl)piperazine-1-carboxylate (**43**) (5.0 g, 23.12 mmol) in tetrahydrofuran (50 mL) was added sodium hydride (1.86 g, 46.24 mmol) and the mixture was stirred for 0.5 h at rt. Then 7-bromo-2,6-dichloro-5-fluoro-quinazolin-4-ol (**S1**) (6 g, 19.24 mmol) was added, and the mixture was stirred for 1 h at rt. The reaction was quenched with water and extracted with ethyl acetate. The solvent was removed and the residue was purified by flash chromatography on silica gel eluting with dichloromethane/methanol (10:90) to afford compound **S2** (7 g, 72% yield) as a brown solid. LCMS (ESI, m/z): calcd. for C₁₈H₂₂BrCl₂N₄O₄⁺ [M+H]⁺ 507.0, found 507.2

Step 2: To a mixture of *tert*-butyl (3S)-3-((7-bromo-2,6-dichloro-4-hydroxyquinazolin-5-yl)oxymethyl)piperazine-1-carboxylate (**S2**) (5 g, 9.84 mmol) in dichloromethane (40 mL) were added *N,N*-diisopropylethylamine (5.12 mL, 29.4 mmol) and PyBOP (7.68 g, 14.76 mmol), and the mixture was stirred overnight at rt. The residue was purified by flash chromatography on silica

gel eluting with petrol ether/ethyl acetate (64:36) to afford compound **S3** (2.18 g, 45% yield) as a brown solid. LCMS (ESI, m/z): calcd. for $C_{18}H_{20}BrCl_2N_4O_3^+$ $[M+H]^+$ 489.0, found 489.2.

Step 3: To a mixture of *tert*-Butyl (S)-10-bromo-7,11-dichloro-3,4,13,13a-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazoline-2(1*H*)-carboxylate (**S3**) (2.2 g, 4.49 mmol) in dimethyl sulfoxide (15 mL) was added cesium fluoride (0.83 mL, 22.38 mmol) and the mixture was stirred for 1 h at 80 °C. The resulting solution was diluted with water and the solids were collected to afford the crude **S4** (2.0 g, 94% yield). LCMS (ESI, m/z): calcd. for $C_{18}H_{20}BrClFN_4O_3^+$ $[M+H]^+$ 473.0, found 473.2.

Steps 4 & 5: Under nitrogen, a solution of *tert*-butyl (S)-10-bromo-11-chloro-7-fluoro-3,4,13,13a-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazoline-2(1*H*)-carboxylate (**S4**) (800 mg, 1.69 mmol), bis(pinacolato)diboron (343 mg, 1.35 mmol), potassium acetate (0.32 mL, 5.05 mmol), [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(II) (123 mg, 0.17 mmol) in 1,4-dioxane (8 mL) was stirred at 90 °C for 1 h. The organic layer was concentrated in vacuum. The crude product was directly used in the next step without purification.

Under nitrogen, to a mixture of the above crude product (800 mg, 1.82 mmol) in acetonitrile (10 mL) and water (2 mL) were added 6-bromo-*N,N*-bis[(4-methoxyphenyl)-methyl]-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**47**) (450 mg, 0.91 mmol), sodium carbonate (1.5 g, 14.44 mmol) and [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(II) (133 mg, 0.18 mmol) at rt. The resulting mixture was stirred at 90 °C for 2 h. The organic layer was concentrated in vacuum. The residue was purified by flash chromatography on silica gel eluting with petrol ether/ethyl acetate (64:36) to afford compound **S5** (740 mg, 50% yield) as a yellow solid. LCMS (ESI, m/z): calcd. for $C_{41}H_{42}ClF_4N_6O_5^+$ $[M+H]^+$ 809.3, found 809.2.

Step 6: To a solution of (S)-(1-methylpyrrolidin-2-yl)methanol (**S6**) (100 mg, 0.87 mmol) in tetrahydrofuran (5 mL) was added sodium hydride (69 mg, 1.72 mmol) and the mixture was stirred for 0.5 h at 0 °C. Then *tert*-butyl (S)-10-(6-(bis(4-methoxybenzyl)amino)-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-11-chloro-7-fluoro-3,4,13,13a-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazoline-2(1*H*)-carboxylate (**S5**) (700 mg, 0.86 mmol) was added and the mixture was stirred at rt for 1h. The resulting solution was quenched with water and extracted with ethyl acetate. The solvent was removed. The residue was purified by flash chromatography on

silica gel eluting with petrol ether/ethyl acetate (56:44) to afford **S7** (500 mg, 64% yield) as a yellow solid. LCMS (ESI, m/z): calcd. for $C_{47}H_{54}ClF_3N_7O_6^+$ $[M+H]^+$ 904.4, found 904.2.

Step 7: To a solution of *tert*-butyl (S)-10-(6-(bis(4-methoxybenzyl)amino)-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-11-chloro-7-(((S)-1-methylpyrrolidin-2-yl)methoxy)-3,4,13,13a-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazoline-2(1*H*)-carboxylate (**S7**) (500 mg, 0.55 mmol) in dichloromethane (5 mL) was added TFA (2.5 mL) and the mixture was stirred for 2 h at rt. The solvent was concentrated in vacuum to afford compound **6a** (400 mg, 90% yield) LCMS (ESI, m/z): calcd for $C_{42}H_{46}ClF_3N_7O_4^+$ $[M+H]^+$ 804.3, found 804.1.

Step 8: A solution of 6-((S)-11-chloro-7-(((S)-1-methylpyrrolidin-2-yl)methoxy)-1,2,3,4,13,13a-hexahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazolin-10-yl)-*N,N*-bis(4-methoxybenzyl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**6a**) (2 g, 2.49 mmol) in TFA (10 mL) was stirred for 4 h at 70 °C. The solvent was removed under vacuum. The residue was dissolved in water and the pH was adjusted to 9-10 with ammonium hydroxide. The solids were collected to afford 6-((S)-11-chloro-7-(((S)-1-methylpyrrolidin-2-yl)methoxy)-1,2,3,4,13,13a-hexahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazolin-10-yl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**6b**) (1 g, 71% yield) as a yellow solid. LCMS (ESI, m/z): calcd. for $C_{26}H_{30}ClF_3N_7O_2^+$ $[M+H]^+$ 564.2, found 564.2.

General scheme for the synthesis of 7-26 and MS data

Analog 7-26 was synthesized following the general scheme below. Mass spectrometric data is listed in Table S1.

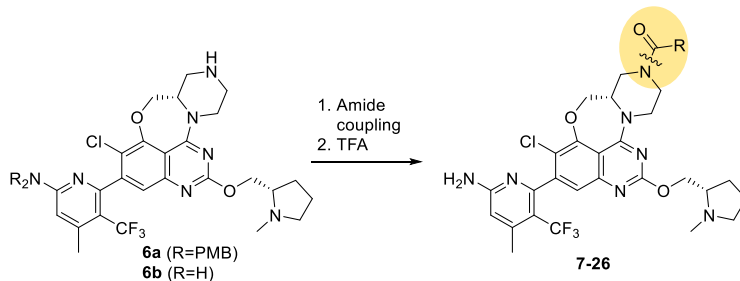
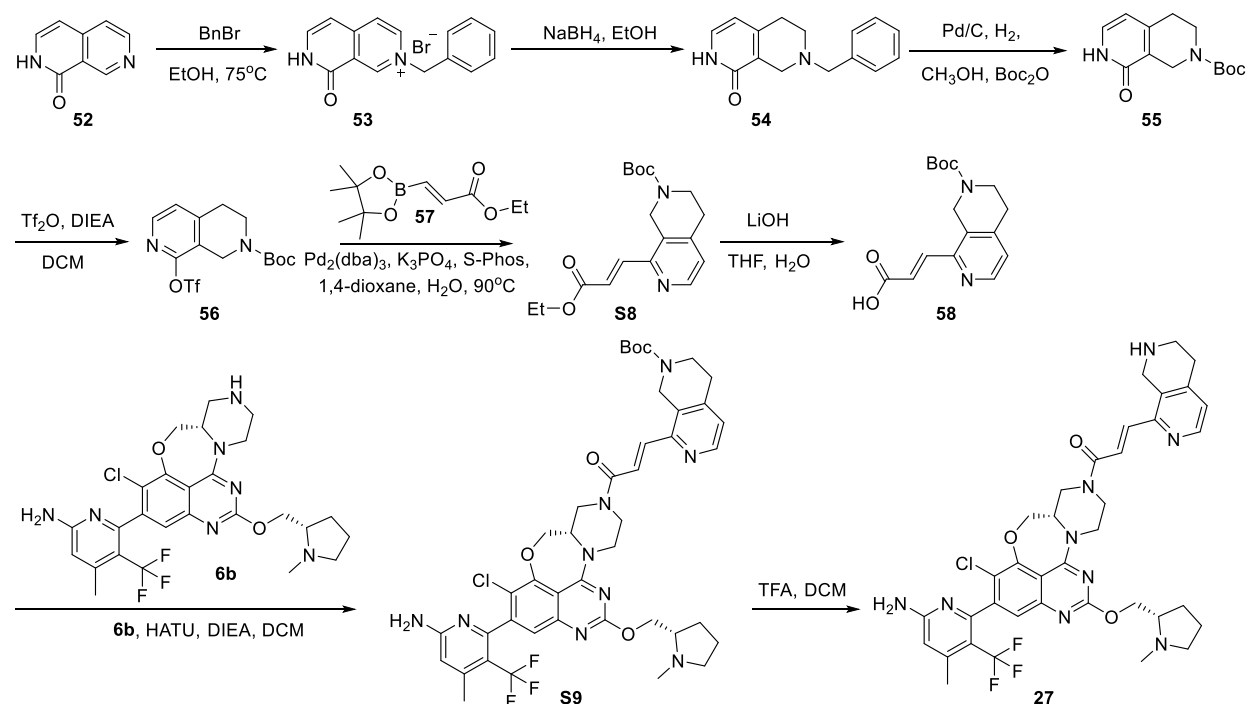


Table S1: LCMS data for analogs 7-26.

Cmpd.	Amide	Molecular formula	Calculated m/z [M+H] ⁺	Observed m/z
7		C ₃₁ H ₃₈ ClF ₃ N ₈ O ₃	663.3	663.3
8		C ₃₀ H ₃₄ ClF ₃ N ₈ O ₃	647.3	647.2
9 ^a		C ₃₂ H ₃₈ ClF ₃ N ₈ O ₄	691.3	691.2
10		C ₂₉ H ₃₄ ClF ₃ N ₈ O ₃	635.4	635.2
11 ^a		C ₃₅ H ₄₂ ClF ₃ N ₈ O ₃	715.3	715.2
12		C ₃₁ H ₃₆ ClF ₃ N ₈ O ₃	661.3	661.3
13 ^a		C ₃₃ H ₄₀ ClF ₃ N ₈ O ₄	705.3	705.3
14		C ₃₄ H ₄₂ ClF ₃ N ₈ O ₃	703.3	703.4
15		C ₃₂ H ₃₈ ClF ₃ N ₈ O ₃	675.3	675.3
16 ^a		C ₃₂ H ₃₈ ClF ₃ N ₈ O ₄	691.3	691.3
17		C ₃₄ H ₄₂ ClF ₃ N ₈ O ₃	703.3	703.4
18		C ₃₄ H ₄₂ ClF ₃ N ₈ O ₃	703.3	703.2
19		C ₃₂ H ₃₈ ClF ₃ N ₈ O ₃	675.3	675.2
20 ^a		C ₃₂ H ₃₆ ClF ₃ N ₈ O ₃	673.3	673.2
21		C ₃₃ H ₄₀ ClF ₃ N ₈ O ₃	689.3	689.3
22 ^a		C ₃₄ H ₄₀ ClF ₃ N ₈ O ₃	701.3	701.2
23 ^a		C ₃₃ H ₄₀ ClF ₃ N ₈ O ₃	689.3	689.3
24		C ₃₄ H ₄₀ ClF ₃ N ₈ O ₃	701.3	701.2
25 ^a		C ₃₃ H ₃₈ ClF ₃ N ₈ O ₄	703.3	703.2
26 ^a		C ₃₃ H ₃₆ ClF ₃ N ₈ O ₃	685.3	685.3

^aMixture of diastereomers.

Synthesis of compound 27



Step 1: To a solution of 2H-2,7-naphthyridin-1-one (**52**) (2.0 g, 13.7 mmol) in ethanol (20 mL) was added benzyl bromide (13 mL, 109 mmol) and the mixture was stirred for 1 h at 75 °C. The solids were collected by filtration to afford **53** (3 g, 69% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{15}H_{13}N_2O^+$ $[M]^+$ 237.1, found 237.1.

Step 2: To a mixture of 2-benzyl-8-oxo-7,8-dihydro-2,7-naphthyridin-2-ium bromide (**53**) (2.6 g, 11 mmol) in ethanol (50 mL) was added sodium borohydride (6.2 g, 164 mmol) and the mixture was stirred for 3 h at rt. The reaction was quenched with water. The ethanol was removed under vacuum. The precipitated solids were collected by filtration to afford compound **54** (1.3 g, 49% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{15}H_{17}N_2O^+$ $[M+H]^+$ 241.1, found 241.2.

Step 3: Under hydrogen, a mixture of 7-benzyl-5,6,7,8-tetrahydro-2,7-naphthyridin-1(2H)-one (**54**) (800 mg, 3.3 mmol) and 10% Pd/C (400 mg) in methanol (20 mL) was stirred for 1.5 h at rt. Then di-*tert*-butyl dicarbonate (1.1 g, 5.05 mmol) was added and the mixture was stirred at room temperature for 1h. The solids were removed by filtration, and the solvent was removed under vacuum. The residue was purified by flash chromatography on silica gel eluting with ethyl

acetate/petroleum ether (0-85%) to afford compound **55** (830 mg, 100% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{13}H_{19}N_2O_3^+$ $[M+H]^+$ 251.1, found 251.1.

Step 4: To a solution of *tert*-butyl 8-oxo-3,4,7,8-tetrahydro-2,7-naphthyridine-2(1H)-carboxylate (**55**) (500 mg, 2 mmol) and pyridine (0.80 mL, 10 mmol) in dichloromethane (3 mL) was added trifluoromethanesulfonic anhydride (0.67 mL, 4 mmol) and the mixture was stirred for 1 h at 0 °C. The reaction was quenched with water and extracted with ethyl acetate. The solvent was removed under vacuum. The residue was purified by flash chromatography on silica gel eluting with ethyl acetate/petroleum ether (0-50%) to afford **56** (400 mg, 52% yield) as a yellow oil. LCMS (ESI): calcd. for $C_{14}H_{18}F_3N_2O_5S^+$ $[M+H]^+$ 383.1, found 383.2.

Step 5: Under nitrogen, a mixture of *tert*-butyl 8-(((trifluoromethyl)sulfonyl)oxy)-3,4-dihydro-2,7-naphthyridine-2(1H)-carboxylate (**56**) (100 mg, 0.26 mmol), ethyl (*E*)-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)prop-2-enoate (**57**) (177 mg, 0.78 mmol), tris(dibenzylideneacetone) dipalladium (24 mg, 0.03 mmol), potassium phosphate (277 mg, 1.3 mmol) and S-Phos (21 mg, 0.05 mmol) in 1,4-dioxane (3 mL) and water (0.5 mL) was stirred for 12 h at 100 °C. The solvent was removed under vacuum. The residue was purified by flash chromatography on silica gel eluting with ethyl acetate/petroleum ether (0-20%) to afford **S8** (70 mg, 81% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{18}H_{25}N_2O_4^+$ $[M+H]^+$ 333.2, found 333.2.

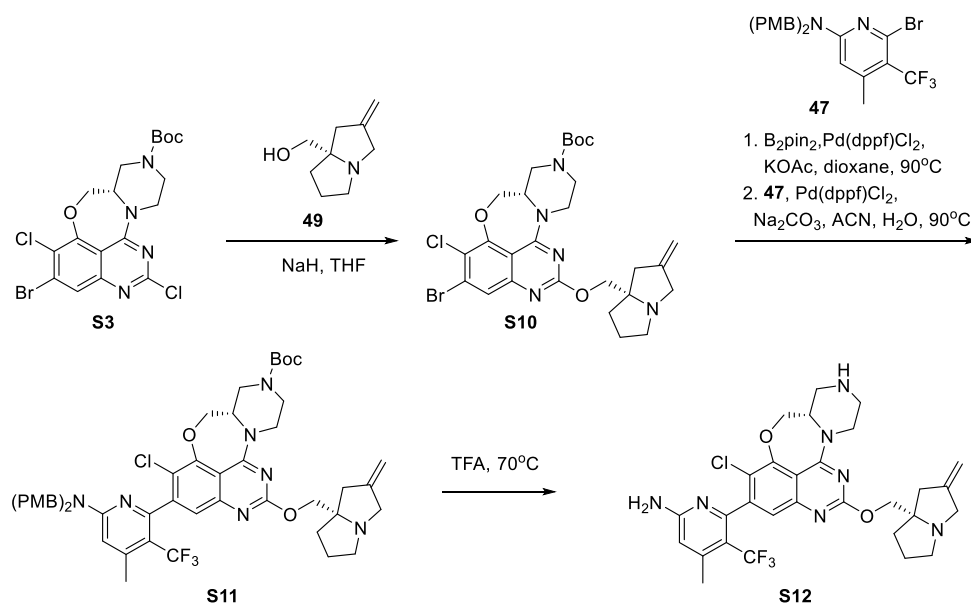
Step 6: A solution of *tert*-butyl (*E*)-8-(3-ethoxy-3-oxoprop-1-en-1-yl)-3,4-dihydro-2,7-naphthyridine-2(1H)-carboxylate (**S8**) (60 mg, 0.18 mmol) and sat. lithium hydroxide in water (0.3 mL) in tetrahydrofuran (1 mL) was stirred at rt for 1 hour. Then pH was adjusted to 6 with 1 M HCl. The solvent was removed under vacuum, the residual was dissolved by dichloromethane (10 mL). After filtration, the solvent was removed under vacuum to afford **58** (50 mg, 91% yield) as a yellow oil. LCMS (ESI): calcd. for $C_{16}H_{21}N_2O_4^+$ $[M+H]^+$ 305.2, found 305.1.

Step 7: A solution of (*E*)-3-(7-(*tert*-butoxycarbonyl)-5,6,7,8-tetrahydro-2,7-naphthyridin-1-yl) acrylic acid (**58**) (70 mg, 0.23 mmol), 6-((*S*)-11-chloro-7-(((*S*)-1-methylpyrrolidin-2-yl)methoxy)-1,2,3,4,13,13a-hexahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazolin-10-yl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**6b**) (260 mg, 0.46 mmol) and DIEA (0.06 mL, 0.35 mmol) in dichloromethane (1 mL) was added HATU (43.73 mg, 0.12 mmol) and stirred at room temperature for 1 hour. The solvent was concentrated under vacuum to afford the crude product. The crude

product was purified by flash chromatography on silica gel eluting with MeOH/dichloromethane (0-10%) to afford compound **S9** (30 mg, 15% yield) as a white solid. LCMS (ESI): calcd. for $C_{42}H_{48}ClF_3N_9O_5^+$ $[M+H]^+$ 850.3, found 850.3.

Step 8: A solution of *tert*-butyl 8-((*E*)-3-((*S*)-10-(6-amino-4-methyl-3-(trifluoromethyl) pyridin-2-yl)-11-chloro-7-(((*S*)-1-methylpyrrolidin-2-yl)methoxy)-3,4,13,13a-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazolin-2(1*H*)-yl)-3-oxoprop-1-en-1-yl)-3,4-dihydro-2,7-naphthyridine-2(1*H*)-carboxylate (**S9**) (30 mg, 0.04 mmol) and trifluoroacetic acid (0.25 mL) in dichloromethane (1 mL) was stirred at room temperature for 1 h. The solvent was concentrated under vacuum. The residue was purified by prep-HPLC to afford compound **27** (15 mg, 53% yield) as a white solid and as FA salt. LCMS (ESI): calcd. for $C_{37}H_{40}ClF_3N_9O_3^+$ $[M+H]^+$ 750.3, found 750.5; 1H NMR (300 MHz, Methanol- d_4) δ 8.55 (s, 1H), 8.39 (d, J = 5.0 Hz, 1H), 7.84 – 7.71 (m, 1H), 7.64 (dd, J = 14.8, 2.1 Hz, 1H), 7.24 (d, J = 5.1 Hz, 1H), 7.16 (d, J = 1.1 Hz, 1H), 6.59 (d, J = 1.3 Hz, 1H), 5.08 – 4.96 (m, 1H), 4.77 – 4.31 (m, 6H), 4.25 (s, 2H), 4.20 – 4.06 (m, 1H), 3.82 – 3.60 (m, 1H), 3.55 – 3.36 (m, 3H), 3.31 – 3.25 (m, 1H), 3.19 (t, J = 6.0 Hz, 2H), 2.96 (t, J = 6.0 Hz, 2H), 2.86 – 2.73 (m, 4H), 2.51 – 2.40 (m, 3H), 2.35 – 2.15 (m, 1H), 2.08 – 1.88 (m, 3H).

Synthesis of compound S12



Step 1: To a solution of (S)-(2-methylenetetrahydro-1H-pyrrolizin-7a(5H)-yl)methanol (**49**) (1.0 g, 6.5 mmol) in tetrahydrofuran (15 mL) was added sodium hydride (0.42 g, 10.6 mmol) at 0 °C and the mixture was stirred for 0.5 h at the same temperature under nitrogen. Then *tert*-butyl (S)-10-bromo-7,11-dichloro-3,4,13,13a-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazoline-2(1H)-carboxylate (**S3**) (2.8 g, 5.7 mmol) was added at 0 °C and stirring was continued for 2 h at rt. Then the reaction was quenched with sat. NH₄Cl (100 mL) and extracted with ethyl acetate (3 × 100 mL). The organic layer was concentrated under vacuum. The residue was purified by flash chromatography on silica gel eluting with DCM/MeOH (10/1) to provide **S10** (3.2 g, 92% yield) as a yellow solid. LCMS (ESI, m/z): calcd. for C₂₇H₃₄BrClN₅O₄⁺ [M+H]⁺ 606.2, found 606.1.

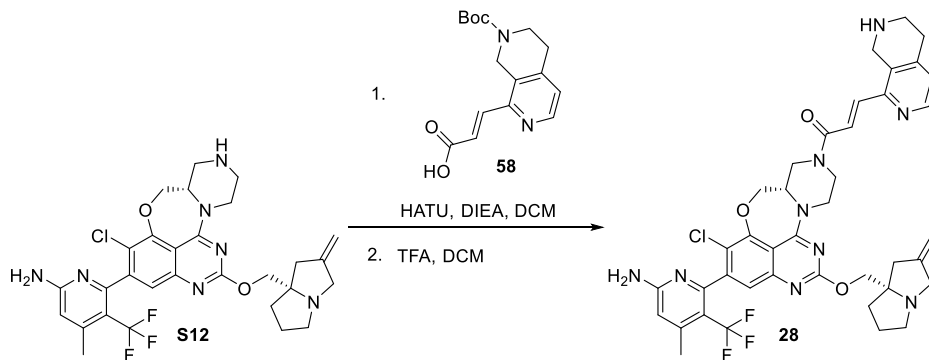
Step 2: Under nitrogen, a mixture of *tert*-butyl (S)-10-bromo-11-chloro-7-(((S)-2-methylenetetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-3,4,13,13a-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazoline-2(1H)-carboxylate (**S10**) (5.0 g, 8.20 mmol), bis(pinacolato)diboron (6.3 g, 24 mmol), potassium acetate (2.4 g, 24 mmol) and (1,1'-bis(diphenylphosphino)ferrocene)dichloropalladium(II) (630 mg, 0.82 mmol) in 1,4-dioxane (100 mL) was stirred for 2 h at 90 °C. The solvent was concentrated under vacuum. The crude product was directly used in the next step without purification.

Step 3: To the above residue in acetonitrile (100 mL) and water (20 mL) was added 6-bromo-*N,N*-bis((4-methoxyphenyl)methyl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**47**) (3.4 g, 6.8 mmol), sodium carbonate (2.7 g, 24 mmol) and (1,1'-bis(diphenylphosphino)ferrocene)dichloropalladium(II) (630 mg, 0.82 mmol) at rt under nitrogen. The resulting mixture was stirred for 2 h at 90 °C. The organic layer was concentrated under vacuum. The residue was purified by flash chromatography on silica gel eluting with DCM/MeOH (9/1) to provide **S11** (1.9 g, 24% yield) as a yellow solid. LCMS (ESI, m/z): calcd. for C₅₀H₅₆ClF₃N₇O₆⁺ [M+H]⁺ 942.4, found 942.

Step 4: A solution of *tert*-butyl (S)-10-(6-(bis(4-methoxybenzyl)amino)-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-11-chloro-7-(((S)-2-methylenetetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-3,4,13,13a-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazoline-2(1H)-carboxylate (**S11**) (8.6 g, 9.13 mmol) in TFA (50 mL) was stirred for 5 h at 70 °C. The solvent was removed under vacuum. The residue was dissolved in water and the pH was adjusted to 9-10 with

ammonium hydroxide. After filtration, the solids were collected to afford 6-((S)-11-chloro-7-(((S)-2-methylenetetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)-1,2,3,4,13,13a-hexahydro pyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazolin-10-yl)-4-methyl-5-(trifluoromethyl) pyridin-2-amine (**S12**) (5.2 g, 95% yield) as a yellow solid. LCMS (ESI, m/z): calcd. for $C_{29}H_{32}ClF_3N_7O_2^+$ $[M+H]^+$ 602.2, found 602.2.

Synthesis of compound 28

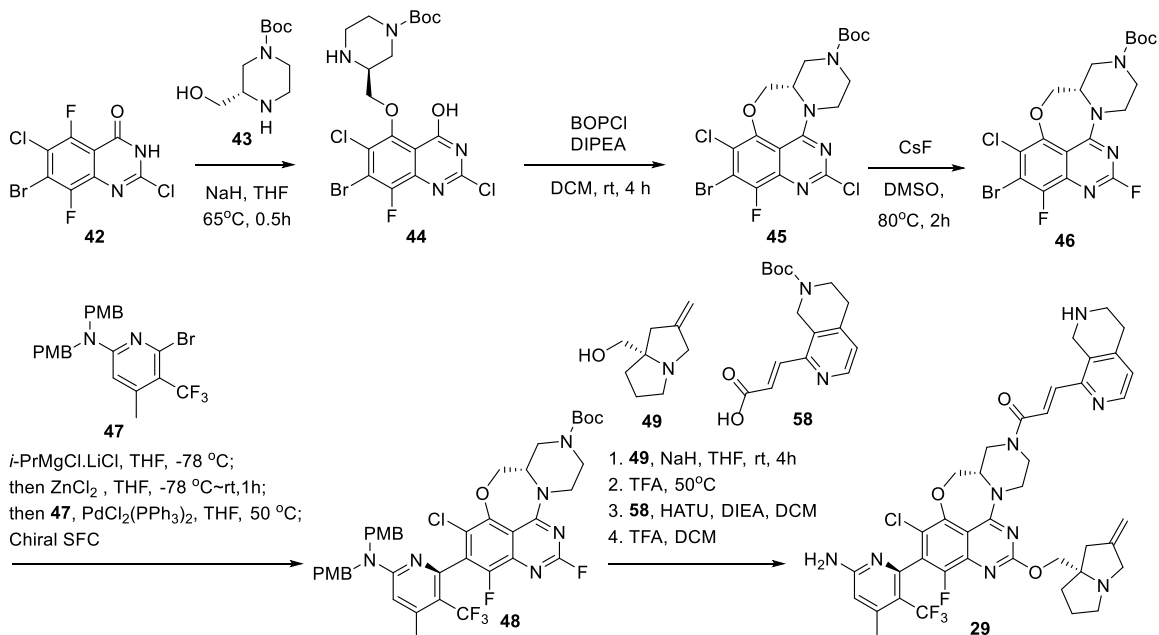


Step 1: A solution of (*E*)-3-(7-tert-butoxycarbonyl-6,8-dihydro- 5*H*-2,7-naphthyridin-1-yl)prop-2-enoic acid (**58**) (50 mg, 0.16 mmol) and 6-((*S*)-11-chloro-7-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)-1,2,3,4,13,13a-hexahydropyrazino[2',1':3,4][1,4]oxazepino [5,6,7-de]quinazolin-10-yl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**S12**) (99 mg, 0.16 mmol) in dichloromethane (1 mL) was added HATU (94 mg, 0.25 mmol) and DIEA (0.09 mL, 0.49 mmol). The solution was stirred for 4 h at 25 °C. The solvent was removed under vacuum. The resulting residue was purified by reverse phase chromatography (0.1% ammonium bicarbonate in water/ACN) to afford the product (80 mg, 55% yield) as a white solid. LCMS (ESI, m/z): calcd. for $C_{45}H_{50}ClF_3N_9O_5^+$ $[M+H]^+$ 888.4, found 888.2.

Step 2: To a solution of *tert*-butyl 8-((*E*)-3-((*S*)-10-(6-amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-11-chloro-7-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)-3,4,13,13a-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazolin-2(1*H*)-yl)-3-oxoprop-1-en-1-yl)-3,4-dihydro-2,7-naphthyridine-2(1*H*)-carboxylate (80 mg, 0.09 mmol) in dichloromethane (1 mL) was added trifluoroacetic acid (0.5 mL) and the mixture was stirred for 1 h at rt. The solvent was concentrated under vacuum. The residue was purified by prep-HPLC to afford the title compound (**28**) (6.0 mg, 9% yield) as a white solid. LCMS (ESI, m/z): calcd. for $C_{40}H_{42}ClF_3N_9O_3^+$ $[M+H]^+$ 788.3, found 788.5. 1H NMR (300 MHz, Methanol- d_4) δ 8.37 (d, *J* =

5.0 Hz, 1H), 7.77 (d, $J = 15.0$ Hz, 1H), 7.62 (d, $J = 15.7$ Hz, 1H), 7.22 (d, $J = 5.0$ Hz, 1H), 7.15 (s, 1H), 6.58 (s, 1H), 5.17 – 4.92 (m, 3H), 4.73 – 4.43 (m, 4H), 4.40 – 4.29 (m, 2H), 4.24 – 4.07 (m, 3H), 3.96 – 3.80 (m, 1H), 3.75 – 3.38 (m, 4H), 3.21 – 3.04 (m, 2H), 2.98 – 2.75 (m, 4H), 2.65 – 2.35 (m, 4H), 2.29 – 2.12 (m, 1H), 2.08 – 1.88 (m, 3H), 1.43 – 1.21 (m, 1H).

Synthesis of compound 29



Step 1: To a solution of *tert*-butyl (S)-3-(hydroxymethyl)piperazine-1-carboxylate (**43**) (1.96 g, 9.06 mmol) in tetrahydrofuran (5 mL) was added sodium hydride (0.73 g, 18.1 mmol) at 0 °C and the mixture was stirred for 0.5 h at the same temperature. Then 7-bromo-2,6-dichloro-5,8-difluoroquinazolin-4-ol (**42**) (3.0 g, 9.09 mmol) was added at 0 °C and the mixture was stirred for 1 h at 65 °C. The resulting solution was quenched with sat. NH₄Cl and extracted with ethyl acetate. The solvent was removed under vacuum to afford crude **44** (4.3 g, 90% yield) as a white solid. LCMS (ESI, m/z): calcd. for C₁₈H₂₁BrCl₂FN₄O₄⁺ [M+H]⁺ 525.0, found 525.1.

Step 2: To a mixture of *tert*-butyl (S)-3-(((7-bromo-2,6-dichloro-8-fluoro-4-hydroxyquinazolin-5-yl)oxy)methyl)piperazine-1-carboxylate (**44**) (4.3g, 8.17 mmol) in dichloromethane (80 mL) were added bis(2-oxo-3-oxazolidinyl)phosphinic chloride (3.2 g, 12.57 mmol) and *N,N*-diisopropylethylamine (4.31 mL, 24.76 mmol), and the mixture was stirred 2 h at rt. The resulting solution was diluted with dichloromethane and extracted with brine, dried over anhydrous sodium sulfate and concentrated. The residue was purified by flash chromatography on silica gel eluting

with petrol ether/ethyl acetate (62:38) to afford **45** (1.4 g, 34% yield) as a yellow solid. LCMS (ESI, m/z): calcd. for $C_{18}H_{19}BrCl_2FN_4O_3^+$ $[M+H]^+$ 507.0, found 507.1.

Step 3: To a mixture of *tert*-butyl (7*S*)-12-bromo-11,16-dichloro-13-fluoro-9-oxa-2,5,15,17-tetrazatetracyclo[8.7.1.0^{2,7}.0^{14,18}]octadeca-1(17),10(18),11,13,15-pentaene-5-carboxylate (**45**) (4.0 g, 7.87 mmol) in dimethyl sulfoxide (15 mL) was added caesium fluoride (6.0 g, 39.5 mmol), and the mixture was stirred for 1h at 80 °C. The resulting solution was diluted with water and extracted with ethyl acetate. The solvent was concentrated. The residue was purified by flash chromatography on silica gel eluting with petrol ether/ethyl acetate (57:43) to afford **46** (3.1 g, 80% yield) as a yellow solid. LCMS (ESI, m/z): calcd. for $C_{18}H_{19}BrClF_2N_4O_3^+$ $[M+H]^+$ 491.0, found 491.2.

Step 4: Under nitrogen, a mixture of *tert*-butyl (5*S*)-10-bromo-11-chloro-7,9-difluoro-3,4,13,13a-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazoline-2(1*H*)-carboxylate (**46**) (1.6 g, 3.25 mmol) in tetrahydrofuran (20 mL) at -78 °C was added 1.3 M isopropylmagnesium chloride lithium chloride complex in THF (3.04 mL, 3.95 mmol), and the mixture was stirred at -78 °C for 30 min. Then 2 M zinc chloride in THF (1.92 mL, 3.84 mmol) was added and the mixture was stirred at -78 °C for 5 min and then stirred at room temperature for 1h. The above solution was transferred into a degassed slurry of 6-bromo-*N,N*-bis[(4-methoxyphenyl)methyl]-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**47**) (1.62 g, 3.26 mmol) and bis(triphenylphosphine)-palladium(II) chloride (0.48 g, 0.68 mmol) in tetrahydrofuran (10 mL) and the mixture was stirred at 50 °C for 16 h. The reaction mixture was quenched with water and extracted with ethyl acetate. The solvent was concentrated under vacuum and the residue was purified by flash eluting with ethyl acetate/PE (0-50%) to afford *tert*-butyl (13*aS*)-10-(6-(bis(4-methoxybenzyl)amino)-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-11-chloro-7,9-difluoro-3,4,13,13a-tetrahydropyrazino-[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazoline-2(1*H*)-carboxylate (2 g, 75% yield) as a yellow solid. LCMS (ESI, m/z): calcd. for $C_{41}H_{41}ClF_5N_6O_5^+$ $[M+H]^+$ 827.3, found 827.1. The crude product (2.0 g, 2.42 mmol) was purified by chiral SFC with the following conditions to afford **48** (0.65 g, 33% yield): Column: CHIRAL ART Cellulose-SB, 3*25 cm, 5 μ m; Mobile Phase A: CO₂, Mobile Phase B: MEOH(0.1% 2M NH₃-MEOH); Flow rate: 70 mL/min; Gradient: isocratic 50% B; Column Temperature(°C): 35; Back Pressure(bar): 100; Wave Length: 220 nm; RT1(min):

9.82; RT2(min): 12.86; Sample Solvent: MeOH: DCM=2: 1; Injection Volume: 2 mL; Number Of Runs: 20.

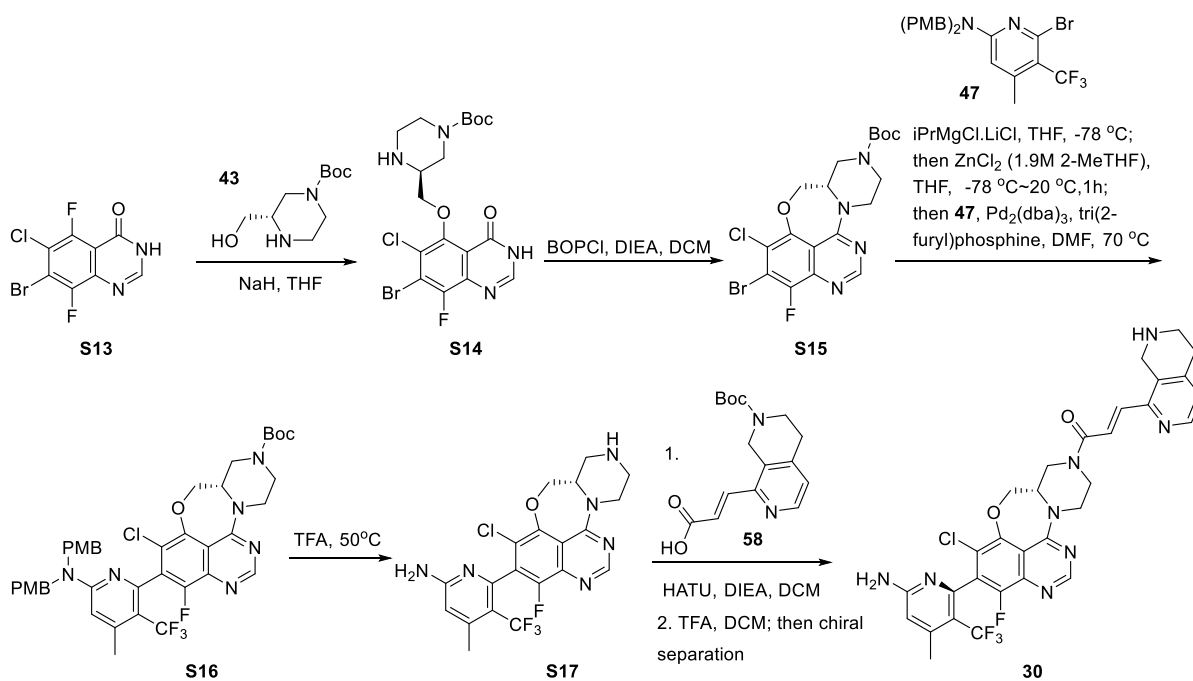
Step 5: To a solution of ((8*S*)-6-methylene-2,3,5,7-tetrahydro-1*H*-pyrrolizin-8-yl)methanol (**49**) (21 mg, 0.14 mmol) in tetrahydrofuran (1 mL) was added sodium hydride (15.0mg, 0.3800mmol) at 0 °C and the mixture was stirred for 15 min at rt. Then *tert*-butyl (10*R*,13*aS*)-10-(6-(bis(4-methoxybenzyl)amino)-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-11-chloro-7,9-difluoro-3,4,13,13*a*-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-*de*]quinazoline-2(1*H*)-carboxylate (**48**) (100 mg, 0.12 mmol) was added and the mixture was stirred at rt for 2 h. The reaction was quenched with water. The reaction mixture was extracted with ethyl acetate and the organic layers were combined. The solvent was concentrated under vacuum to afford product **50** (110 mg, 95% yield) as a yellow solid. LCMS (ESI): calcd. for C₅₀H₅₅ClF₄N₇O₆⁺ [M+H]⁺ 960.4, found 960.4.

Step 6: A solution of *tert*-butyl (10*S*,13*aS*)-10-(6-(bis(4-methoxybenzyl)amino)-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-11-chloro-9-fluoro-7-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)-3,4,13,13*a*-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-*de*]quinazoline-2(1*H*)-carboxylate (100 mg, 0.10 mmol) in TFA (3 mL) was stirred at 70 °C for 2 h. The solvent was concentrated under vacuum to afford product **51** (60 mg, 93% yield) as a yellow oil. LCMS (ESI): calcd. for C₂₉H₃₁ClF₄N₇O₂⁺ [M+H]⁺ 620.2, found 620.

Step 7: A solution of 6-((10*R*,13*aS*)-11-chloro-9-fluoro-7-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)-1,2,3,4,13,13*a*-hexahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-*de*]quinazolin-10-yl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (60 mg, 0.10 mmol), (*E*)-3-(7-*tert*-butoxycarbonyl-6,8-dihydro-5*H*-2,7-naphthyridin-1-yl)prop-2-enoic acid (**58**) (45 mg, 0.15 mmol), HATU (80 mg, 0.21 mmol) and DIEA (0.2 mL, 1.15 mmol) in dichloromethane (1mL) was stirred at rt for 2 h. The solvent was concentrated under vacuum. The residue was purified by reverse phase chromatography (acetonitrile/0.1% ammonium bicarbonate in water) to afford *tert*-butyl 8-(((*E*)-3-((10*R*,13*aS*)-10-(6-amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-11-chloro-9-fluoro-7-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)-3,4,13,13*a*-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-*de*]quinazolin-2(1*H*)-yl)-3-oxoprop-1-en-1-yl)-3,4-dihydro-2,7-naphthyridine-2(1*H*)-carboxylate (80 mg, 91% yield) as a yellow oil. LCMS (ESI): calcd. for C₄₅H₄₉ClF₄N₉O₅⁺ [M+H]⁺ 906.4, found 906.1.

Step 8: A solution of *tert*-butyl 8-((*E*)-3-((10*R*,13*aS*)-10-(6-amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-11-chloro-9-fluoro-7-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)-3,4,13,13*a*-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-*de*]quinazolin-2(1*H*)-yl)-3-oxoprop-1-en-1-yl)-3,4-dihydro-2,7-naphthyridine-2(1*H*)-carboxylate (80 mg, 0.09 mmol) and trifluoroacetic acid (1 mL) in dichloromethane (1 mL) was stirred at rt for 2 h. The solvent was concentrated under vacuum. The crude product was purified by prep-HPLC to afford **29** (10 mg, 14% yield) as a white solid and as FA salt. LCMS (ESI): calcd. for C₄₀H₄₁ClF₄N₉O₃⁺ [M+H]⁺ 806.3, found 806.4; ¹H NMR (300 MHz, Methanol-*d*₄) δ 8.51 (s, 1H), 8.45 (d, *J* = 5.0 Hz, 1H), 7.84 – 7.53 (m, 2H), 7.28 (d, *J* = 5.0 Hz, 1H), 6.63 (s, 1H), 5.27 (s, 2H), 5.07 (d, *J* = 13.2 Hz, 1H), 4.95 – 4.90 (m, 1H), 4.83 – 4.48 (m, 6H), 4.43 – 4.26 (m, 4H), 4.25 – 4.11 (m, 1H), 3.86 (d, *J* = 14.4 Hz, 1H), 3.73 (d, *J* = 16.4 Hz, 2H), 3.59 – 3.39 (m, 2H), 3.23 – 3.14 (m, 1H), 3.10 – 2.95 (m, 3H), 2.77 (d, *J* = 16.0 Hz, 1H), 2.46 (s, 3H), 2.39 (s, 1H), 2.28 – 1.88 (m, 3H).

Synthesis of compound 30



Step 1: To a solution of *tert*-butyl (*S*)-3-(hydroxymethyl)piperazine-1-carboxylate (**43**) (0.92 g, 4.26 mmol) in tetrahydrofuran (10 mL) was added sodium hydride (0.51 g, 21.32 mmol) and the mixture was stirred at 0 °C for 10 min. Then 7-bromo-6-chloro-5,8-difluoro-3*H*-quinazolin-4-one (**S13**) (2.1 g, 7.11 mmol) was added and the mixture was stirred at rt for 1 h. The reaction was

quenched with water and extracted with ethyl acetate. The solvent was concentrated under vacuum. The residue was purified by flash chromatography on silica gel eluting with dichloromethane/methanol (10%) to afford **S14** (1.80 g, 52% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{18}H_{22}BrClFN_4O_4^+$ $[M+H]^+$ 491.1, found 491.0.

Step 2: A solution of *tert*-butyl (S)-3-(((7-bromo-6-chloro-8-fluoro-4-oxo-3,4-dihydroquinazolin-5-yl)oxy)methyl)piperazine-1-carboxylate (**S14**) (380 mg, 0.77 mmol), bis(2-oxo-3-oxazolidinyl) phosphinic chloride (294 mg, 1.16 mmol) and DIEA (0.4 mL) in dichloromethane (2 mL) was stirred at rt for 2 h. The reaction was quenched with water and extracted with dichloromethane, and the organic layers were combined. The solvent was concentrated under vacuum. The residue was purified by flash chromatography on silica gel eluting with ethyl acetate/petroleum ether (45%) to afford **S15** (240 mg, 66% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{18}H_{20}BrClFN_4O_3^+$ $[M+H]^+$ 473.0, found 473.0.

Step 3: Under nitrogen, to a solution of *tert*-butyl (S)-5-bromo-6-chloro-4-fluoro-8a,9,11,12-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazoline-10(8*H*)-carboxylate (**S15**) (210 mg, 0.44 mmol) in tetrahydrofuran (1mL) was added 1.3 M *i*-PrMgCl-LiCl in THF (0.4 mL, 0.53mmol) and the mixture was stirred for 0.5h at -78°C. Then 2 M Zinc Chloride in 2-Methyltetrahydrofuran (0.27 mL, 0.53 mmol) was added and the mixture was stirred at rt for 1 h. Under nitrogen, this solution was added to a solution of 6-bromo-*N,N*-bis (4-methoxybenzyl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**47**) (263 mg, 0.53 mmol), tris(dibenzylideneacetone)-dipalladium (40 mg, 0.04 mmol) and tri(2-furyl)phosphine (20 mg, 0.09mmol) in *N,N*-Dimethylformamide (1 mL) and the resulting mixture was stirred at 70 °C for 2 h. The reaction mixture was quenched with water and extracted with ethyl acetate, then concentrated under vacuum. The residue was purified by flash chromatography on silica gel eluting with ethyl acetate/petroleum ether (65%) to afford **S16** (180 mg, 50% yield) as a yellow oil. LCMS (ESI): calcd. for $C_{41}H_{42}ClF_4N_6O_5^+$ $[M+H]^+$ 809.3, found 809.3.

Step 4: A solution of *tert*-butyl (8a*S*)-5-(6-(bis(4-methoxybenzyl)amino)-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-4-fluoro-8a,9,11,12-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-de]quinazoline-10(8*H*)-carboxylate (**S16**) (170 mg, 0.21mmol) and trifluoroacetic acid (1 mL) in dichloromethane (1 mL) was stirred at 50 °C for 4 h. The solvent

was concentrated under vacuum. The residue was purified by reverse phase chromatography (acetonitrile/10 mM ammonium bicarbonate in water) to afford **S17** (60mg, 61% yield) as a yellow solid. LCMS (ESI, m/z): calcd. for $C_{20}H_{18}ClF_4N_6O^+$ $[M+H]^+$ 469.1, found 469.1.

Step 5: A solution of (*E*)-3-(7-*tert*-butoxycarbonyl-6,8-dihydro-5*H*-2,7-naphthyridin-1-yl) prop-2-enoic acid (**58**) (34 mg, 0.13 mmol), 6-((8*aS*)-6-Chloro-4-fluoro-8*a*,9,10,11,12-hexahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-*de*]quinazolin-5-yl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**S17**) (60 mg, 0.13 mmol), HATU (58 mg, 0.15 mmol) and DIEA (0.5 mL) in *N,N*-dimethylformamide (1 mL) was stirred at rt for 2 h. The residue was directly purified by reverse phase chromatography (acetonitrile/10 mM Ammonium bicarbonate in water) to afford the product (60 mg, 62% yield) as a yellow solid. LCMS (ESI, m/z): calcd. for $C_{36}H_{36}ClF_4N_8O_4^+$ $[M+H]^+$ 755.3, found 755.2.

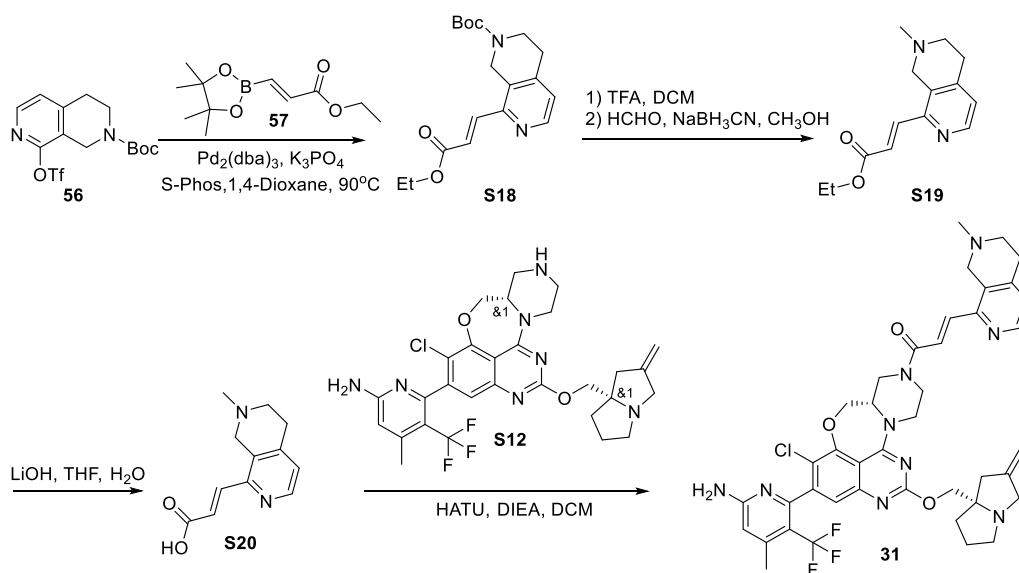
Step 6: A solution of *tert*-butyl 8-((*E*)-3-((8*aS*)-5-(6-amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-4-fluoro-8*a*,9,11,12-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-*de*]quinazolin-10(8*H*)-yl)-3-oxoprop-1-en-1-yl)-3,4-dihydro-2,7-naphthyridine-2(1*H*)-carboxylate (**57** mg, 0.08 mmol) and trifluoroacetic acid (1 mL) in dichloromethane (1 mL) was stirred at rt for 1 h. The solvent was concentrated under vacuum. The residue was purified by chiral HPLC to afford compound **30** and its atropisomer **30'**.

(*E*)-1-((5*R*,8*aS*)-5-(6-Amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-4-fluoro-8*a*,9,11,12-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-*de*]quinazolin-10(8*H*)-yl)-3-(5,6,7,8-tetrahydro-2,7-naphthyridin-1-yl)prop-2-en-1-one **30** (16.4 mg, 38% yield) as a white solid; LCMS (ESI): calcd. for $C_{31}H_{28}ClF_4N_8O_2^+$ $[M+H]^+$ 655.2, found 655.2; 1H NMR (300 MHz, Methanol-*d*₄) δ 8.60 (s, 1H), 8.36 (d, *J* = 5.0 Hz, 1H), 7.77 (d, *J* = 14.5 Hz, 1H), 7.62 (d, *J* = 14.9 Hz, 1H), 7.21 (d, *J* = 5.0 Hz, 1H), 6.63 (d, *J* = 1.4 Hz, 1H), 5.14 (d, *J* = 13.5 Hz, 1H), 4.66 (s, 3H), 4.57 – 4.36 (m, 1H), 4.26 – 4.00 (m, 3H), 3.86 – 3.58 (m, 1H), 3.53 – 3.36 (m, 2H), 3.31 – 3.20 (m, 1H), 3.10 (t, *J* = 5.9 Hz, 2H), 2.91 (t, *J* = 5.8 Hz, 2H), 2.55 – 2.35 (m, 3H), 1.39 (dd, *J* = 35.6, 6.8 Hz, 2H); tR = 2.509 min (CHIRALPAK IF 3, 4.6*50 mm, 3 μ m, (Hex:dichloromethane=3:1)(0.1%DEA):EtOH=85:15, 1 ml/min, 254 nm).

(*E*)-1-((5*S*,8*aS*)-5-(6-Amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-4-fluoro-8*a*,9,11,12-tetrahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-*de*]quinazolin-10(8*H*)-yl)-3-

(5,6,7,8-tetrahydro-2,7-naphthyridin-1-yl)prop-2-en-1-one **30'** (16.1 mg, 37% yield) as a white solid; LCMS (ESI): calcd. for $C_{31}H_{28}ClF_4N_8O_2^+$ $[M+H]^+$ 655.2, found 655.2; 1H NMR (300 MHz, Methanol- d_4) δ 8.60 (s, 1H), 8.36 (d, J = 5.0 Hz, 1H), 7.78 (d, J = 14.5 Hz, 1H), 7.62 (d, J = 14.9 Hz, 1H), 7.21 (d, J = 5.0 Hz, 1H), 6.63 (s, 1H), 5.19 – 5.02 (m, 1H), 4.75 – 4.55 (m, 3H), 4.53 – 4.32 (m, 1H), 4.28 – 4.09 (m, 3H), 3.68 (t, J = 11.8 Hz, 1H), 3.58 – 3.36 (m, 2H), 3.10 (t, J = 5.9 Hz, 2H), 2.91 (t, J = 5.9 Hz, 2H), 2.60 – 2.29 (m, 3H); tR = 3.571 min (CHIRALPAK IF 3, 4.6*50 mm, 3 μ m, MTBE(0.1%DEA):MeOH=50:50, 1 ml/min, 254 nm).

Synthesis of compound 31



Step 1: Under nitrogen, a solution of *tert*-butyl 8-(((trifluoromethyl)sulfonyl)oxy)-3,4-dihydro-2,7-naphthyridine-2(1*H*)-carboxylate (**56**) (100 mg, 0.26 mmol), ethyl (*E*)-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)prop-2-enoate (**57**) (177 mg, 0.78 mmol), tris(dibenzylideneacetone)dipalladium (24 mg, 0.03 mmol), potassium phosphate (277 mg, 1.3 mmol) and SPhos (21 mg, 0.05 mmol) in 1,4-Dioxane (3 mL) was stirred for 12 h at 100 °C. The solvent was removed under vacuum. The residue was purified by flash chromatography on silica gel eluting with petrol ether/ethyl acetate (0-20%) to afford **S18** (70 mg, 81% yield) as a yellow solid. LCMS (ESI, m/z): calcd. for $C_{18}H_{25}N_2O_4^+$ $[M+H]^+$ 333.2, found 333.2.

Step 2: To a solution of *tert*-butyl (*E*)-8-(3-ethoxy-3-oxoprop-1-en-1-yl)-3,4-dihydro-2,7-naphthyridine-2(1*H*)-carboxylate (**S18**) (100 mg, 0.30 mmol) in dichloromethane (1 mL) was added TFA (0.5 mL) and the mixture was stirred for 1 h at rt. The solvent was removed to afford

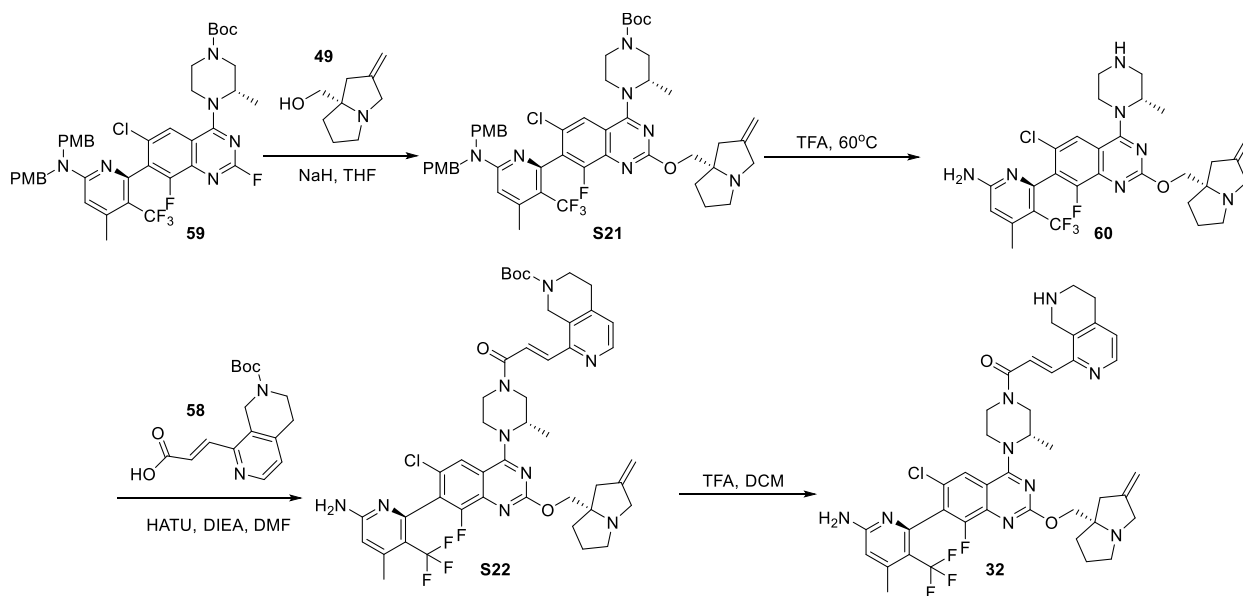
crude ethyl (*E*)-3-(5,6,7,8-tetrahydro-2,7-naphthyridin-1-yl)acrylate (50 mg, 72% yield) as a yellow oil.

Step 3: To a solution of ethyl (*E*)-3-(5,6,7,8-tetrahydro-2,7-naphthyridin-1-yl)acrylate (50 mg, 0.22 mmol) in methanol (2 mL) were added 33% formaldehyde in water (0.1 mL) and sodium cyanoborohydride (68 mg, 1.08 mmol) and the mixture was stirred for 2 h at rt. The resulting solution was directly purified by reverse phase chromatography (acetonitrile/0.1% formic acid in water) to afford **S19** (30 mg, 57% yield) as a yellow oil. LCMS (ESI, *m/z*): calcd. for $C_{14}H_{19}N_2O_2^+$ $[M+H]^+$ 247.1, found 247.2.

Step 4: To a solution of ethyl (*E*)-3-(7-methyl-6,8-dihydro-5*H*-2,7-naphthyridin-1-yl)prop-2-enoate (**S19**) (30 mg, 0.12 mmol) in tetrahydrofuran (0.5 mL) was added saturated aqueous lithium hydroxide (0.5 mL) and the mixture was stirred for 4 h at rt. Then 1 M HCl was added to adjust the pH to 6. The resulting solution was directly purified by reverse phase chromatography (acetonitrile/10 mmol NH_4HCO_3 in water) to afford **S20** (20 mg, 75% yield) as a yellow oil. LCMS (ESI, *m/z*): calcd. for $C_{12}H_{15}N_2O_2^+$ $[M+H]^+$ 219.1, found 219.2.

Step 5: To a solution of 6-((*S*)-11-chloro-7-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)-1,2,3,4,13,13*a*-hexahydropyrazino[2',1':3,4][1,4]oxazepino[5,6,7-*de*]quinazolin-10-yl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**S12**) (55 mg, 0.03 mmol) and (*E*)-3-(7-methyl-6,8-dihydro-5*H*-2,7-naphthyridin-1-yl)prop-2-enoic acid (**S20**) (20 mg, 0.09 mmol) in dichloromethane (1 mL) was added DIEA (35 mg, 0.27 mmol) and HATU (35 mg, 0.27 mmol) and the mixture was stirred at rt for 1 h. The solvent was concentrated and the crude product was purified by prep-HPLC to afford **31** (12 mg, 16% yield) as a white solid. LCMS (ESI, *m/z*): calcd. for $C_{41}H_{44}ClF_3N_9O_3^+$ $[M+H]^+$ 802.3, found 802.3. 1H NMR (400 MHz, Methanol-*d*₄) δ 8.56 (s, 0.38H), 8.44 – 8.29 (m, 1H), 7.86 – 7.76 (m, 1H), 7.68 – 7.60 (m, 1H), 7.25 (d, *J* = 5.0 Hz, 1H), 7.16 – 7.10 (m, 1H), 6.59 (s, 1H), 5.54 – 4.93 (m, 2H), 4.77 – 4.56 (m, 3H), 4.54 – 4.42 (m, 1H), 4.40 – 4.23 (m, 2H), 4.21 – 4.02 (m, 2H), 3.90 – 3.79 (m, 2H), 3.76 – 3.60 (m, 1H), 3.57 – 3.36 (m, 2H), 3.28 – 3.20 (m, 1H), 3.06 – 2.97 (m, 2H), 2.95 – 2.73 (m, 4H), 2.57 (s, 3H), 2.45 (d, *J* = 2.2 Hz, 3H), 2.29 – 2.05 (m, 1H), 2.04 – 1.86 (m, 3H), 1.84 – 1.75 (m, 2H).

Synthesis of compound 32



Step 1: To a solution of (*S*)-(2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methanol (**49**) in tetrahydrofuran (5 mL) was added sodium hydride (31 mg, 1.29 mmol) at 0 °C and the mixture was stirred for 30 min at rt. Then *tert*-butyl (*S*)-4-(7-((*R*)-6-(bis(4-methoxybenzyl)amino)-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-2,8-difluoroquinazolin-4-yl)-3-methyl-piperazine-1-carboxylate (**59**) (300 mg, 0.37 mmol) was added and the mixture was stirred at rt for 2h. The reaction was quenched with sat. NH₄Cl, then purified by reverse flash chromatography acetonitrile/10 mmol NH₄HCO₃ in water (10-70) to afford **S21** (300 mg, 86% yield) as a yellow oil. LCMS (ESI): calcd. for C₅₀H₅₇ClF₄N₇O₅⁺ [M+H]⁺ 946.4, found 946.2.

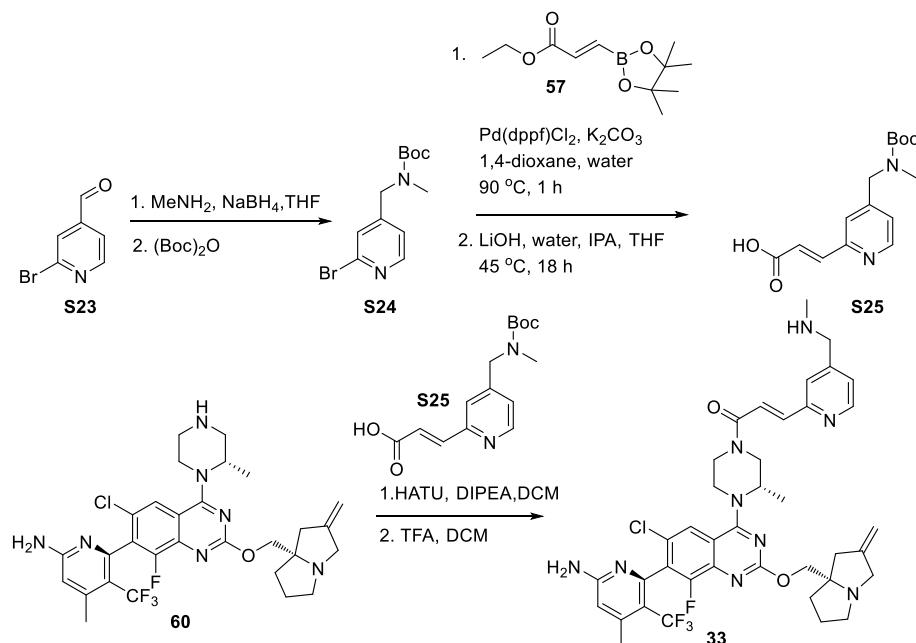
Step 2: A solution of *tert*-butyl (*S*)-4-(7-((*R*)-6-(bis(4-methoxybenzyl)amino)-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)quinazolin-4-yl)-3-methylpiperazine-1-carboxylate (**S21**) (430 mg, 0.45 mmol) in trifluoroacetic acid (1 mL) was stirred at 60°C for 3 h. The residue was purified by reverse flash chromatography acetonitrile/0.1% of FA in water to afford **60** (230 mg, 84% yield) as light yellow oil. LCMS (ESI): calcd. for C₂₉H₃₃ClF₄N₇O⁺ [M+H]⁺ 606.2, found 606.2.

Step 3: To a solution of (*R*)-6-(6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)-4-(((*S*)-2-methylpiperazin-1-yl)quinazolin-7-yl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**60**) (150 mg, 0.25 mmol) and (*E*)-3-(7-*tert*-butoxycarbonyl-6,8-dihydro-

5H-2,7-naphthyridin-1-yl)prop-2-enoic acid (**58**) (225 mg, 0.74 mmol) in tetrahydrofuran (2 mL) were added HATU (290 mg, 0.76 mmol) and DIEA (0.5 mL). The resulting solution was stirred for 3 h at rt. The solvent was removed and the residue was purified by reverse flash chromatography acetonitrile /10 mmol/L of NH_4HCO_3 in water (40-80) to afford **S22** (50 mg, 23% yield) as a light yellow oil. LCMS (ESI): calcd. for $\text{C}_{45}\text{H}_{51}\text{ClF}_4\text{N}_9\text{O}_4^+$ $[\text{M}+\text{H}]^+$ 892.4, found 892.4.

Step 4: To a solution of *tert*-butyl 8-((*E*)-3-((*S*)-4-((*R*)-7-(6-amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7a(5*H*)-yl)methoxy)quinazolin-4-yl)-3-methylpiperazin-1-yl)-3-oxoprop-1-en-1-yl)-3,4-dihydro-2,7-naphthyridine-2(1*H*)-carboxylate (**S22**) (50 mg, 0.06 mmol) in dichloromethane (1 mL) was added trifluoroacetic acid (1 mL) and the mixture was stirred for 1 h at rt. The solvent was concentrated under vacuum. The crude product was purified by prep-HPLC to afford compound **32** (12.7 mg, 28% yield) as a white solid. LCMS (ESI): calcd. for $\text{C}_{40}\text{H}_{43}\text{ClF}_4\text{N}_9\text{O}_2^+$ $[\text{M}+\text{H}]^+$ 792.3, found 792.4; ^1H NMR (300 MHz, Methanol- d_4) δ 8.36 (d, J = 5.0 Hz, 1H), 7.92 – 7.71 (m, 2H), 7.68 – 7.51 (m, 1H), 7.22 (d, J = 5.0 Hz, 1H), 6.63 (s, 1H), 5.07 (s, 2H), 4.95 (s, 1H), 4.63 – 4.33 (m, 4H), 4.18 (s, 2H), 3.93 – 3.69 (m, 3H), 3.48 (d, J = 14.2 Hz, 2H), 3.32 – 3.26 (m, 2H), 3.19 – 3.05 (m, 2H), 2.97 – 2.81 (m, 4H), 2.69 – 2.45 (m, 4H), 2.32 – 2.13 (m, 1H), 2.11 – 1.85 (m, 3H), 1.48 (t, J = 6.4 Hz, 3H).

Synthesis of compound 33



Step 1: A mixture of 2-bromopyridinecarboxaldehyde (**S23**) (0.5 g, 2.61 mmol) and methyl amine (2.6 mL 2M, 6.20 mmol) in THF was stirred at 25 °C for 18 h. The reaction mixture was concentrated and redissolved in methanol-THF (5 mL 1:1 mixture) and sodium borohydride (0.1 g, 2.61 mmol) was added. The reaction mixture was stirred at 25 °C for 20 min and was quenched with water. The reaction mixture was extracted with iPrOAc and the organic layer was washed with brine, dried over sodium sulfate and concentrated. The residue was purified by column chromatography (silica gel, 0-5% MeOH/DCM) to obtain 1-(2-bromopyridin-4-yl)-N-methylmethanamine (0.26 g, 50%): ¹H NMR (400 MHz, CDCl₃) δ 8.36 – 8.20 (m, 1H), 7.55 – 7.42 (m, 1H), 7.23 – 7.21 (m, 1H), 3.75 (s, 2H), 2.45 (s, 3H).

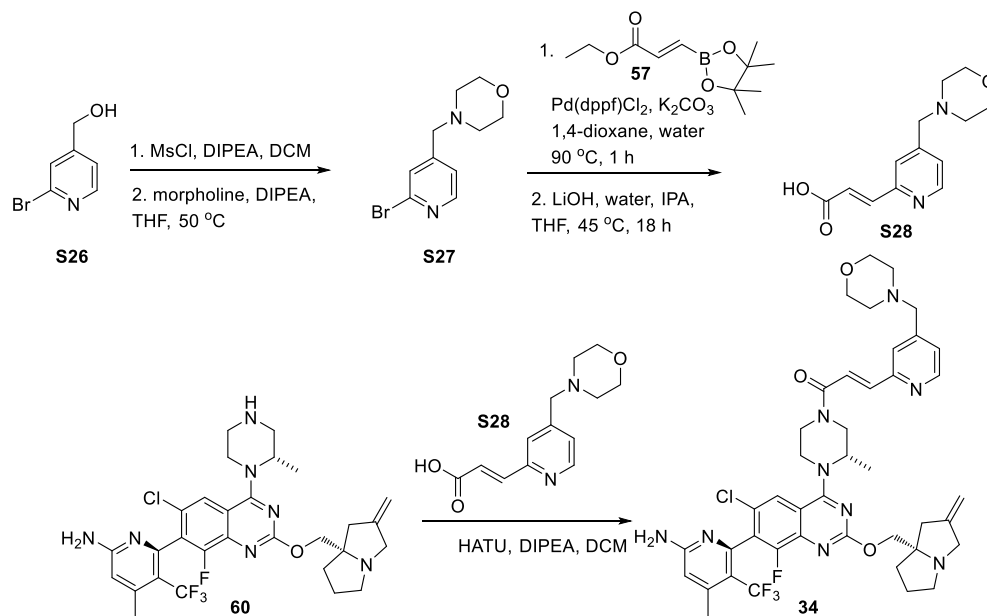
Step 2: To 1-(2-bromopyridin-4-yl)-N-methylmethanamine (0.26 g, 1.3 mmol) in THF (5 ml) was added tert-butyl dicarbonate (310 mg, 1.42 mmol) and the mixture was stirred at 25 °C for 18 h. The reaction mixture was concentrated and purified by column chromatography (silica gel, 0-50% iPrOAc/heptane) to obtain tert-butyl ((2-bromopyridin-4-yl)methyl)(methyl)carbamate (**S24**) (353 mg, 90.6%). ¹H NMR (400 MHz, CDCl₃) δ 8.30 (brs, 1H), 7.37 – 7.30 (m, 1H), 7.08 (brs, 1H), 4.38 (brs, 2H), 2.85 br(s, 3H), 1.53 – 1.33 (brs, 9H).

Step 3: A mixture of tert-butyl ((2-bromopyridin-4-yl)methyl)(methyl)carbamate (**S24**) (350 mg, 1.6 mmol), ethyl (E)-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)acrylate (**57**) (0.45 g, 1.71 mmol), potassium carbonate (0.44 g, 4.7 mmol), and Pd(dppf)Cl₂ (0.04 g, 0.08 mmol) in 1,4-dioxane/water (3 mL:1 mL) was heated at 90 °C for 1h. The reaction mixture was cooled, and diluted with iPrOAc. The organic layer was washed with brine, dried over sodium sulfate and concentrated. The residue was purified by column chromatography (silica gel, 0-100% iPrOAc/heptane) to obtain ethyl (E)-3-(4-(((tert-butoxycarbonyl)(methyl)amino)methyl)pyridin-2-yl)acrylate (0.17 g, 47%): ¹H NMR (400 MHz, CDCl₃) δ 7.66 (d, *J* = 15.7 Hz, 1H), 7.25 (dttt, *J* = 3.5, 2.0, 1.0, 0.5 Hz, 1H), 7.11 (d, *J* = 5.0 Hz, 1H), 6.92 (d, *J* = 15.7 Hz, 1H), 4.44 (s, 2H), 4.28 (q, *J* = 7.1 Hz, 2H), 2.91 (s, 1H), 2.86 (s, 1H), 1.55 – 1.39 (m, 9H), 1.34 (t, *J* = 7.1 Hz, 3H).

Step 4: Ethyl (E)-3-(4-(((tert-butoxycarbonyl)(methyl)amino)methyl)pyridin-2-yl)acrylate (0.17 g, 0.54 mmol) was treated with LiOH as described for **34** (below) and the reaction mixture was acidified with dilute HCl and extracted with iPrOAc. The organic layer was washed with brine, dried over sodium sulfate and concentrated to obtain (E)-3-(4-(((tert-butoxycarbonyl)(methyl)amino)methyl)pyridin-2-yl)acrylic acid (**S25**) (0.09 g, 58%). ¹H NMR (400 MHz, MeOD) δ 8.54 (d, *J* = 5.1 Hz, 1H), 7.65 (d, *J* = 15.9 Hz, 1H), 7.49 (d, *J* = 1.8 Hz, 1H), 7.31 – 7.17 (m, 1H), 6.84 (d, *J* = 15.9 Hz, 1H), 4.50 (s, 2H), 2.93 (s, 3H), 1.61 – 1.35 (m, 9H).

Steps 5 and 6: (E)-3-(4-(((tert-Butoxycarbonyl)(methyl)amino)methyl)pyridin-2-yl)acrylic acid was coupled with **60** as described for **34** (below) and deprotected with TFA at 25 °C in DCM. The product was purified using a SPE column to obtain **33** (8.1 mg, 31% yield) over two steps. Purity 94%; LC/MS (ESI): calcd. for C₃₉H₄₃ClF₄N₉O₂⁺ [M+H]⁺ 780.3, found 780.3. ¹H NMR (400 MHz, DMSO-*d*₆) δ 8.59 (d, *J* = 5.0 Hz, 1H), 7.84 (s, 1H), 7.70 (s, 1H), 7.66 – 7.47 (m, 2H), 7.37 (dd, *J* = 5.1, 1.4 Hz, 1H), 6.86 (s, 2H), 6.50 (s, 1H), 4.91 (s, 2H), 4.81 (s, 1H), 4.46 – 4.19 (m, 2H), 4.19 – 3.95 (m, 4H), 3.81 (s, 3H), 3.63 – 3.53 (m, 1H), 3.20 (s, 1H), 3.07 – 2.98 (m, 1H), 2.61 (d, *J* = 14.9 Hz, 2H), 2.37 (d, *J* = 5.3 Hz, 8H), 2.06 – 1.92 (m, 1H), 1.87 (s, 1H), 1.85 – 1.76 (m, 1H), 1.70 (dd, *J* = 12.1, 7.9 Hz, 1H), 1.30 (t, *J* = 6.9 Hz, 4H).

Synthesis of compound 34



Steps 1 and 2: To an ice-cold solution of 2-bromopyridine-4-methanol (**S26**) (0.50 g, 2.66 mmol) in DCM (10 mL) were added methanesulfonyl chloride (0.36 g, 3.13 mmol) and DIPEA (0.67 g, 5.22 mmol). The mixture was stirred for 15 min and allowed to warm to 25 °C. The reaction mixture was diluted with DCM, washed with water, brine, dried over sodium sulfate and concentrated. The residue was dissolved in THF (5 mL) and morpholine (0.27 g, 3.13 mmol) and DIPEA (0.67 g, 5.22 mmol) were added and the mixture was heated 50 °C for 18 h. The reaction mixture was diluted with iPrOAc, washed with water, brine, dried over sodium sulfate and concentrated. The residue was purified by column chromatography (silica gel, 0-100% iPrOAc/heptane) to obtain 4-((2-bromopyridin-4-yl)methyl)morpholine (**S27**) (0.42 g, 62%). LC/MS (ESI): calcd. for C₁₀H₁₄BrN₂O⁺ [M+H]⁺ 257.0, found 257. ¹HNMR (400 MHz, CDCl₃) δ 8.30 (d, *J* = 5.0 Hz, 1H), 7.53 – 7.49 (m, 1H), 7.26 – 7.23 (m, 1H), 3.77 – 3.70 (m, 4H), 3.47 (s, 2H), 2.48 – 2.41 (m, 4H).

Step 3: A mixture of the above bromide **S27** (0.40 g, 1.55 mmol), ethyl (E)-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)acrylate (**57**) (0.41 g, 1.71 mmol) potassium carbonate (0.65 g, 4.7 mmol), Pd(dppf)Cl₂ (0.06 g, 0.08 mmol) in 1,4-dioxane/water (3 mL:1 mL) was heated at 90 °C for 1 h. The reaction mixture was cooled, diluted with iPrOAc. The organic layer was separated, washed with, brine, dried over sodium sulfate and concentrated. The residue was purified by column

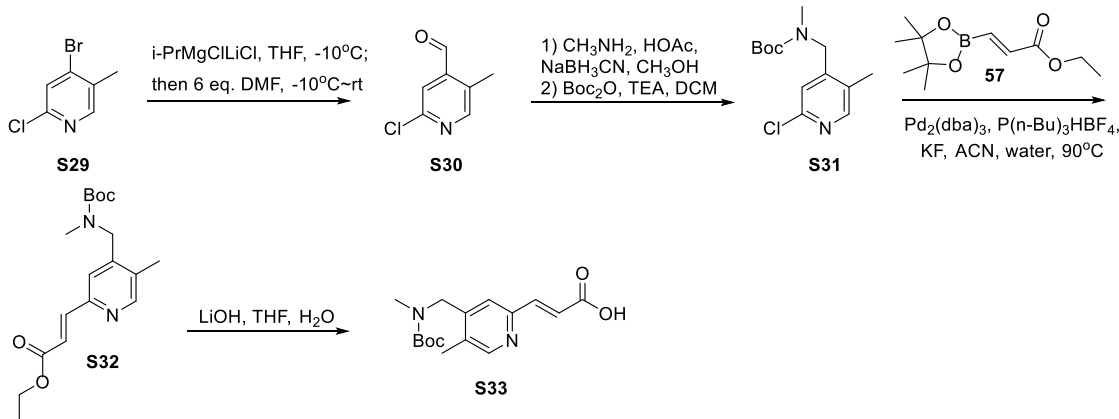
chromatography (silica gel, 0-100% iPrOAc/heptane) to obtain the acrylate (350 mg, 81%). LC/MS (ESI): calcd. for $C_{15}H_{21}N_2O_3^+$ $[M+H]^+$ 277.2, found 277. 1H NMR (400 MHz, $CDCl_3$) δ 8.57 (dd, $J = 4.9, 0.8$ Hz, 1H), 7.68 (d, $J = 15.8$ Hz, 1H), 7.49 – 7.39 (m, 1H), 7.26 – 7.24 (m, 1H), 6.92 (d, $J = 15.7$, 1H), 4.36 – 4.16 (m, 2H), 3.75 – 3.71 (m, 4H), 3.51 (s, 2H), 2.51 – 2.40 (m, 4H), 1.38 – 1.30 (m, 3H).

Step 4: The above acrylate (250 mg, 0.9 mmol) was dissolved in a mixture of IPA/MeOH (2 mL:2 mL) and LiOH (1 mL, 1M) was added and the mixture was heated at 45 °C for 18 h. The reaction mixture was cooled, acidified with dil. HCl, and concentrated. The residue was loaded on an SPE column and eluted with 5% ammonium hydroxide. The elute was concentrated and acidified with HCl-dioxane and concentrated to get acid **S28** as a solid (0.1 g, 38%). 1H NMR (400 MHz, DMSO) δ 8.79 (d, $J = 5.2$ Hz, 1H), 8.35 (s, 1H), 7.87 (dd, $J = 5.1, 1.6$ Hz, 1H), 7.63 (d, $J = 15.9$ Hz, 1H), 6.95 (d, $J = 15.8$ Hz, 1H), 4.48 (s, 2H), 4.02 – 3.85 (m, 4H), 3.40 – 3.05 (m, 4H).

Step 5: **60** was treated with acrylic acid **S28** under standard amide coupling conditions (1.2 eq. **S28**, 1.2 eq. HATU, 3 eq. DIPEA, rt, 1h) to obtain the compound **34** in 34% yield after HPLC purification: Purity 100%; LCMS (ESI): calcd. for $C_{42}H_{47}ClF_4N_9O_3^+$ $[M+H]^+$ 836.3, found 836.3. 1H NMR (400 MHz, DMSO- d_6) δ 8.57 (s, 1H), 7.86 (s, 1H), 7.70 – 7.55 (m, 2H), 7.55 (s, 2H), 7.38 – 7.30 (m, 1H), 6.85 (s, 3H), 6.50 (s, 1H), 5.00 (s, 3H), 4.83 (s, 1H), 4.46 – 4.04 (m, 5H), 3.77 (s, 2H), 3.65 – 3.51 (m, 8H), 3.49 (s, 1H), 3.12 (s, 1H), 2.43 – 2.34 (m, 2H), 2.38 (s, 7H), 1.30 (s, 4H).

Synthesis of compound 35

a) Synthesis of intermediate S33



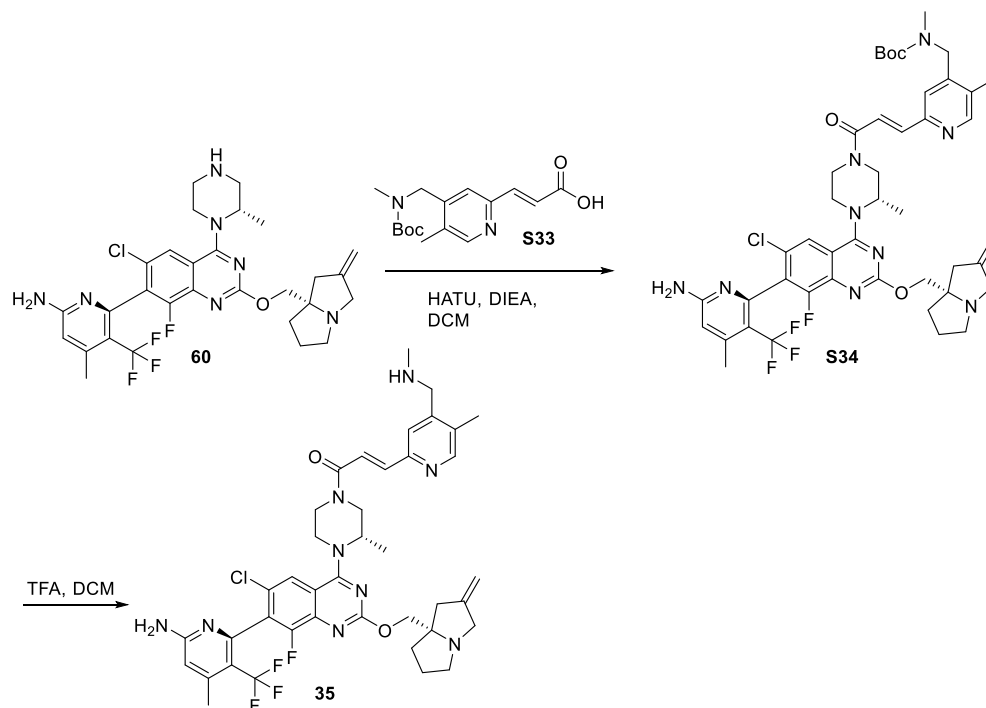
Step 1: Under nitrogen, a mixture of 4-bromo-2-chloro-5-methylpyridine (**S29**) (15 g, 73 mmol) in tetrahydrofuran (300 mL) was added 1.3 M isopropylmagnesium chloride lithium chloride complex in tetrahydrofuran (75 mL, 97.5 mmol) at -10 °C. The resulting mixture was stirred for 1 h at -10 °C. Then *N,N*-dimethylformamide (30 mL, 387 mmol) was added at -10 °C. The reaction was gradually warmed to rt and stirred for overnight. The reaction was quenched with saturated ammonia chloride and extracted with ethyl acetate. The organic layers were combined. The organic layer was dried over anhydrous sodium sulfate and concentrated under vacuum. The residue was purified by flash chromatography on silica gel eluting with ethyl acetate/PE (5-20%) to afford **S30** (6.8 g, 60% yield) as a yellow solid. LCMS (ESI, *m/z*): calcd. for $C_7H_7ClNO^+$ $[M+H]^+$ 156.0, found 156.3.

Steps 2 and 3: To a solution of 2-chloro-5-methyl-pyridine-4-carbaldehyde (**S30**) (1.0 g, 6.43 mmol), acetic acid (1.0 mL) and 33% methylamine in ethanol (4 mL) in methanol (20 mL) was added sodium cyanoborohydride (0.81 g, 13 mmol). The reaction was stirred at rt for 1h. The solvent was removed under vacuum. Then the residue was dissolved in DCM, trimethylamine (1 mL) and di-*tert*-butyl dicarbonate (7.0 g, 32 mmol) were added, and the mixture was stirred at rt for 1h. The reaction mixture was diluted with water and extracted with dichloromethane. The organic layers were combined. The organic layer was dried over anhydrous sodium sulfate and concentrated under vacuum. The residue was purified by flash chromatography on silica gel eluting with ethyl acetate/PE (10-40%) to afford **S31** (1.2 g, 69% yield) as a yellow solid. LCMS (ESI, *m/z*): calcd for $C_{13}H_{20}ClN_2O_2^+$ $[M+H]^+$ 271.1, found 271.1.

Step 4: Under nitrogen, to a solution of *tert*-butyl ((2-chloro-5-methylpyridin-4-yl)methyl)(methyl)carbamate (**S31**) (1.2 g, 4.4 mmol) in acetonitrile (16 mL) and water (4 mL) were added 2-ethoxycarbonylvinylboronic acid pinacol ester (**57**) (3.0 g, 13 mmol), tris(dibenzylideneacetone)dipalladium (406 mg, 0.44 mmol), tri-*tert*-butylphosphine tetrafluoroborate (258 mg, 0.89 mmol) and potassium fluoride (514 mg, 8.86 mmol), and the mixture was stirred at 90 °C for 2 h. The reaction was quenched with water and extracted with ethyl acetate. The organic layers were combined, dried over anhydrous sodium sulfate and concentrated under vacuum. The residue was purified by flash chromatography on silica gel eluting with petrol ether/ethyl acetate (1:1) to afford **S32** (1.2 g, 81% yield) as a yellow solid. LCMS (ESI, *m/z*): calcd. for $C_{18}H_{27}N_2O_4^+$ $[M+H]^+$ 335.2, found 335.2.

Step 5: To a solution of ethyl (E)-3-(4-(((tert-butoxycarbonyl)(methyl)amino)methyl)-5-methylpyridin-2-yl)acrylate (**S32**) (1.2 g, 3.6 mmol) in tetrahydrofuran (7 mL) were added lithium hydroxide (260 mg, 11 mmol) and water (3 mL) and the mixture was stirred for 2h at 50 °C. The residue was purified by reverse phase chromatography (ACN/0.1% HCOOH in water) to afford **S33** (1.0 g, 91% yield) as a yellow oil. LCMS (ESI, m/z): calcd. for $C_{16}H_{23}N_2O_4^+$ $[M+H]^+$ 307.1, found 307.2.

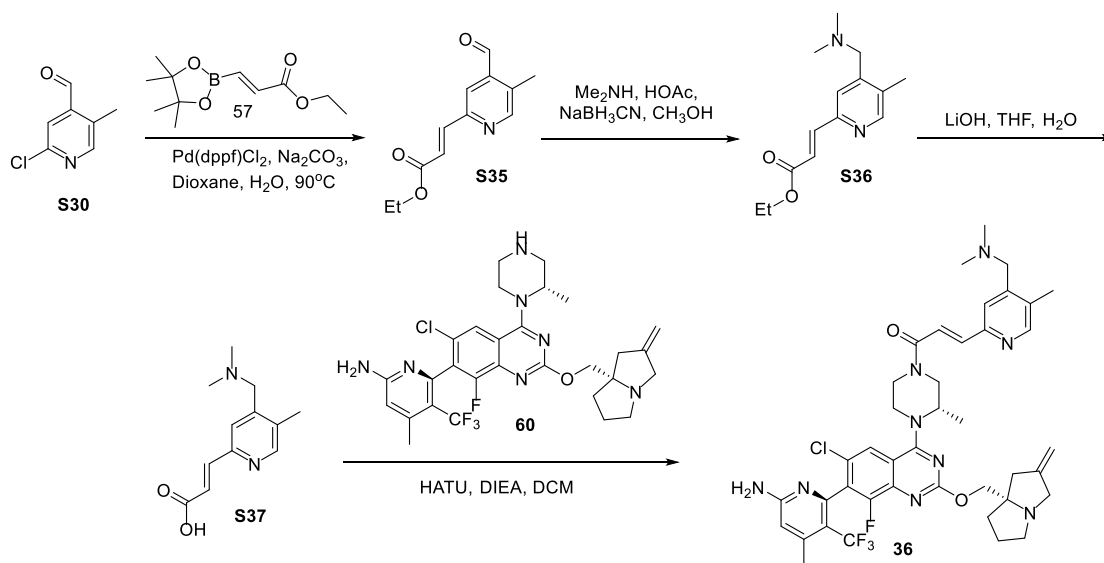
b) Synthesis of compound 35



Step 1: A solution of 6-((*R*)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)-4-(((*S*)-2-methylpiperazin-1-yl)quinazolin-7-yl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**60**) (50 mg, 0.08 mmol), (*E*)-3-(4-(((tert-butoxycarbonyl)(methyl)amino)methyl)-5-methylpyridin-2-yl)acrylic acid (**S33**) (25 mg, 0.08 mmol), HATU (37 mg, 0.10mmol) and DIEA (31 mg, 0.25 mmol) in dichloromethane (1 mL) was stirred for 2 h at rt. The solvent was concentrated under vacuum. The residue was purified by reverse phase chromatography (acetonitrile/0.1% ammonium bicarbonate in water) to afford **S34** (30 mg, 41% yield) as a yellow oil. LCMS (ESI): calcd. for $C_{45}H_{53}ClF_4N_9O_4^+$ $[M+H]^+$ 894.4, found 894.2.

Step 2: A solution of *tert*-butyl ((2-((*E*)-3-((*S*)-4-((*R*)-7-(6-amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1H-pyrrolizin-7a(5H)-yl)methoxy)quinazolin-4-yl)-3-methylpiperazin-1-yl)-3-oxoprop-1-en-1-yl)-5-methylpyridin-4-yl)methyl(methyl)carbamate (**S34**) (30 mg, 0.03 mmol) in dichloromethane (1 mL) was added trifluoroacetic acid (1 mL) and the mixture was stirred for 2 h at rt. The solvent was concentrated under vacuum. The crude product was purified by prep-HPLC to afford **35** (15 mg, 53% yield) as a white solid and as FA salt. LCMS (ESI, *m/z*): calcd. for $C_{40}H_{45}ClF_4N_9O_2^+ [M+H]^+$ 794.3, found 794.4. 1H NMR (300 MHz, Methanol- d_4) δ 8.42 (s, 1H), 7.89 (d, J = 1.7 Hz, 1H), 7.74 – 7.45 (m, 3H), 6.73 – 6.49 (m, 1H), 5.08 (s, 2H), 4.65 – 4.55 (m, 1H), 4.50 – 4.27 (m, 4H), 4.24 – 4.10 (m, 1H), 3.97 – 3.64 (m, 5H), 3.58 – 3.40 (m, 2H), 3.33 – 3.25 (m, 1H), 2.88 (d, J = 13.6 Hz, 2H), 2.63 – 2.54 (m, 1H), 2.53 (s, 3H), 2.50 – 2.45 (m, 3H), 2.39 (s, 3H), 2.28 – 2.13 (m, 1H), 2.10 – 1.83 (m, 3H), 1.48 (t, J = 7.0 Hz, 3H).

Synthesis of compound 36



Step 1: Under nitrogen, to a mixture of 2-chloro-5-methylpyridine-4-carbaldehyde (**S30**) (300 mg, 1.93 mmol), 2-ethoxycarbonylvinylboronic acid pinacol ester (**57**) (480 mg, 2.12 mmol), sodium carbonate (618 mg, 5.83 mmol) and water (0.8 mL) in 1,4-dioxane (4 mL) was added [1,1'-bis(diphenylphosphino)ferrocene]dichloropalladium(II) (150 mg, 0.20 mmol) and the mixture was stirred for 2 h at 90 °C. The solvent was removed under vacuum. The residue was purified by flash

chromatography on silica gel eluting with ethyl acetate/petroleum ether (15%) to afford **S35** (177 mg, 42% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{12}H_{14}NO_3^+$ $[M+H]^+$ 220.1, found 220.0

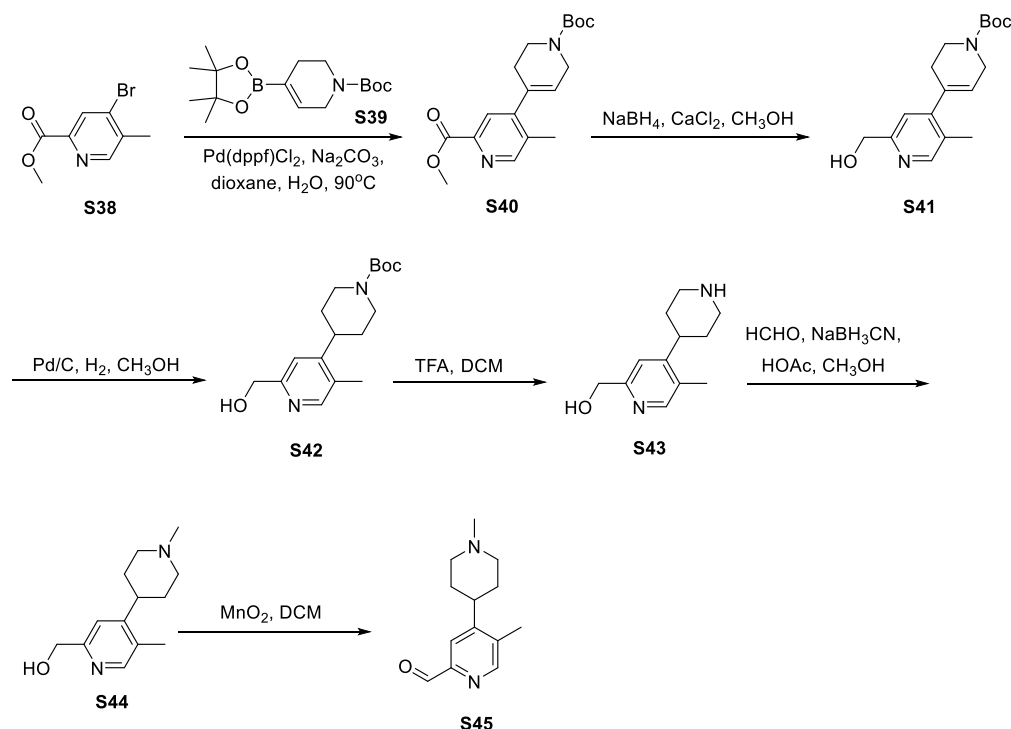
Step 2: A mixture of ethyl (*E*)-3-(4-formyl-5-methyl-2-pyridyl)prop-2-enoate (**S35**) (127 mg, 0.58 mmol), 2 M dimethylamine in tetrahydrofuran (2.9 mL, 5.8 mmol) and acetic acid (1.3 mL, 22.71 mmol) in methanol (5 mL) was stirred for 30 min at rt. Then sodium cyanoborohydride (76 mg, 1.11 mmol) was added, and the mixture was stirred for 1 h at rt. The solvent was concentrated under vacuum. The resulting residue was purified by reverse phase chromatography (acetonitrile/10 mM ammonium bicarbonate in water) to afford **S36** (85 mg, 59% yield) as a yellow oil. LCMS (ESI): calcd. for $C_{14}H_{21}N_2O_2^+$ $[M+H]^+$ 249.2, found 249.0.

Step 3: A mixture of ethyl (*E*)-3-(4-((dimethylamino)methyl)-5-methylpyridin-2-yl)acrylate (**S36**) (70 mg, 0.28 mmol) in tetrahydrofuran (1 mL) was treated with sat. aqueous lithium hydroxide solution (1.5 mL) and stirred at 50 °C for 2 h. Then the reaction mixture pH was adjusted to pH = 5 with 1 M hydrochloric acid. The solvent was concentrated under vacuum. The resulting residue was purified by reverse phase chromatography (acetonitrile 0-5%/10 mM ammonium bicarbonate in water) to afford **S37** (51mg, 82% yield) as a yellow oil. LCMS (ESI): calcd. for $C_{12}H_{17}N_2O_2^+$ $[M+H]^+$ 221.1, found 221.0.

Step 4: A mixture of 6-((*R*)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)-4-((*S*)-2-methylpiperazin-1-yl)quinazolin-7-yl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**60**) (99 mg, 0.16 mmol), (*E*)-3-(4-((dimethylamino)methyl)-5-methylpyridin-2-yl)acrylic acid (**S37**) (45 mg, 0.20 mmol), DIEA (63 mg, 0.49 mmol) and HATU (99 mg, 0.26 mmol) in dichloromethane (5 mL) was stirred for 2 h at rt. The solvent was concentrated under vacuum. The product was purified by prep-HPLC to afford compound **36** (33 mg, 25% yield) as a white solid. LCMS (ESI) calcd. for $C_{41}H_{47}ClF_4N_9O_2^+$ $[M+H]^+$ 808.4, found 808.4; HRMS calc. for $C_{41}H_{47}ClF_4N_9O_2^+$ $[M+H]^+$, 808.3472; found, 808.3472; 1H NMR (300 MHz, Methanol- d_4) δ 8.41 (s, 1H), 7.87 (d, J = 1.7 Hz, 1H), 7.74 – 7.43 (m, 3H), 6.62 (s, 1H), 5.04 – 4.95 (m, 2H), 4.93 – 4.89 (m, 1H), 4.66 – 4.39 (m, 1H), 4.38 – 4.24 (m, 3H), 4.23 – 4.11 (m, 1H), 3.93 – 3.63 (m, 3H), 3.54 (s, 2H), 3.48 – 3.39 (m, 1H), 3.32 – 3.26 (m, 1H), 3.20 – 3.06 (m, 1H), 2.85 – 2.62 (m, 2H), 2.54 – 2.43 (m, 4H), 2.41 (s, 3H), 2.32 (s, 6H), 2.23 – 2.04 (m, 1H), 2.04 – 1.77 (m, 3H), 1.59 – 1.32 (m, 3H).

Synthesis of compound 37

a) Synthesis of intermediate S45



Step 1: Under nitrogen, a mixture of methyl 4-methyl-4-bromo-5-methylpicolinate (**S38**) (500 mg, 2.2 mmol), *tert*-butyl 4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3,6-dihydropyridine-1(2H)-carboxylate (**S39**) (739 mg, 2.4 mmol), sodium carbonate (925 mg, 8.7 mmol) and 1,1'-bis(diphenylphosphino)ferrocene-palladium(II)dichloride dichloromethane complex (168 mg, 0.22 mmol) in 1,4-dioxane (6 mL) and water (1.2 mL) was stirred for 2 h at 90°C . The resulting residue was purified by reverse phase chromatography (acetonitrile/0.1% Ammonium bicarbonate in water) to afford **S40** (325 mg, 45% yield) as a yellow oil. LCMS (ESI, m/z): calcd. for $\text{C}_{18}\text{H}_{25}\text{N}_2\text{O}_4^+$ $[\text{M}+\text{H}]^+$ 333.2, found 333.2.

Step 2: A mixture of 1-(*tert*-butyl) 2'-methyl 5'-methyl-3,6-dihydro-[4,4'-bipyridine]-1,2'(2H)-dicarboxylate (**S40**) (315 mg, 0.95 mmol) and calcium chloride (527 mg, 4.8 mmol) in methanol (5 mL) was treated with sodium borohydride (721 mg, 19 mmol) and stirred for 2 h at rt. The solvent was concentrated under vacuum. The residue was purified by flash chromatography on silica gel eluting with dichloromethane/methanol (20%) to afford compound **S41** (266 mg, 57% yield) as a yellow oil. LCMS (ESI, m/z): calcd. for $\text{C}_{17}\text{H}_{25}\text{N}_2\text{O}_3^+$ $[\text{M}+\text{H}]^+$ 305.2, found 305.2.

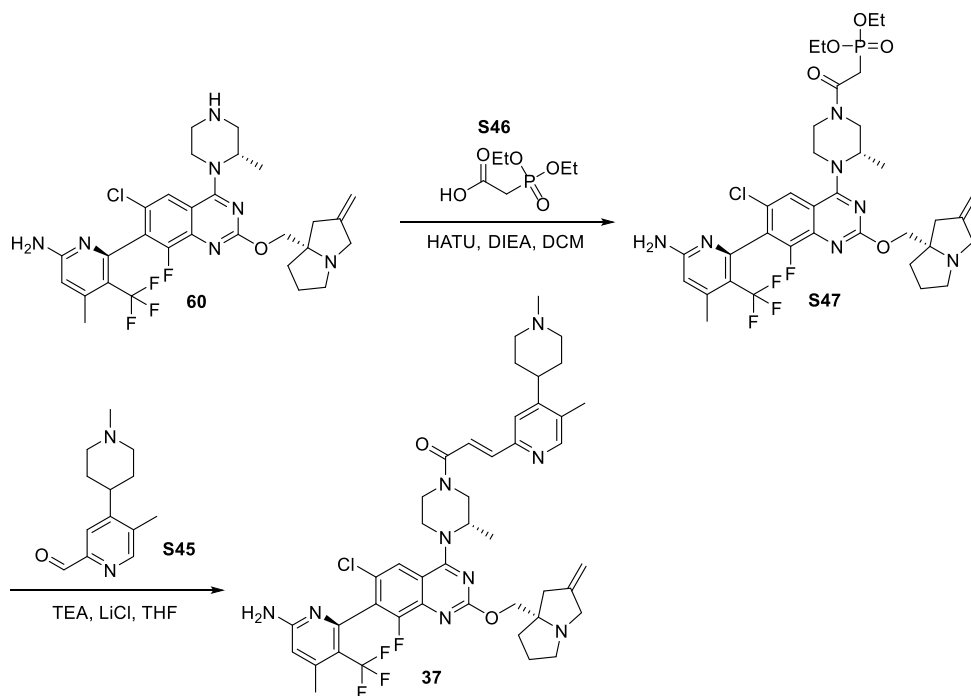
Step 3: Under hydrogen, a mixture of *tert*-Butyl 2'-(hydroxymethyl)-5'-methyl-3,6-dihydro-[4,4'-bipyridine]-1(2H)-carboxylate (**S41**) (266 mg, 0.86 mmol) and 10% palladium on carbon (100 mg) in methanol (8 mL) was stirred at rt for 30 minutes. After filtration, the filtrate was concentrated under reduced pressure to afford **S42** (253 mg, 96% yield) as a yellow oil. LCMS (ESI, m/z): calcd. for $C_{17}H_{27}N_2O_3^+$ [M+H]⁺ 307.2, found 307.2.

Step 4: A mixture of *tert*-butyl 4-(2-(hydroxymethyl)-5-methylpyridin-4-yl)piperidine-1-carboxylate (**S42**) (250 mg, 0.81 mmol) in dichloromethane (2 mL) was treated with TFA (2 mL) and stirred for 1 h at rt. The solvent was concentrated under vacuum. The residue was purified by reverse phase chromatography (ACN/10 mmol NH_4HCO_3) to afford **S43** (160 mg, 95% yield) as a yellow oil. LCMS (ESI, m/z): calcd. for $C_{12}H_{19}N_2O^+$ [M+H]⁺ 207.2, found 207.1.

Step 5: A solution of (5-methyl-4-(piperidin-4-yl)pyridin-2-yl)methanol (**S43**) (160 mg, 0.77 mmol), 33% formaldehyde in water (0.5 mL, 5.5 mmol), and acetic acid (0.2 mL) in methanol (3 mL) was stirred at rt for 10 min, then sodium cyanoborohydride (97 mg, 1.5 mmol) was added and the mixture was stirred at rt for 1 h. The organic layer was concentrated in vacuum. The residue was purified by reverse phase chromatography (ACN/10 mmol NH_4HCO_3) to afford **S44** (100 mg, 59% yield) as a yellow oil. LCMS (ESI, m/z): calcd. for $C_{13}H_{21}N_2O^+$ [M+H]⁺ 221.2, found 221.2.

Step 6: A mixture of (5-methyl-4-(1-methylpiperidin-4-yl)pyridin-2-yl)methanol (**S44**) (100 mg, 0.45 mmol) and manganese dioxide (393 mg, 4.5 mmol) in dichloromethane (5 mL) was stirred for 2 h at rt. After filtration, the filtrate was concentrated under reduced pressure to afford **S45** (50 mg, 51% yield) as a yellow oil. LCMS (ESI): calcd. for $C_{13}H_{19}N_2O^+$ [M+H]⁺ 219.2, found 219.1.

b) Synthesis of compound 37

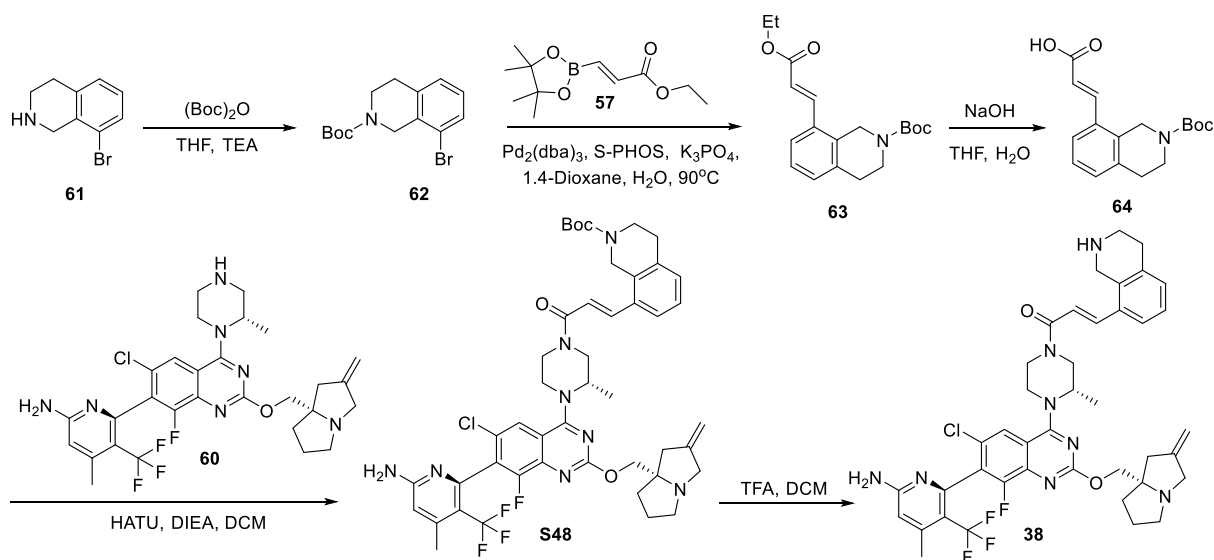


Step 1: A solution of 6-((*R*)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7a(5*H*)-yl)methoxy)-4-((*S*)-2-methylpiperazin-1-yl)quinazolin-7-yl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**60**) (200 mg, 0.33 mmol), diethylphosphonoacetic acid (**S46**) (0.05 mL, 0.33 mmol), HATU (150 mg, 0.40 mmol) and DIEA (0.2 mL) in dichloromethane (2 mL) was stirred for 2 h at rt. The solvent was concentrated under vacuum. The residue was purified by reverse phase chromatography (acetonitrile/10 mM ammonium bicarbonate in water) to afford **S47** (150 mg, 58% yield) as a yellow oil. LCMS (ESI, *m/z*): calcd. for $C_{35}H_{44}ClF_4N_7O_5P^+$ [$M+H$] $^+$ 784.3, found 784.2.

Step 2: A solution of diethyl (2-((*S*)-4-((*R*)-7-(6-amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7a(5*H*)-yl)methoxy)quinazolin-4-yl)-3-methylpiperazin-1-yl)-2-oxoethyl)phosphonate (**S47**) (100 mg, 0.13 mmol), 5-methyl-4-(1-methyl-4-piperidyl)pyridine-2-carbaldehyde (**S45**) (27 mg, 0.13 mmol), lithium chloride (16 mg, 0.38 mmol) and triethylamine (38 mg, 0.38 mmol) in tetrahydrofuran (1 mL) was stirred at rt for 2 h. The residue was purified by reverse phase chromatography (acetonitrile/0.1% ammonium bicarbonate in water) and the solvent was concentrated under vacuum. The residue was purified by prep-HPLC to afford **37** (30 mg, 28% yield) as a white solid. LCMS (ESI): calcd.

for $C_{44}H_{51}ClF_4N_9O_2^+ [M+H]^+$ 848.4, found 848.4; HRMS calc. for $C_{44}H_{51}ClF_4N_9O_2^+ [M+H]^+$, 848.3785; found, 848.3773; 1H NMR (400 MHz, Methanol- d_4) δ 8.38 (s, 1H), 7.87 (d, J = 1.6 Hz, 1H), 7.74 – 7.45 (m, 3H), 6.63 (s, 1H), 5.00 (s, 2H), 4.64 – 4.40 (m, 1H), 4.37 – 4.11 (m, 4H), 3.88 – 3.64 (m, 3H), 3.51 – 3.33 (m, 2H), 3.22 – 3.11 (m, 2H), 3.07 (d, J = 11.8 Hz, 2H), 2.91 – 2.64 (m, 3H), 2.53 – 2.45 (m, 4H), 2.40 (d, J = 8.7 Hz, 6H), 2.30 – 2.05 (m, 3H), 2.04 – 1.68 (m, 7H), 1.47 (t, J = 8.0 Hz, 3H).

Synthesis of compound 38



Step 1: A mixture of 8-bromo-1,2,3,4-tetrahydro-isoquinoline (**61**) (5.00 g, 23.57 mmol), di-tert-butyl dicarbonate (7.68 g, 35.23 mmol) and triethylamine (7.18 g, 70.95 mmol) in tetrahydrofuran (100 mL) was stirred at 25°C for 1 hour. The solvent was concentrated under vacuum. The residue was purified by flash chromatography on silica gel eluting with ethyl acetate/petroleum ether (20%) to afford tert-butyl 8-bromo-3,4-dihydro-1H-isoquinoline-2-carboxylate (**62**) (6.58 g, 89% yield) as a yellow oil. LCMS (ESI): calcd. for $C_{14}H_{19}BrNO_2^+ [M+H]^+$ 312.1, found 312.0.

Step 2: Under nitrogen, a mixture of 2-ethoxycarbonylvinylboronic acid pinacol ester (**57**) (2.90 g, 12.83 mmol), *tert*-butyl 8-bromo-3,4-dihydro-1H-isoquinoline-2-carboxylate (2.00 g, 6.41 mmol), 2-dicyclohexylphosphino-2',6'-dimethoxy-1,1'-biphenyl (0.52 g, 1.27 mmol), potassium phosphate tribasic (5.14 g, 19.32 mmol), tris(dibenzylideneacetone)dipalladium (0.60 g, 0.66 mmol) in water (10 mL) and 1,4-dioxane (50 mL) was stirred at 90 °C for 2 h. The solvent was concentrated under vacuum. The residue was purified by flash chromatography on silica gel

eluting with dichloromethane/petroleum ether (70%) to afford **63** (973 mg, 46% yield) as a yellow oil. LCMS (ESI): calcd. for $C_{19}H_{25}NNaO_4^+$ $[M+Na]^+$ 354.2, found 354.1

Step 3: A mixture of *tert*-butyl (*E*)-8-(3-ethoxy-3-oxoprop-1-en-1-yl)-3,4-dihydroisoquinoline-2(1*H*)-carboxylate (**63**) (200 mg, 0.60 mmol), water (2.5 mL) and sodium hydroxide (400 mg, 10.00 mmol) in tetrahydrofuran (2 mL) was stirred at 50 °C for overnight. Then the pH of the reaction mixture was adjusted to pH = 6.0 with 1M hydrochloric acid. The resulting mixture was extracted with dichloromethane and the organic layers were combined. The solvent was concentrated under vacuum to afford compound **64** (140 mg, 77% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{17}H_{21}NNaO_4^+$ $[M+Na]^+$ 326.1, found 326.0.

Step 4: A solution of 6-((*R*)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)-4-((*S*)-2-methylpiperazin-1-yl)quinazolin-7-yl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**60**) (100 mg, 0.17 mmol), (*E*)-3-(2-(*tert*-butoxycarbonyl)-1,2,3,4-tetrahydroisoquinolin-8-yl)acrylic acid (**64**) (60 mg, 0.20 mmol), HATU (100 mg, 0.26 mmol) and *N,N*-diisopropylethylamine (120 mg, 0.93 mmol) in dichloromethane (5 mL) was stirred at rt for 2 h. The solvent was concentrated under vacuum. The residue was purified by flash chromatography on silica gel eluting with MeOH/DCM (20%) to afford **S48** (93 mg, 63% yield) as a white solid. LCMS (ESI): calcd. for $C_{46}H_{52}ClF_4N_8O_4^+$ $[M+H]^+$ 891.4, found 891.3.

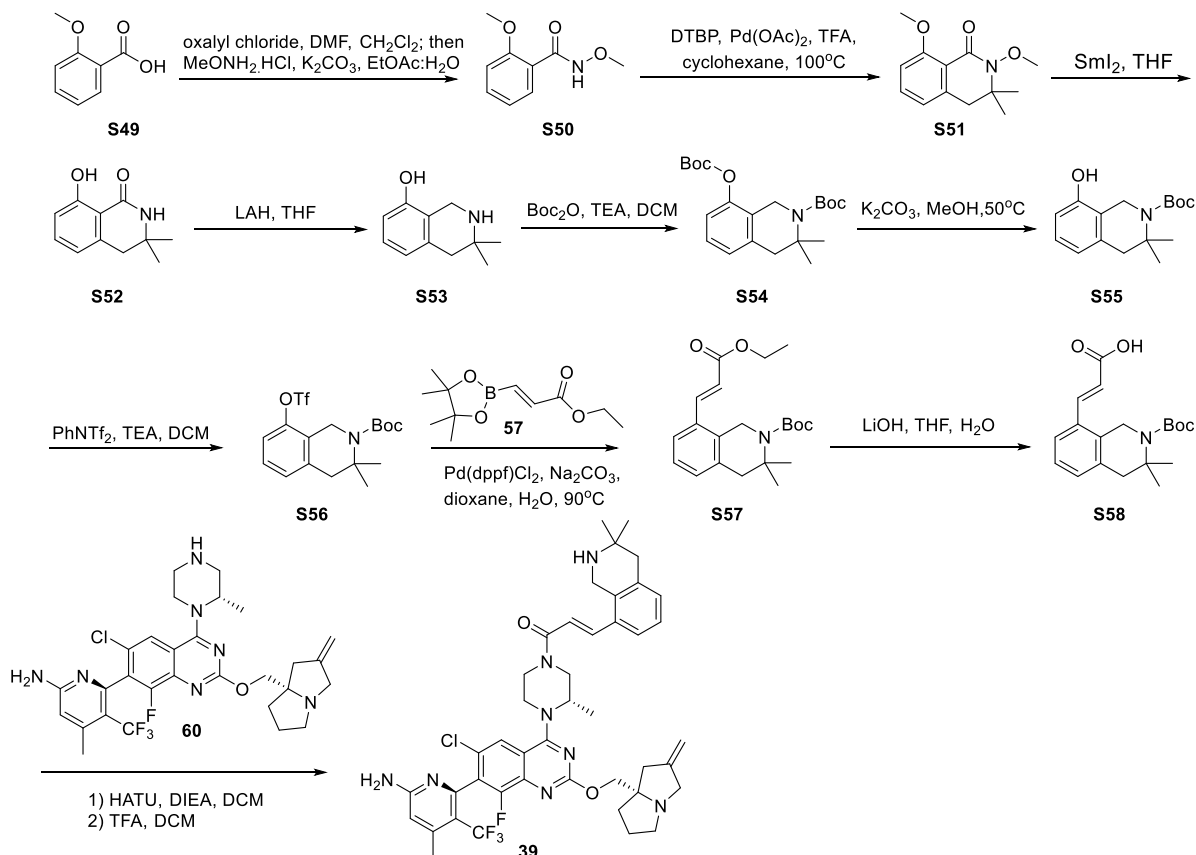
Step 5: A mixture of *tert*-butyl 8-((*E*)-3-((*S*)-4-((*R*)-7-(6-amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)-methoxy)quinazolin-4-yl)-3-methylpiperazin-1-yl)-3-oxoprop-1-en-1-yl)-3,4-dihydroisoquinoline-2(1*H*)-carboxylate (**S48**) (93 mg, 0.10 mmol) in dichloromethane (1 mL) was added trifluoroacetic acid (0.5 mL) and stirred at rt for 1 hour. The solvent was concentrated under vacuum. The product was purified by Prep-HPLC with the following condition to afford compound **38** (50 mg, 53% yield) as a white solid and as TFA salt.

Column: Xselect CSH C18 OBD Column 30*150mm 5 μ m, n; Mobile Phase A: Water (0.05%TFA), Mobile Phase B: ACN; Flow rate: 60 mL/min; Gradient: 17% B to 41% B in 8 min, 41% B; Wave Length: 254/220 nm; RT1(min): 7.28; Number Of Runs: 2.

LCMS (ESI): calcd. for $C_{41}H_{44}ClF_4N_8O_2^+$ $[M+H]^+$ 791.3, found 791.3; HRMS calc. for $C_{41}H_{44}ClF_4N_8O_2^+$ $[M+H]^+$, 791.3206; found, 791.3190; 1H NMR (300 MHz, Methanol-*d*₄) δ 7.96

(d, $J = 1.7$ Hz, 1H), 7.88 – 7.69 (m, 2H), 7.46 – 7.31 (m, 2H), 7.28 – 7.10 (m, 1H), 6.76 – 6.62 (m, 1H), 5.32 (s, 2H), 4.98 (s, 1H), 4.72 (d, $J = 4.3$ Hz, 2H), 4.63 – 4.46 (m, 3H), 4.45 – 4.16 (m, 3H), 4.02 – 3.90 (m, 2H), 3.89 – 3.40 (m, 5H), 3.31 – 3.23 (m, 1H), 3.18 (t, $J = 6.3$ Hz, 2H), 3.09 (d, $J = 16.0$ Hz, 1H), 2.83 (d, $J = 16.1$ Hz, 1H), 2.55 – 2.46 (m, 3H), 2.46 – 2.37 (m, 1H), 2.33 – 2.05 (m, 3H), 1.56 – 1.36 (m, 3H).

Synthesis of compound 39



Step 1: Under nitrogen, to a mixture of 2-methoxybenzoic acid (**S49**) (10 g, 65.7 mmol) in dichloromethane (50 mL) was added oxalyl chloride (6.7 mL, 78.8 mmol) and DMF (0.53 mL, 6.8 mmol) dropwise at 0°C and the mixture was stirred at for 4 h at the same temperature. The solvent was removed under vacuum. The crude product was directly used in the next step without purification. To *o*-methylhydroxylamine (3.4 g, 72.3 mmol) in ethyl acetate (100 mL) and water (50 mL) was added potassium carbonate (18 g, 130.2 mmol) and the mixture was stirred at 0°C for 5min. Then the crude acyl chloride was dissolved in ethyl acetate (100 mL) and added dropwise to the above solution at 0°C and stirred overnight at rt. The resulting solution was quenched with

brine and extracted with ethyl acetate, dried over anhydrous sodium sulfate and concentrated. The residue was purified by flash chromatography on silica gel eluting with petroleum ether/ethyl acetate (51/49) to afford **S50** (7.2 g, 61% yield) as a white solid. LCMS (ESI): calcd. for $C_9H_{12}NO_3^+$ $[M+H]^+$ 182.1, found 182.1.

Step 2¹: To a mixture of *N*,2-dimethoxybenzamide (**S50**) (3 g, 16.6 mmol) in cyclohexane (30 mL) were added palladium(II) acetate (378 mg, 1.67 mmol), trifluoroacetic acid (25.54 mL, 331.5 mmol) and tert-butyl peroxide (12 mL, 65.7 mmol), and the mixture was stirred for 1 h at 100 °C. The reaction was quenched with sat. $NaHCO_3$ and extracted with ethyl acetate. The solvent was removed under vacuum. The residue was purified by flash chromatography on silica gel eluting with petroleum ether/ethyl acetate(73/27) to afford **S51** (1.8 g, 46% yield) as a brown oil. LCMS (ESI): calcd. for $C_{13}H_{18}NO_3^+$ $[M+H]^+$ 236.1, found 236.2.

Step 3: Under nitrogen, to a solution of 2,8-dimethoxy-3,3-dimethyl-4*H*-isoquinolin-1-one (**S51**) (1.5 g, 6.38 mmol) in tetrahydrofuran (5 mL) was added 0.1 M samarium(II) iodide in tetrahydrofuran (200 mL, 20 mmol) at rt. The resulting solution was stirred for 5 h at rt. The reaction was quenched with sat. $Na_2S_2O_3$ and extracted with ethyl acetate. The organic layers were washed with brine, dried over anhydrous sodium sulfate and concentrated. The residue was purified by flash chromatography on silica gel eluting with dichloromethane/methanol (80/20) to afford **S52** (460 mg, 35% yield) as a brown oil. LCMS (ESI, *m/z*): calcd. for $C_{11}H_{14}NO_2^+$ $[M+H]^+$ 192.1, found 192.2.

Step 4: A solution of 8-hydroxy-3,3-dimethyl-2,4-dihydroisoquinolin-1-one (**S52**) (550 mg, 2.9 mmol) in tetrahydrofuran (40 mL) was treated with 1 M LAH in tetrahydrofuran (2.95 mL, 3.0 mmol) at 0 °C and stirred at 70 °C for 5 h. The reaction was quenched with water (3.6 mL), 15% NaOH solution (3.6 mL) and water (10.8 mL). The solids were filtered out. The solvent was concentrated under vacuum to afford crude **S53** (750 mg). The crude product was directly used in the next step without purification. LCMS (ESI, *m/z*): calcd. for $C_{11}H_{16}NO^+$ $[M+H]^+$ 178.1, found 178.2.

¹ Liu, W.; Yu, Q.; Hu, L.; Chen, Z.; Huang, J. Modular synthesis of dihydro-isoquinolines: palladium-catalyzed sequential $C(sp^2)$ -H and $C(sp^3)$ -H bond activation. *Chem. Sci.* **2015**, 6, 5768-5772.

Step 5: To a mixture of 3,3-dimethyl-2,4-dihydro-1*H*-isoquinolin-8-ol (**S53**) (750 mg, 4.23 mmol) in 1,4-dioxane (0.50 mL) and water (0.50 mL) were added sodium bicarbonate (375 mg, 4.4 mmol) and di-*tert*-butyldicarbonate (1380 mg, 6.32 mmol), the mixture was stirred for 1 h at rt. The reaction was extracted with ethyl acetate and washed with brine, dried over anhydrous sodium sulfate and concentrated. The residue was purified by flash chromatography on silica gel eluting with petroleum ether/ethyl acetate (71/29) to afford **S54** (750 mg, 47% yield) as a white solid. LCMS (ESI): calcd. for $C_{21}H_{32}NO_5^+$ [M+H]⁺ 378.2, found 378.3.

Step 6: To a mixture of *tert*-butyl 8-*tert*-butoxycarbonyloxy-3,3-dimethyl-1,4-dihydroisoquinoline-2-carboxylate (**S54**) (700 mg, 1.85 mmol) in methanol (10 mL) was added potassium carbonate (511 mg, 3.7 mmol), and the mixture was stirred for 2 h at 50 °C. The organic layer was concentrated under vacuum. The resulting solution was diluted with water and extracted with ethyl acetate. The organic layers were combined and washed with brine, dried over anhydrous sodium sulfate and concentrated to afford crude **S55** (500 mg), which was directly used in the next step without purification. LCMS (ESI, m/z): calcd. for $C_{16}H_{24}NO_3^+$ [M+H]⁺ 278.2, found 278.2.

Step 7: To a mixture of *tert*-butyl (3,3-dimethyl-2,4-dihydro-1*H*-isoquinolin-8-yl) carbonate (**S55**) (200 mg, 0.72 mmol) in dichloromethane (5 mL) were added *N*-phenylbis-(trifluoromethanesulfonimide) (510 mg, 1.43 mmol) and triethylamine (0.48 mL, 3.46 mmol), and the mixture was stirred for 1 h at rt. The resulting solution was diluted with water and extracted with ethyl acetate. The organic layers were combined and washed with brine, dried over anhydrous sodium sulfate and concentrated. The residue was purified by flash chromatography on silica gel eluting with petroleum ether/ethyl acetate (94/6) to afford **S56** (250 mg, 85% yield) as a white solid. LCMS (ESI): calcd. for $C_{17}H_{23}F_3NO_5S^+$ [M+H]⁺ 410.1, found 410.0.

Step 8: To a solution of *tert*-butyl 3,3-dimethyl-8-(trifluoromethylsulfonyloxy)-1,4-dihydroisoquinoline-2-carboxylate (**S56**) (250 mg, 0.61 mmol) in acetonitrile (3 mL) and water (0.60 mL) were added 2-ethoxycarbonylvinylboronic acid pinacol ester (**57**) (207 mg, 0.92 mmol), tri-*tert*-butylphosphonium tetrafluoroborate (35 mg, 0.12 mmol), tris(dibenzylideneacetone)-dipalladium (56 mg, 0.06 mmol) and potassium fluoride (106 mg, 1.8 mmol) at rt under nitrogen. The resulting mixture was stirred at 90 °C for 2 h. The solvent was removed under vacuum. The residue was purified by flash chromatography on silica gel eluting with petroleum ether/ethyl

acetate (24/76) to afford **S57** (200 mg, 91% yield) as a white solid. LCMS (ESI): calcd. for $C_{21}H_{30}NO_4^+$ $[M+H]^+$ 360.2, found 360.3.

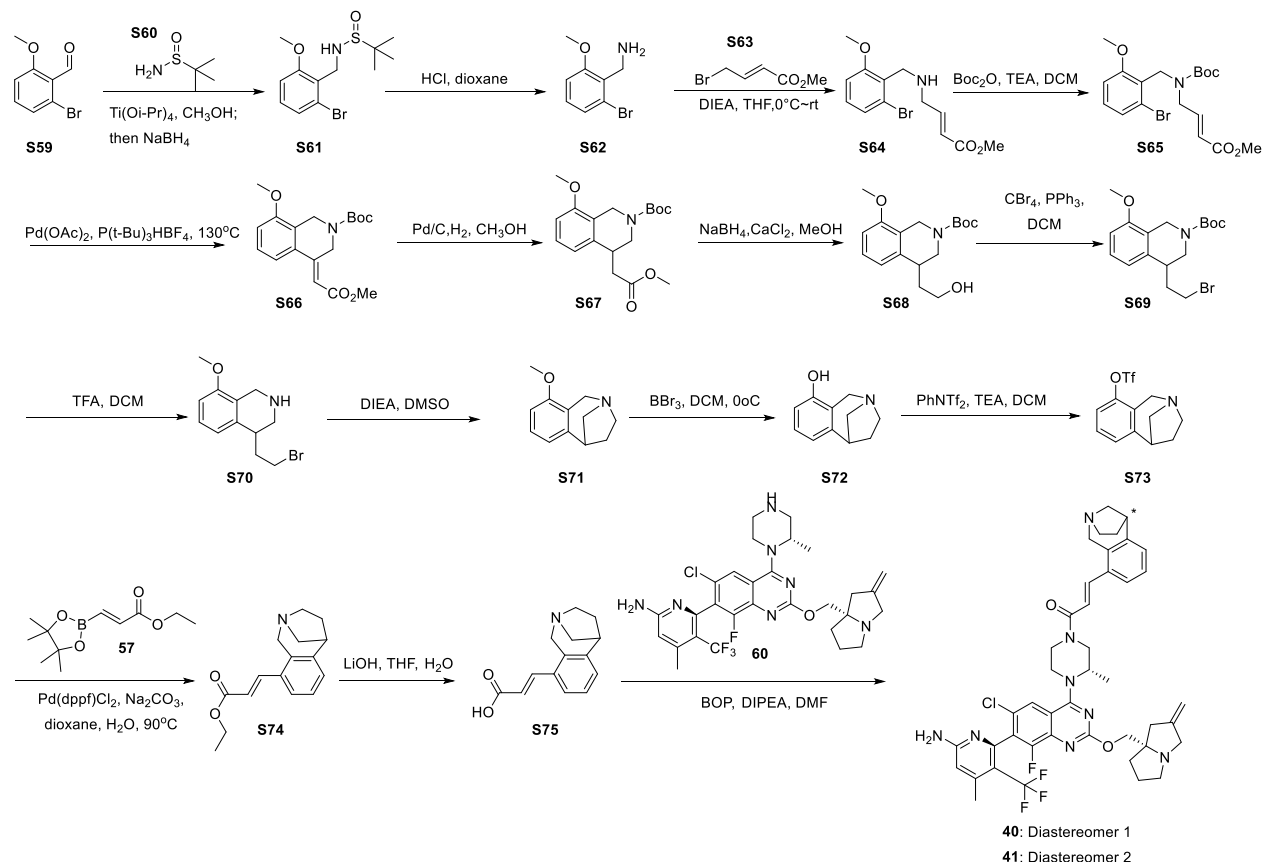
Step 9: To a mixture of *tert*-butyl (*E*)-8-(3-ethoxy-3-oxoprop-1-en-1-yl)-3,3-dimethyl-3,4-dihydroisoquinoline-2(1*H*)-carboxylate (**S57**) (190 mg, 0.53 mmol) in tetrahydrofuran (2 mL) was added sat. lithium hydroxide (2 mL), and the mixture was stirred for 2 h at 50 °C. The solvent was concentrated in vacuum. The residue was purified by reverse phase chromatography eluting with 10 mM FA in water/ACN (60:40) to provide **S58** (105 mg, 60% yield) as a white solid. LCMS (ESI): calcd. for $C_{19}H_{26}NO_4^+$ $[M+H]^+$ 332.2, found 332.0.

Step 10: To a solution of 6-((*R*)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)-4-(((*S*)-2-methylpiperazin-1-yl)quinazolin-7-yl)-4-methyl-5-(trifluoromethyl)pyridin-2-amine (**60**) (50 mg, 0.08 mmol) in dichloromethane (1 mL) was added HATU (50 mg, 0.13 mmol), DIEA (0.05 mL, 0.27 mmol) and (*E*)-3-(2-*tert*-butoxycarbonyl-3,3-dimethyl-1,4-dihydroisoquinolin-8-yl) prop-2-enoic acid (**S58**) (30 mg, 0.09 mmol), the mixture was stirred for 1 h at rt. The solvent was removed in vacuum. The residue was purified by reverse phase chromatography eluting with 10 mM FA in water/ACN (30/70) to afford *tert*-butyl 8-((*E*)-3-(((*S*)-4-((*R*)-7-(6-amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)methoxy)quinazolin-4-yl)-3-methylpiperazin-1-yl)-3-oxoprop-1-en-1-yl)-3,3-dimethyl-3,4-dihydroisoquinoline-2(1*H*)-carboxylate (70 mg, 92% yield) as a white solid. LCMS (ESI, *m/z*): calcd. for $C_{48}H_{56}ClF_4N_8O_4^+$ $[M+H]^+$ 919.4, found 919.5.

Step 11: To a solution of *tert*-butyl 8-((*E*)-3-(((*S*)-4-((*R*)-7-(6-amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1*H*-pyrrolizin-7*a*(5*H*)-yl)-methoxy)quinazolin-4-yl)-3-methylpiperazin-1-yl)-3-oxoprop-1-en-1-yl)-3,3-dimethyl-3,4-dihydroisoquinoline-2(1*H*)-carboxylate (60 mg, 0.07 mmol) in dichloromethane (1 mL) was added trifluoroacetic acid (2 mL), and the solution was stirred for 1h at rt. The organic layer was concentrated in vacuum. The residue was purified by prep-HPLC to afford **39** (16.2 mg, 30% yield) as a white solid. LCMS (ESI, *m/z*): calcd. for $C_{43}H_{48}ClF_4N_8O_2^+$ $[M+H]^+$ 819.4, found 819.4. 1H NMR (400 MHz, Methanol-*d*₄) δ 7.94 – 7.78 (m, 2H), 7.67 – 7.52 (m, 1H), 7.23 (t, *J* = 7.6 Hz, 1H), 7.17 – 6.96 (m, 2H), 6.65 – 6.58 (m, 1H), 5.03 – 4.93 (m, 2H), 4.69 – 4.39 (m, 1H), 4.37 –

4.09 (m, 7H), 3.90 – 3.61 (m, 3H), 3.50 – 3.39 (m, 1H), 3.20 – 3.07 (m, 1H), 2.83 – 2.65 (m, 4H), 2.53 – 2.40 (m, 4H), 2.14 (s, 1H), 2.05 – 1.76 (m, 3H), 1.47 (d, $J = 6.6$ Hz, 3H), 1.21 (s, 6H).

Synthesis of compounds 40 and 41



Step 1: To a mixture of 2-bromo-6-methoxy-benzaldehyde (**S59**) (10.00 g, 46.50 mmol) in methanol (20 mL) were added tetrapropyl titanate (66 g, 232.51 mmol) and 2-methyl-2-propanesulfonamide (**S60**) (28.18 g, 232.51 mmol). The mixture was stirred for 12 h at 80 °C, then sodium borohydride (5301 mg, 139.51 mmol) was added at 0 °C and the mixture was stirred for 2 h at rt. The reaction was quenched with water, extracted with ethyl acetate, dried over anhydrous sodium sulfate and concentrated under vacuum. The crude product was purified by silica gel column eluting with petroleum ether/ethyl acetate (0-50%) to afford **S61** (14.00 g, 94% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{12}H_{19}BrNO_2S^+$ $[M+H]^+$ 320.0, found 320.0.

Step 2: A mixture of *N*-((2-bromo-6-methoxy-phenyl)methyl)-2-methyl-propane-2-sulfinamide (**S61**) (10.00 g, 31.23 mmol) and 4 M HCl in 1,4-Dioxane (150 mL) was stirred for 2 h at 25 °C. After filtration, the solids were collected and washed with DCM to afford **S62** (6.00 g, 89% yield) as a yellow solid. LCMS (ESI): calcd. for $C_8H_{11}BrNO^+$ $[M+H]^+$ 216.0, found 216.0.

Step3: To a mixture of (2-bromo-6-methoxy-phenyl)methanamine (**S62**) (6.00 g, 27.77 mmol) in tetrahydrofuran (10 mL) were added methyl 4-bromocrotonate (**S63**) (1.63 mL, 13.88 mmol) and *N,N*-diisopropylethylamine (19.18 g, 138.84 mmol). The mixture was stirred for 12h at rt. The resulting residue was purified by reverse phase chromatography (acetonitrile/0.1% ammonium bicarbonate in water) to afford **S64** (7.20 g, 79% yield) as a white solid. LCMS (ESI): calcd. for $C_{13}H_{17}BrNO_3^+$ $[M+H]^+$ 314.0, found 314.1.

Step 4: To a mixture of methyl (*E*)-4-((2-bromo-6-methoxy-phenyl)methylamino) but-2-enoate (**S64**) (6.60 g, 21.01 mmol) in tetrahydrofuran (10 mL) were added *N,N*-diisopropylethylamine (13.57 g, 105.04 mmol) and di-*tert*-butyl dicarbonate (5.49 g, 25.21 mmol), and the mixture was stirred for 2 h at rt. The resulting residue was purified by reverse phase chromatography (acetonitrile/0.1% Ammonium bicarbonate in water) to afford compound **S65** (5.10 g, 59% yield) as a white solid. LCMS (ESI): calcd. for $C_{18}H_{25}BrNO_5^+$ $[M+H]^+$ 414.1, found 414.1.

Step 5: To a mixture of methyl (*E*)-4-((2-bromo-6-methoxy-phenyl) methyl-*tert*-butoxycarbonyl-amino) but-2-enoate (**S65**) (4.30 g, 7.97 mmol) in *N,N*-dimethylacetamide (20 mL) were added palladium(II)acetate (357 mg, 1.59 mmol), tri-*tert*-butylphosphine tetrafluoroborate (462 mg, 1.54 mmol), and potassium carbonate (3302 mg, 23.90 mmol). The mixture was stirred for 2h at 130 °C. The reaction mixture was diluted with water and extracted with ethyl acetate and the organic layers were combined. The organic layer was concentrated under vacuum and purified by silica gel chromatography eluting with petroleum ether/ethyl acetate (0-50%) to afford **S66** (3.20 g, 91% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{18}H_{24}NO_5^+$ $[M+H]^+$ 334.2, found 334.2.

Step 6: To a mixture of *tert*-butyl (*Z*)-8-methoxy-4-(2-methoxy-2-oxoethylidene)-3,4-dihydroisoquinoline-2(1*H*)-carboxylate (**S66**) (3.00 g, 9.00 mmol) in methanol (10 mL) was added Pd/C (4.7 g), and the mixture was stirred for 2 h at rt. After filtration, the solvent was removed under vacuum. The residue was purified by silica gel column eluting with petroleum ether/ethyl

acetate (0-50%) to provide **S67** (2.00 g, 66% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{18}H_{26}NO_5^+$ $[M+H]^+$ 336.2, found 336.1.

Step 7: To a mixture of *tert*-butyl 8-methoxy-4-(3-methoxy-3-oxo-propyl)-3,4-dihydro-1*H*-isoquinoline-2-carboxylate (**S67**) (2.00 g, 5.72 mmol) in tetrahydrofuran (10 mL) were added sodium borohydride (652 mg, 17.17 mmol) and calcium chloride (3.17 g, 28.62 mmol). The mixture was stirred for 2 h at rt. The reaction mixture was diluted with water and extracted with ethyl acetate and the organic layers were combined. The organic layer was concentrated under vacuum and the residue was purified by silica gel column eluting with petroleum ether/ethyl acetate (0-50%) to provide **S68** (1.50 g, 85% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{17}H_{26}NO_4^+$ $[M+H]^+$ 308.2, found 308.2.

Step 8: To a mixture of *tert*-butyl 4-(2-hydroxyethyl)-8-methoxy-3,4-dihydro-1*H*-isoquinoline-2-carboxylate (**S68**) (800 mg, 1.30 mmol) in dichloromethane (3.0 mL) was added triphenylphosphine (681 mg, 2.60 mmol), carbon tetrabromide (993 mg, 3.00 mmol). The mixture was stirred for 2h at rt. The reaction mixture was diluted with water and extracted with ethyl acetate. The organic layers were combined. The residue was purified by silica gel column eluting with petroleum ether/ethyl acetate (0-50%) to afford the title compound (**S69**) (700 mg, 73% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{17}H_{25}BrNO_3^+$ $[M+H]^+$ 370.1, found 370.1.

Step 9: To a mixture of *tert*-butyl 4-(2-bromoethyl)-8-methoxy-3,4-dihydro-1*H*-isoquinoline-2-carboxylate (**S69**) (700 mg, 1.89 mmol) in dichloromethane (3.0 mL) was added trifluoroacetic acid (1077 mg, 9.45 mmol), and the mixture was stirred for 2h at 25°C. The reaction was concentrated under vacuum to afford crude **S70** (600 mg). LCMS (ESI): calcd. for $C_{12}H_{17}BrNO^+$ $[M+H]^+$ 270.1, found 270.0.

Step 10: To a mixture of 4-(2-bromoethyl)-8-methoxy-1,2,3,4-tetrahydroisoquinoline (**S70**) (500 mg, 1.85 mmol) in tetrahydrofuran (2.0 mL) was added *N,N*-diisopropylethylamine (1195 mg, 9.25 mmol), and the mixture was stirred for 2h at rt. The resulting residue was purified by reverse phase chromatography (acetonitrile/0.1% ammonium bicarbonate in water) to afford **S71** (300 mg, 86% yield) as a white solid. LCMS (ESI): calcd. for $[M+H]^+$ 190.1, found 190.1.

Step 11: To a mixture of 6-methoxy-9-azatricyclo (7.2.1.02,7) dodeca-2,4,6-triene (**S71**) (400 mg, 2.11 mmol) in dichloromethane (5.0 mL) was added boron tribromide (3.7 g, 14.80 mmol), and the mixture was stirred for 2h at rt. The reaction was quenched with methanol and concentrated under vacuum to afford **S72** (200 mg, 54% yield). LCMS (ESI): calcd. for $C_{11}H_{14}NO^+$ $[M+H]^+$ 176.1, found 176.1.

Step 12: To a mixture of 9-azatricyclo (7.2.1.02,7) dodeca-2,4,6-trien-6-ol (**S72**) (350 mg, 2mmol) in dichloromethane (3.0mL) were added triethylamine (1.39 mL, 9.99 mmol) and *N*-phenyl-bis(trifluoromethanesulfonimide) (1.4 mg, 3.99 mmol), and the mixture was stirred for 2 h at rt. The reaction mixture was diluted with water (100 mL) and extracted with dichlormethane (3 x 100 mL), and the organic layer was concentrated under vacuum and purified by silica gel column eluting with dichlormethane/methanol (0-9%) to provide compound **S73** (150 mg, 24% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{12}H_{13}F_3NO_3S^+$ $[M+H]^+$ 308.1, found 308.1.

Step 13: To a mixture of 9-azatricyclo(7.2.1.02,7)dodeca-2(7),3,5-trien-6-yl trifluoromethanesulfonate (**S73**) (250 mg, 0.81 mmol) in 1,4-dioxane (1.0 mL) and water (0.2 mL) was added Sodium carbonate (258 mg, 2.44 mmol), (1,1-Bis(diphenylphosphino) ferrocene) dichloropalladium(II) (132 mg, 0.16 mmol), 2-ethoxycarbonylvinylboronic acid pinacol ester (**57**) (367 mg, 1.63 mmol), the mixture was stirred for 2h at 90 °C. The reaction mixture was diluted with water and extracted with dichlormethane, the organic layer was concentrated under vacuum and purified by silica gel column eluting with dichlormethane/methanol (0-9%) to afford **S74** (140 mg, 67% yield) as a yellow solid. LCMS (ESI): calcd. for $C_{16}H_{20}NO_2^+$ $[M+H]^+$ 258.2, found 258.1.

Step 14: To a mixture of ethyl (*E*)-3-(9-azatricyclo (7.2.1.02,7) dodeca-2(7),3,5-trien-6-yl) prop-2-enoate (**S74**) (240 mg, 0.93 mmol) in tetrahydrofuran (3.0 mL) were added lithium hydroxide monohydrate (191 mg, 4.66 mmol) and water (1.5 mL), and the mixture was stirred for 4 h at rt. The resulting residue was purified by reverse phase chromatography (acetonitrile/0.1% ammonium bicarbonate in water) to afford **S75** (140 mg, 66% yield) as a white solid. LCMS (ESI): calcd. for $C_{14}H_{16}NO_2^+$ $[M+H]^+$ 230.1, found 230.1.

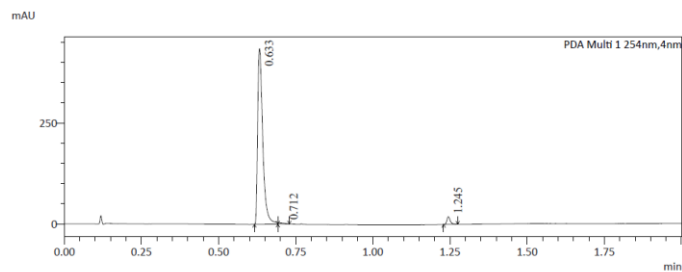
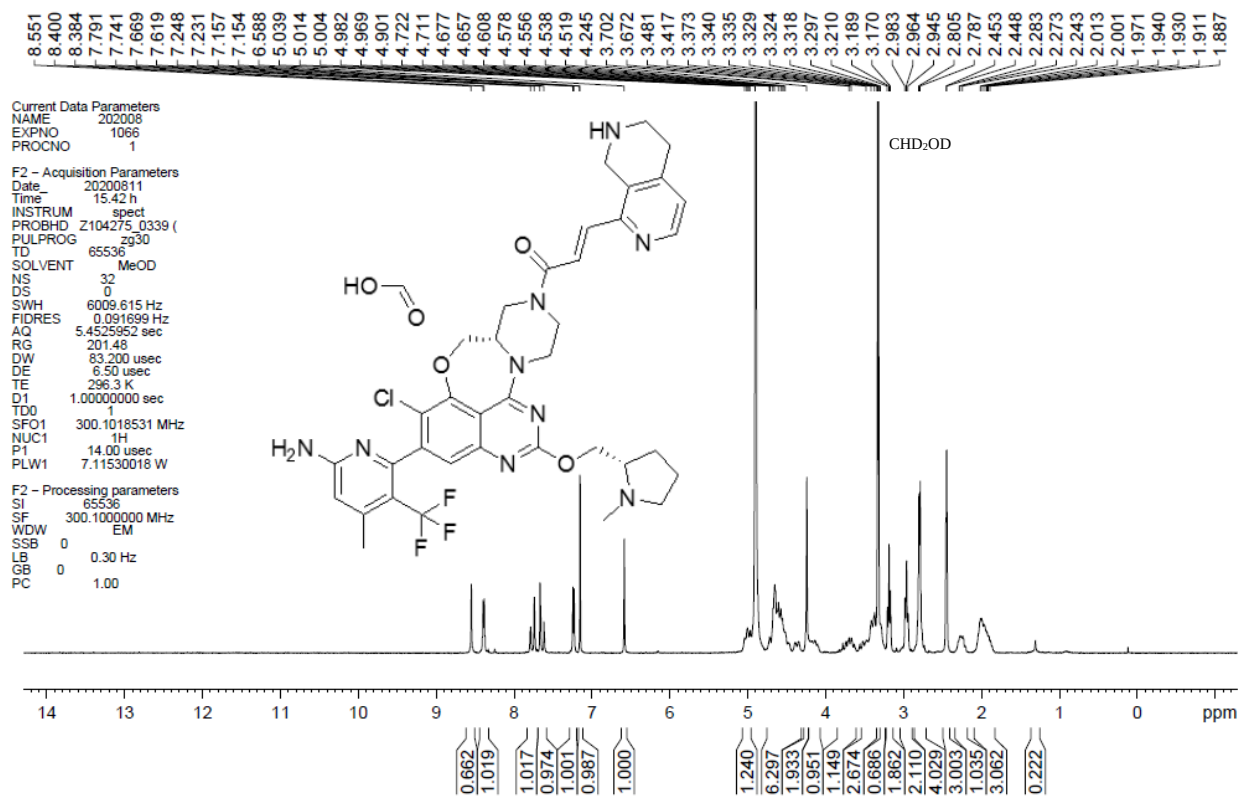
Step 15: To a mixture of (*E*)-3-(9-azatricyclo(7.2.1.02,7)dodeca-2(7),3,5-trien-6-yl)prop-2-enoic acid (**S75**) (40 mg, 0.17 mmol) in *N,N*-dimethylformamide (2.0 mL) were added *N,N*-diisopropylethylamine (112 mg, 0.87 mmol), 1*H*-benzotriazol-1-yloxytris (dimethylamino)

phosphonium hexafluorophosphate (154 mg, 0.35 mmol), and 6-(6-chloro-8-fluoro- 2-(((8S)-6-methylene-2,3,5,7-tetrahydro-1H-pyrrolizin-8-yl)methoxy)-4-((2S)-2-methylpiperazin-1-yl)quinazolin-7-yl)-4-methyl-5-(trifluoromethyl) pyridin-2-amine (**60**) (80 mg, 0.13 mmol), and the mixture was stirred for 2h at rt. The reaction was purified by prep-HPLC and prep-chiral HPLC to afford diastereomers **40** and **41**. *Note: The absolute configuration of both diastereomers was not assigned.*

Diastereomer 1 (40): (*E*)-1-((*S*)-4-((*R*)-7-(6-Amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1H-pyrrolizin-7a(5*H*)-yl)methoxy)quinazolin-4-yl)-3-methylpiperazin-1-yl)-3-(1,3,4,5-tetrahydro-2,5-methanobenzo[*c*]azepin-9-yl)prop-2-en-1-one (4.7 mg, 3% yield). LCMS (ESI): calcd. for $C_{43}H_{46}ClF_4N_8O_2^+$ $[M+H]^+$ 817.3, found 817.3; 1H NMR (400 MHz, Methanol-*d*₄) δ 7.86 (d, *J* = 1.6 Hz, 1H), 7.78 – 7.66 (m, 1H), 7.60 (t, *J* = 7.2 Hz, 1H), 7.21 (t, *J* = 7.6 Hz, 1H), 7.16 – 6.99 (m, 2H), 6.62 (s, 1H), 5.08 – 4.95 (m, 2H), 4.66 – 4.36 (m, 2H), 4.35 – 4.23 (m, 3H), 4.28 – 4.13 (m, 1H), 4.05 (d, *J* = 17.2 Hz, 1H), 3.87 – 3.61 (m, 3H), 3.53 – 3.38 (m, 1H), 3.28 – 3.15 (m, 3H), 3.11 (d, *J* = 11.4 Hz, 1H), 3.00 – 2.93 (m, 1H), 2.93 – 2.86 (m, 2H), 2.84 – 2.61 (m, 2H), 2.54 – 2.34 (m, 4H), 2.27 – 2.07 (m, 2H), 2.04 – 1.96 (m, 2H), 1.96 – 1.70 (m, 2H), 1.56 – 1.37 (m, 3H). *t*_R = 2.656 min (CHIRALPAK IC-3, 4.6*50 mm, 3 μ m, (Hex:DCM=3:1)(0.1%DEA):EtOH=50:50, 1 ml/min, 254 nm).

Diastereomer 2 (41): (*E*)-1-((*S*)-4-((*R*)-7-(6-Amino-4-methyl-3-(trifluoromethyl)pyridin-2-yl)-6-chloro-8-fluoro-2-(((*S*)-2-methylenetetrahydro-1H-pyrrolizin-7a(5*H*)-yl)methoxy)quinazolin-4-yl)-3-methylpiperazin-1-yl)-3-(1,3,4,5-tetrahydro-2,5-methanobenzo[*c*]azepin-9-yl)prop-2-en-1-one (11.7 mg, 8% yield) as a white solid. LCMS (ESI): calcd. for $C_{43}H_{46}ClF_4N_8O_2^+$ $[M+H]^+$ 817.3, found 817.3; 1H NMR (400 MHz, Methanol-*d*₄) δ 7.86 (d, *J* = 1.6 Hz, 1H), 7.78 – 7.66 (m, 1H), 7.60 (t, *J* = 7.2 Hz, 1H), 7.21 (t, *J* = 7.6 Hz, 1H), 7.16 – 6.99 (m, 2H), 6.62 (s, 1H), 5.08 – 4.95 (m, 2H), 4.66 – 4.36 (m, 2H), 4.35 – 4.23 (m, 3H), 4.28 – 4.13 (m, 1H), 4.05 (d, *J* = 17.2 Hz, 1H), 3.87 – 3.61 (m, 3H), 3.53 – 3.38 (m, 1H), 3.28 – 3.15 (m, 3H), 3.11 (d, *J* = 11.4 Hz, 1H), 3.00 – 2.93 (m, 1H), 2.93 – 2.86 (m, 2H), 2.84 – 2.61 (m, 2H), 2.54 – 2.34 (m, 4H), 2.27 – 2.07 (m, 2H), 2.04 – 1.96 (m, 2H), 1.96 – 1.70 (m, 2H), 1.56 – 1.37 (m, 3H). *t*_R = 3.459 min (CHIRALPAK IC-3, 4.6*50 mm, 3 μ m, (Hex:DCM=3:1)(0.1%DEA):EtOH=50:50, 1 ml/min, 254 nm).

Compound 27 (¹H-NMR, LCMS)

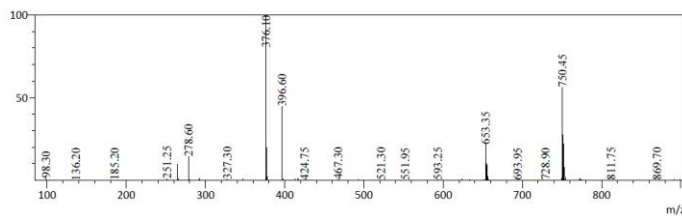


Peak#	Ret. Time	Height	Height%	Area	Area%
7	0.633	434973	95.326	498568	96.294
8	0.712	2099	0.460	4386	0.847
13	1.245	19230	4.214	14803	2.859
Total		456302	100.000	517757	100.000

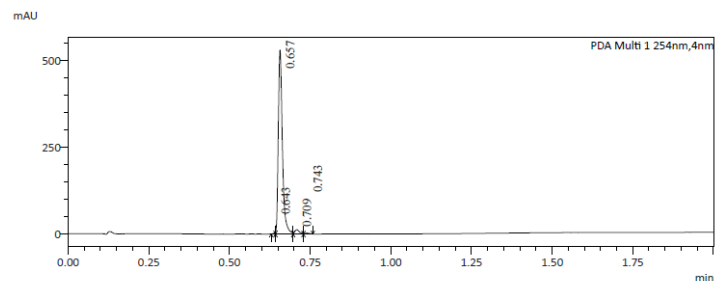
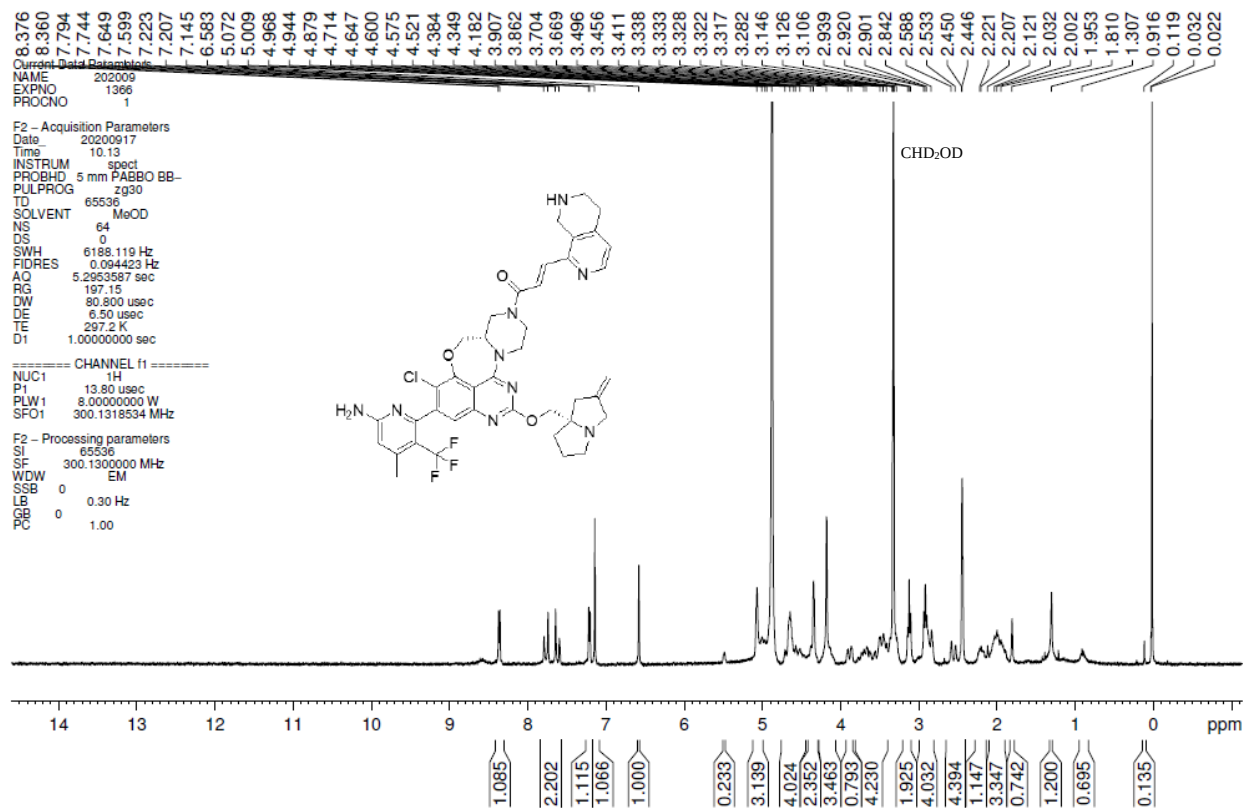
Retention time: 0.629

Spectrum Mode: Averaged 0.621-0.637(79-81) Base Peak: 376.10(182853)

BG Mode:Calc Segment 1 - Event 1



Compound 28 (¹H-NMR, LCMS)

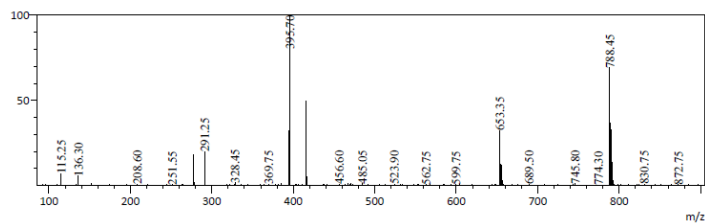


Peak Table
PDA Ch1 254nm

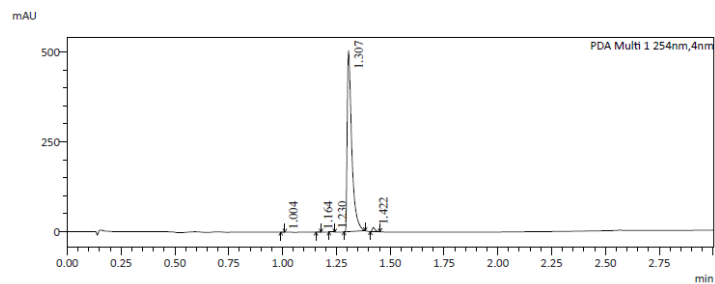
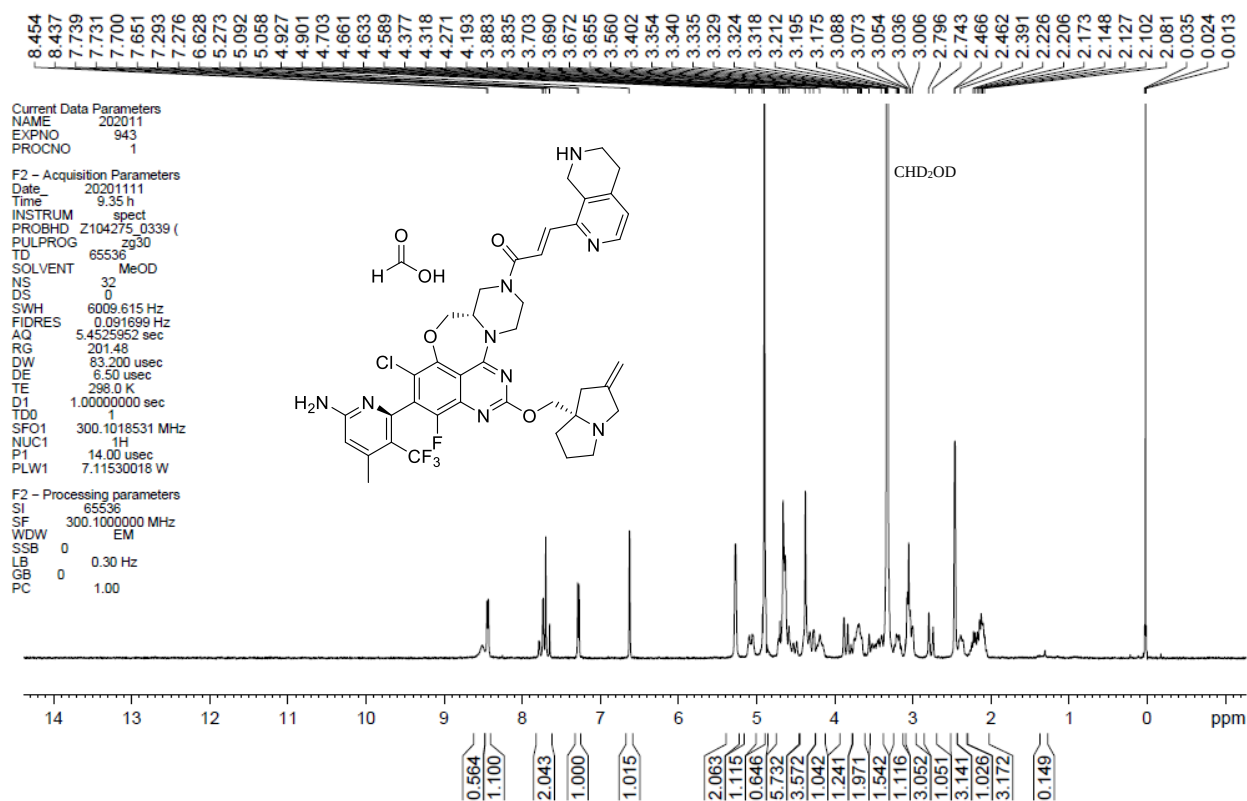
Peak#	Ret. Time	Height	Height%	Area	Area%
6	0.643	4391	0.801	1463	0.302
7	0.657	529144	96.498	466867	96.520
8	0.709	12334	2.249	12712	2.628
9	0.743	2477	0.452	2659	0.550
Total		548346	100.000	483701	100.000

Retention time: 0.650

Spectrum Mode:Single 0.650(83) Base Peak:395.70(46268)
BG Mode:Averaged 0.634-0.675(81-86) Segment 1 - Event 1



Compound 29 (¹H-NMR, LCMS)

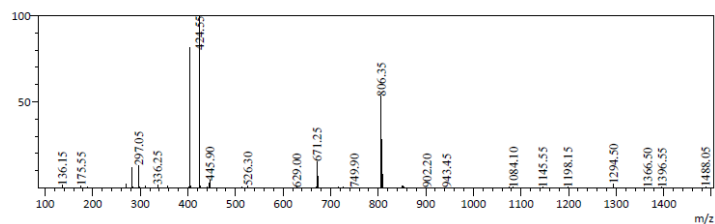


Peak Table

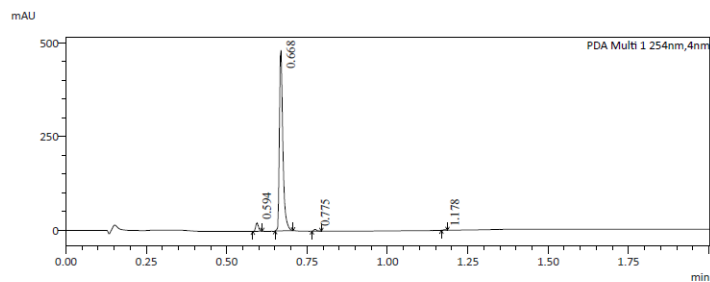
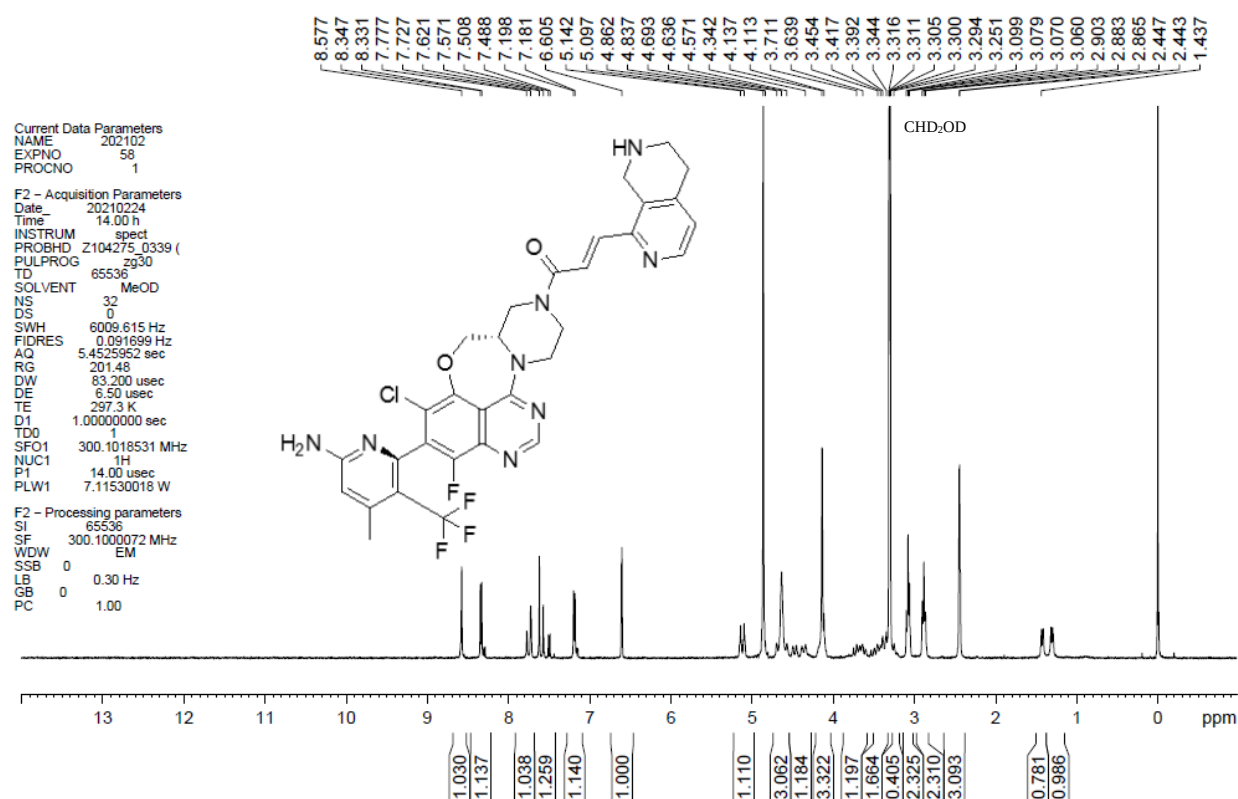
Peak#	Ret. Time	Height	Height%	Area	Area%
1	1.004	436	0.084	239	0.032
2	1.164	624	0.120	441	0.058
3	1.230	1231	0.238	908	0.120
4	1.307	504868	97.377	742385	98.382
5	1.422	11310	2.181	10622	1.408
Total		518469	100.000	754595	100.000

Retention time: 1.312

Spectrum Mode:Single 1.312(163) Base Peak:424.55(430868)
 BG Mode:Averaged 1.270-1.387(158-172) Segment 1 - Event 1



Compound 30 (¹H-NMR, LCMS)

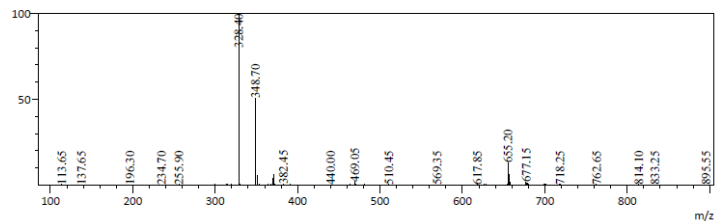


Peak Table

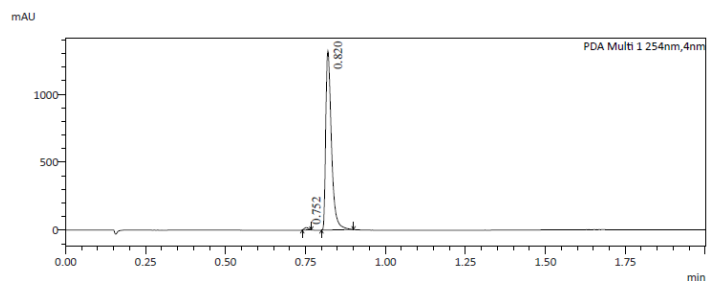
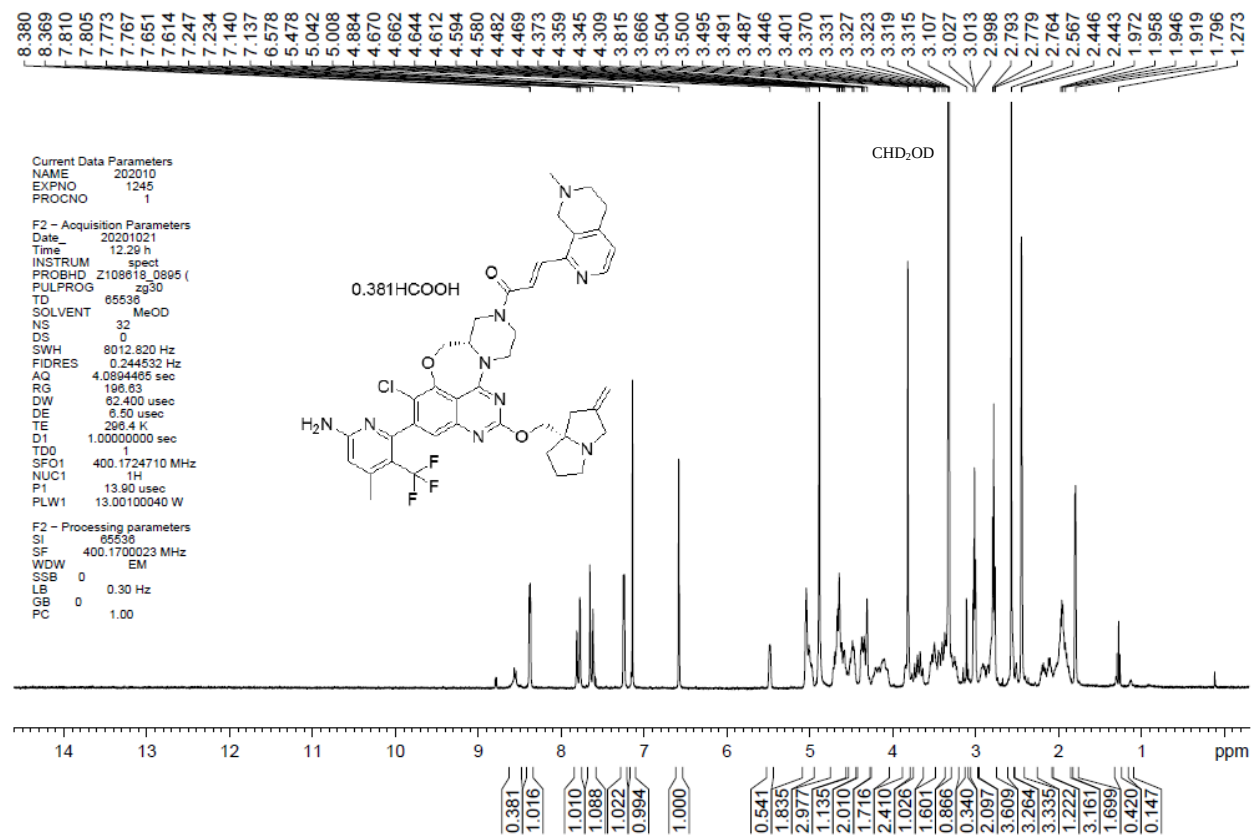
Peak#	Ret. Time	Height	Height%	Area	Area%
1	0.594	22323	4.376	13757	3.610
2	0.668	481457	94.389	363379	95.345
3	0.775	4591	0.900	3037	0.797
4	1.178	1706	0.334	948	0.249
Total		510078	100.000	381121	100.000

Retention time: 0.667

Spectrum Mode:Single 0.667(85) Base Peak:328.40(205231)
 BG Mode:Averaged 0.634-0.717(81-91) Segment 1 - Event 1



Compound 31 (¹H-NMR, LCMS)

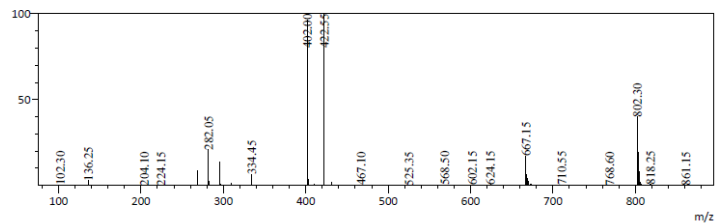


Peak Table
PDA Ch1 254nm

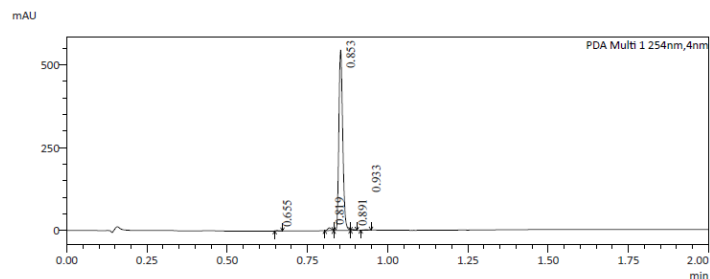
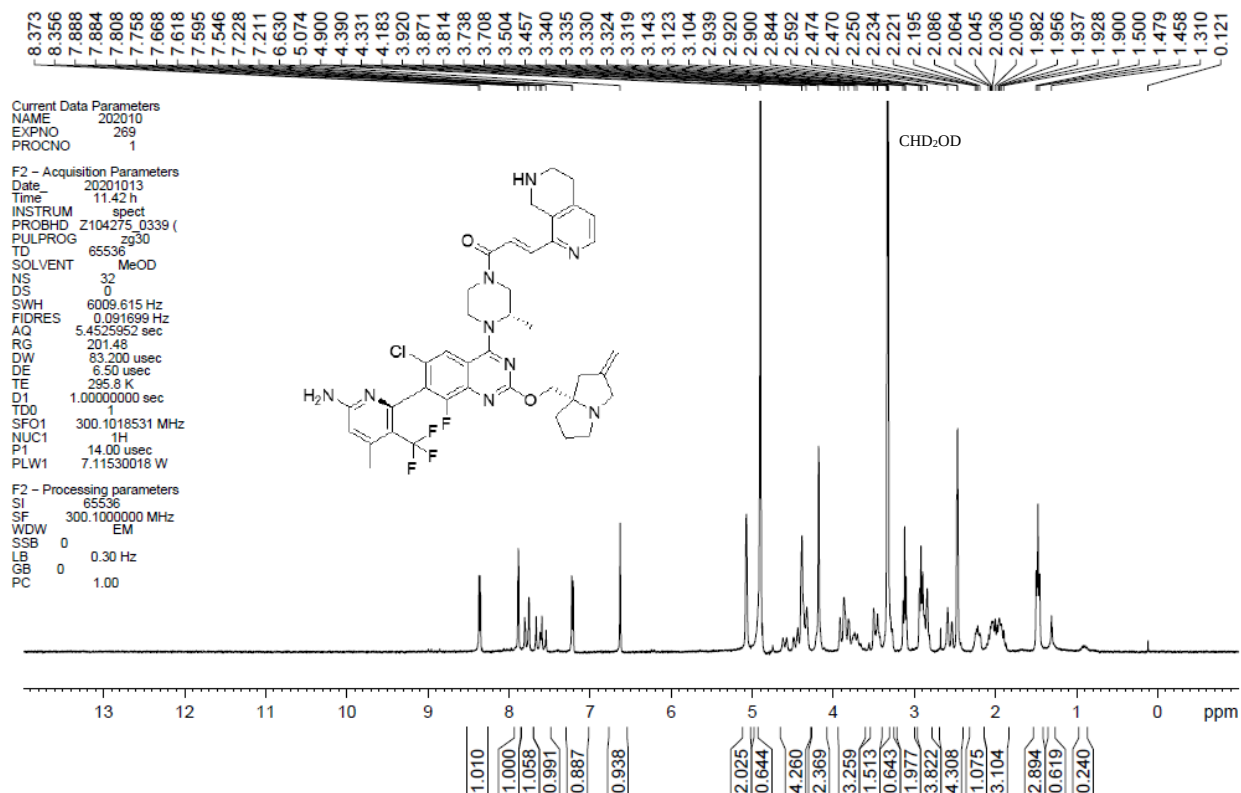
Peak#	Ret. Time	Height	Height%	Area	Area%
1	0.752	16921	1.297	15653	0.949
2	0.820	1287949	98.703	1633992	99.051
Total		1304870	100.000	1649646	100.000

Retention time: 0.816

Spectrum Mode:Single 0.816(171) Base Peak:422.55(760814)
BG Mode:Averaged 0.806-0.836(169-175) Segment 1 - Event 1



Compound 32 (¹H-NMR, LCMS)



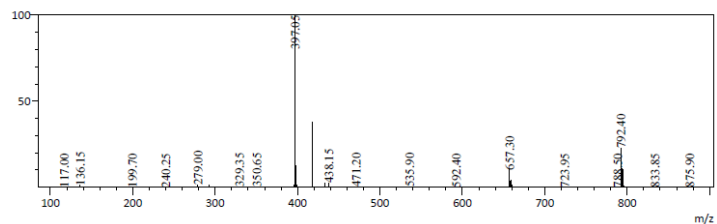
Peak Table

PDA Ch1 254nm

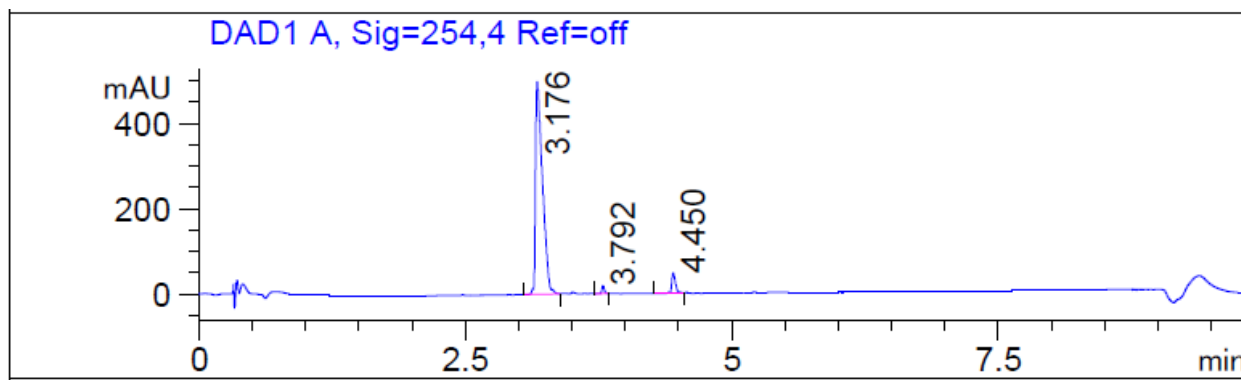
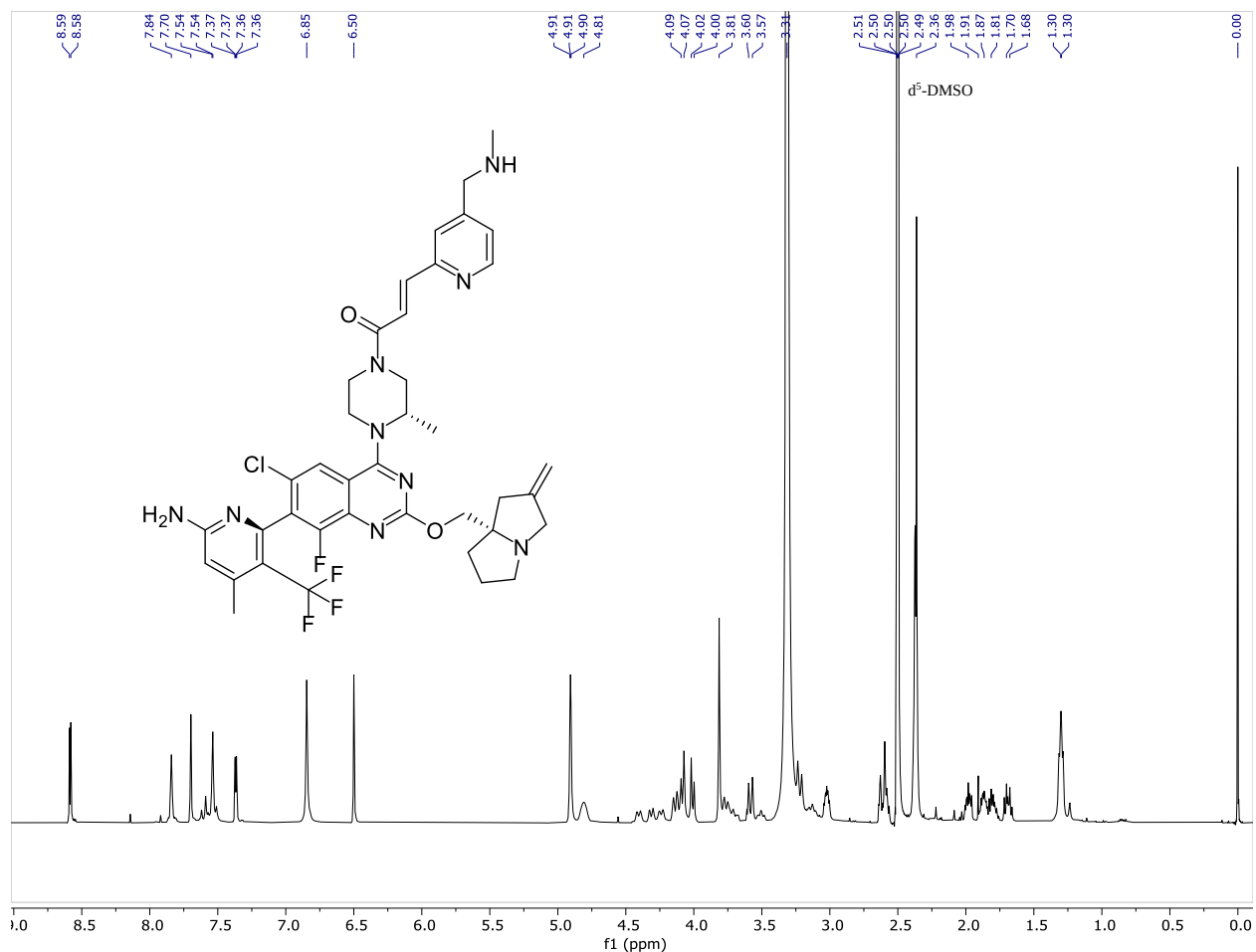
Peak#	Ret. Time	Height	Height%	Area	Area%
1	0.655	1944	0.253	574	0.123
2	0.819	7727	1.455	6693	1.435
3	0.853	518357	97.580	456132	97.760
4	0.891	2615	0.492	1821	0.390
5	0.933	1171	0.220	1365	0.292
Total		531214	100.000	466585	100.000

Retention time: 0.856

Spectrum Mode: Single 0.856(179) Base Peak: 397.05(880822)
BG Mode: Averaged 0.826-0.886(173-185) Segment 1 - Event 1



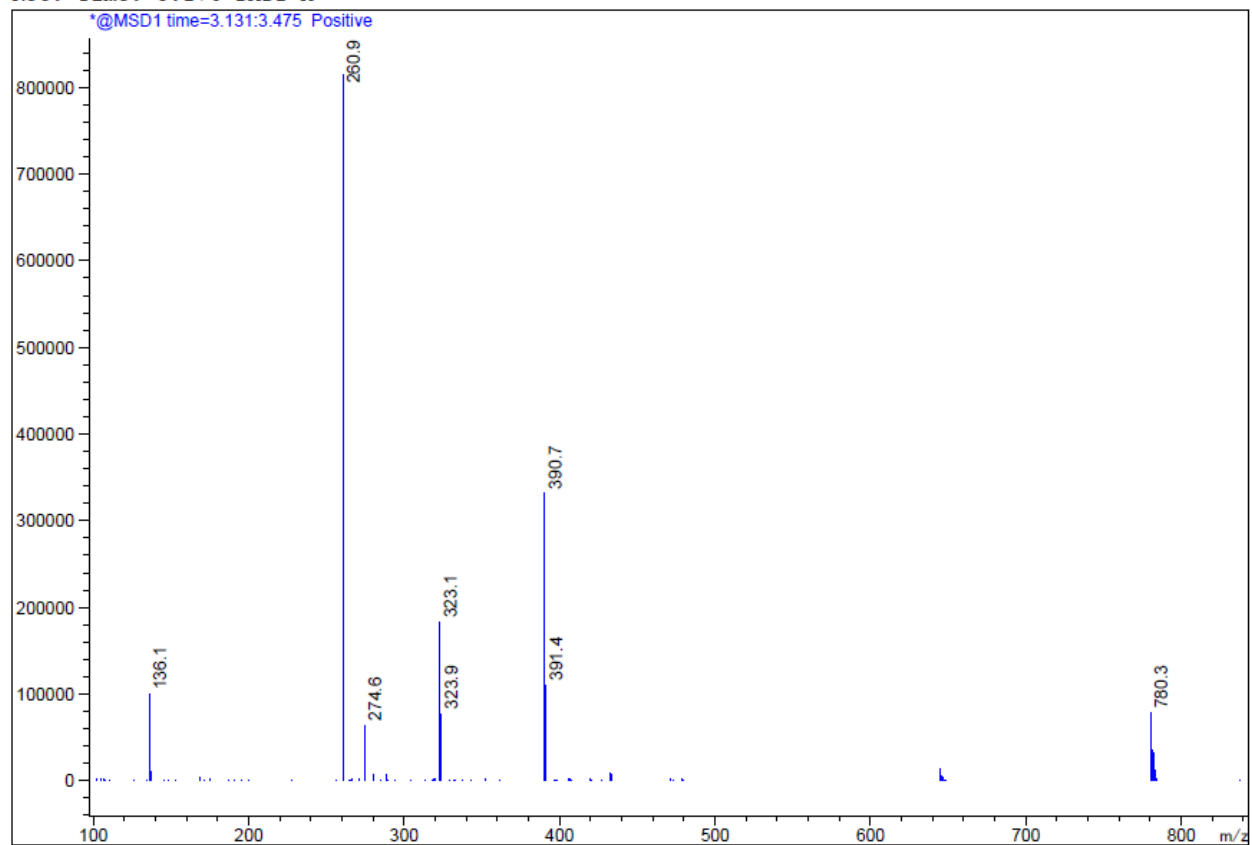
Compound 33 (¹H-NMR, LCMS)



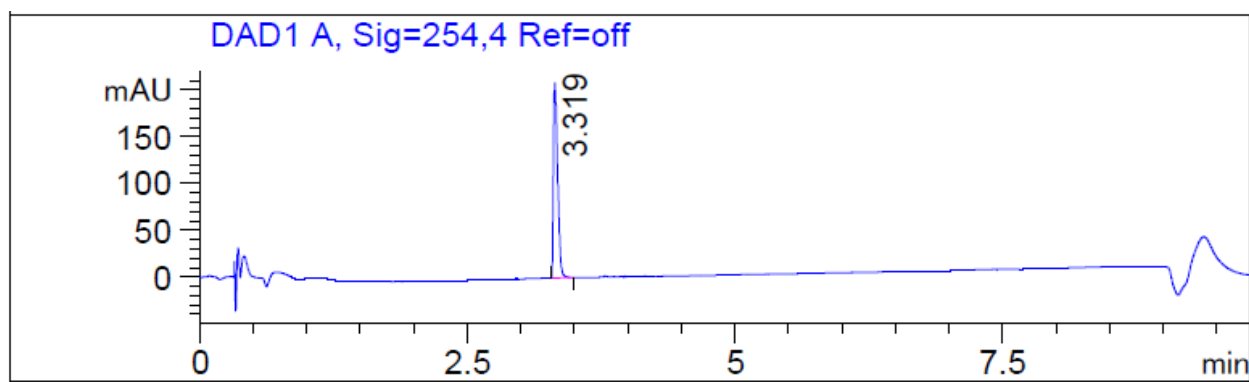
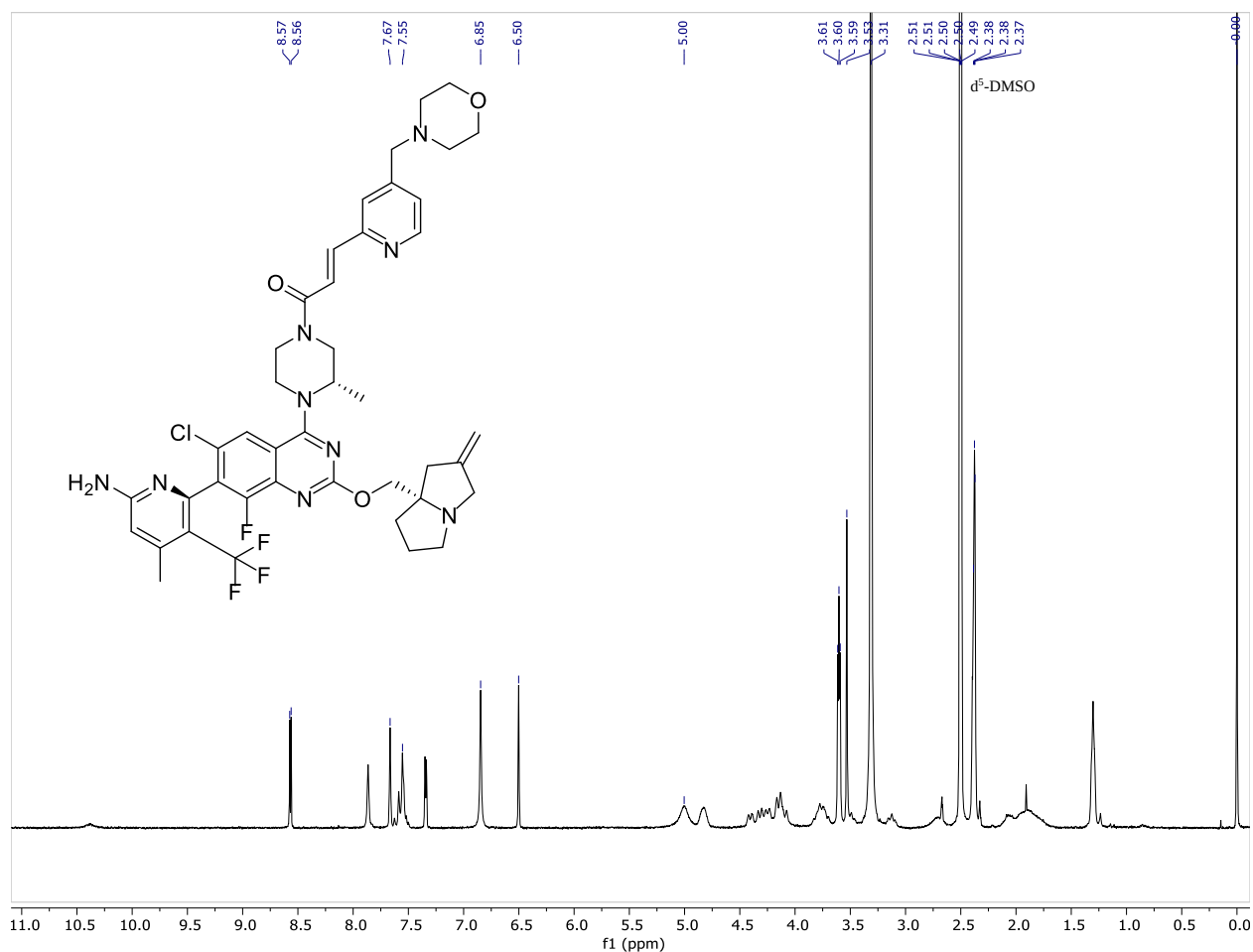
Integration for DAD1 A, Sig=254,4 Ref=off

R.T.	Width	Area	Height	Area%	MS+
3.18	0.06	2270.97	501.21	93.67	260.9
3.79	0.03	30.27	18.49	1.25	324.6
4.45	0.04	123.21	48.32	5.08	309.1

Ret. Time: 3.176 DAD1 A



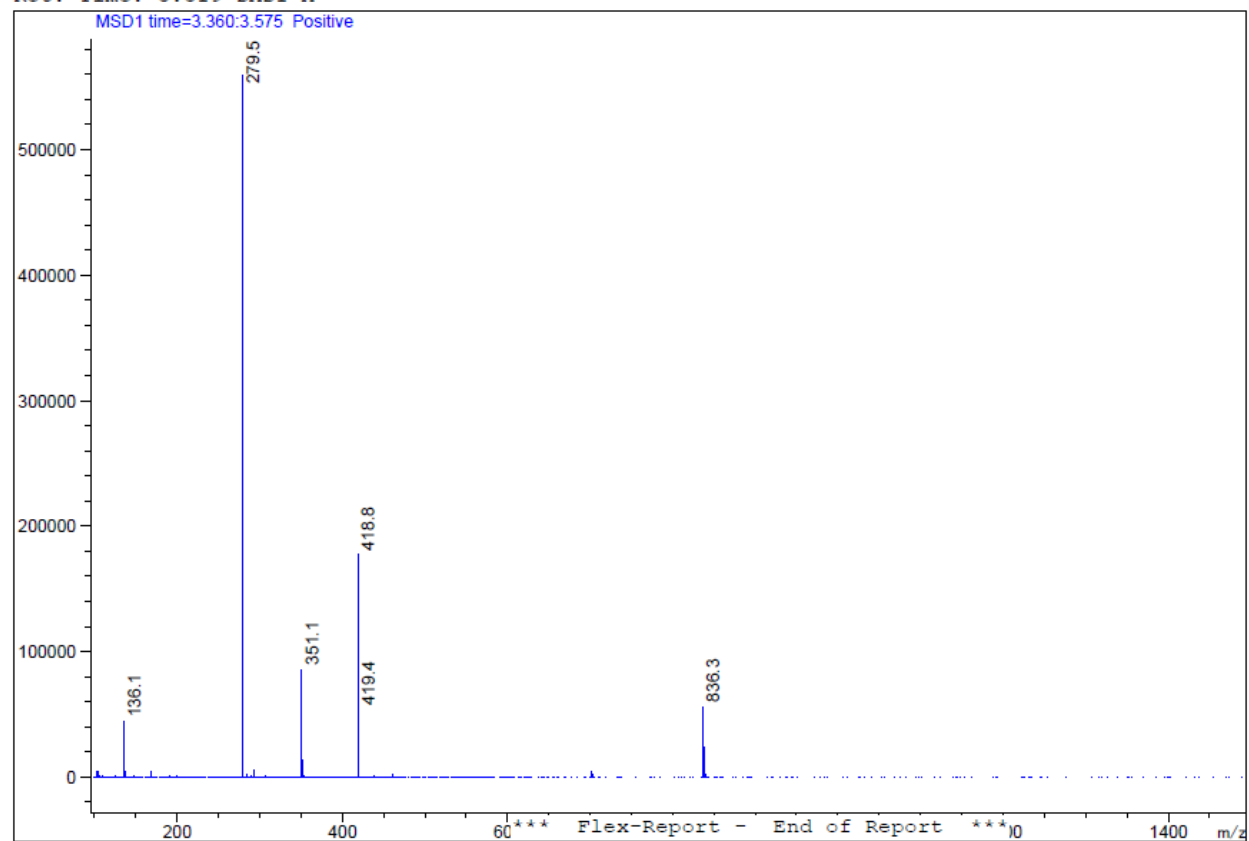
Compound 34 (¹H-NMR, LCMS)



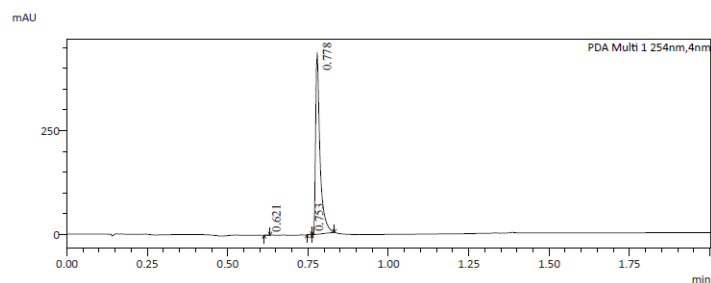
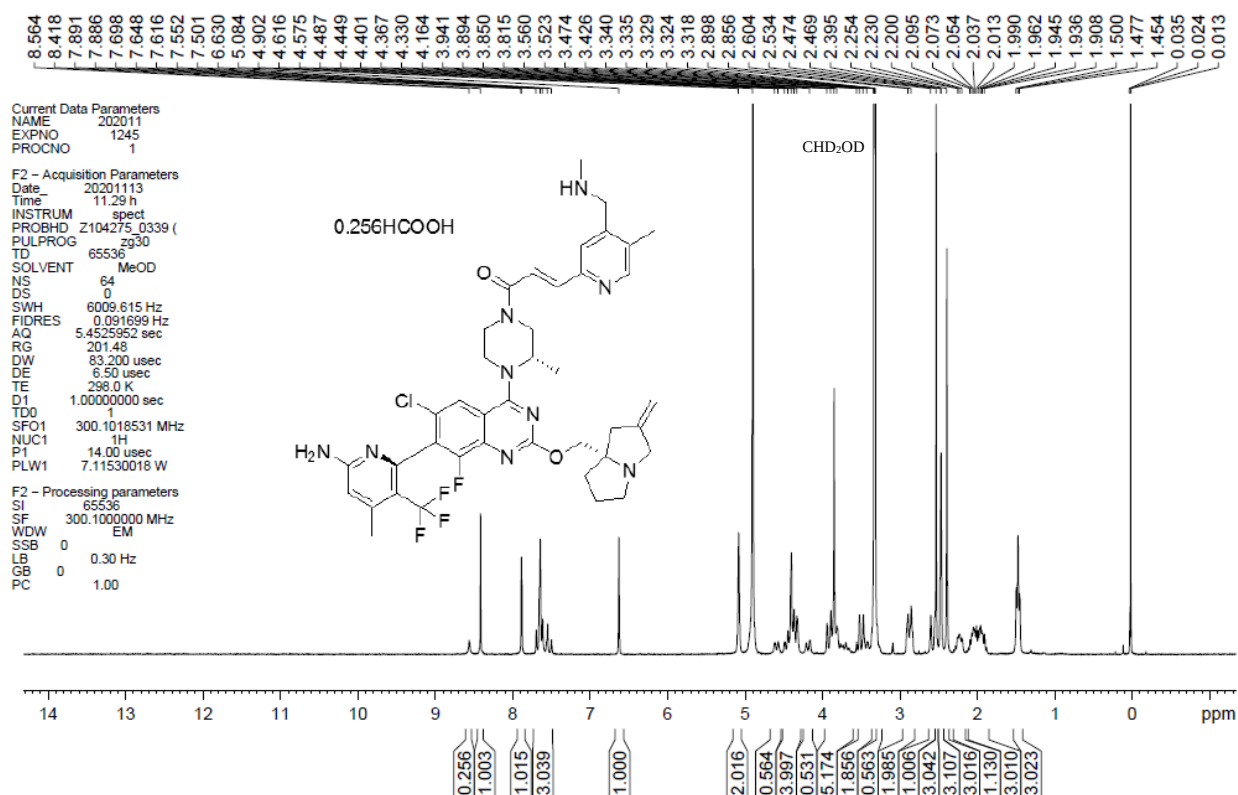
Integration for DAD1 A, Sig=254,4 Ref=off

R.T.	Width	Area	Height	Area%	MS+
3.32	0.04	576.34	208.27	100.00	279.6

Ret. Time: 3.319 DAD1 A



Compound 35 (¹H-NMR, LCMS)

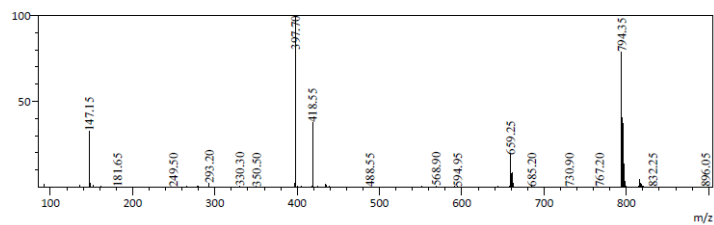


Peak Table

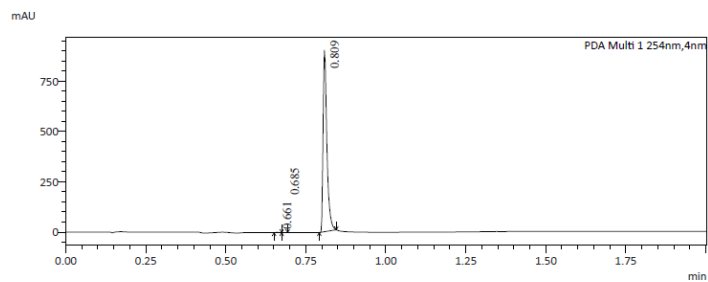
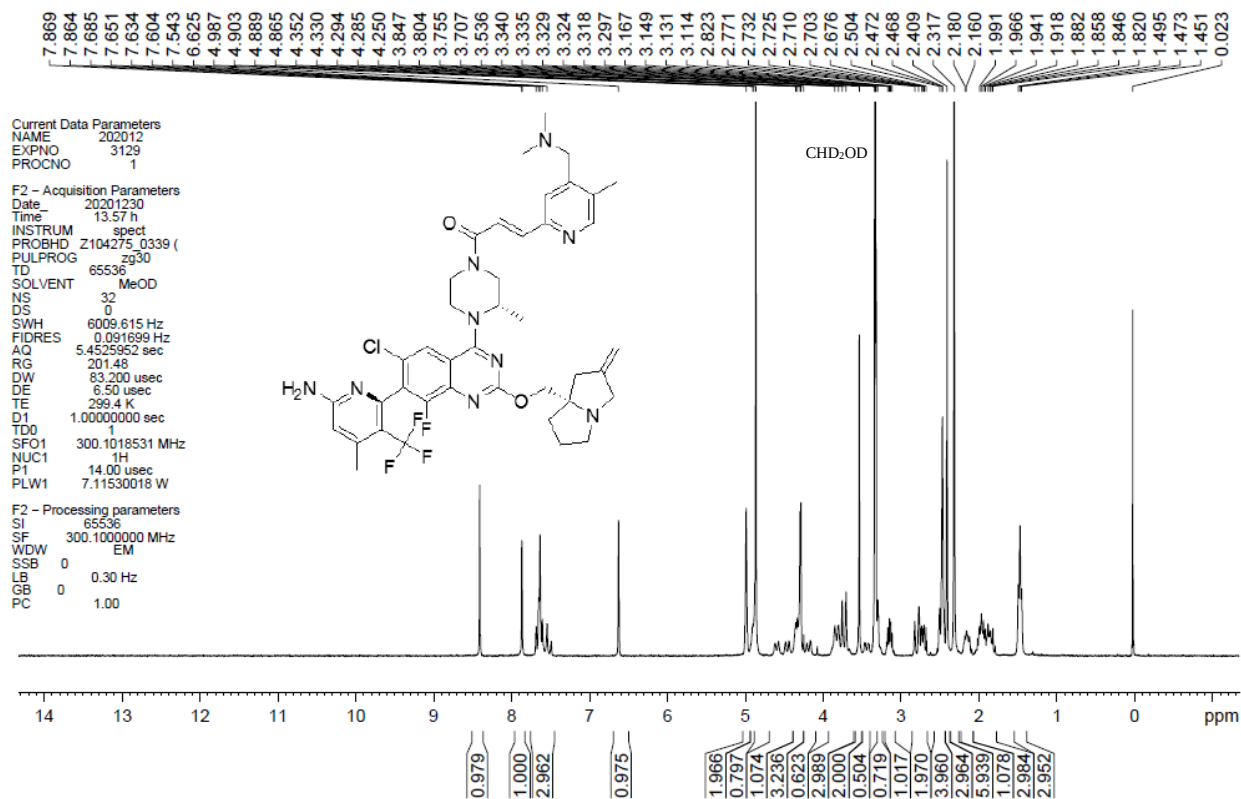
Peak#	Ret. Time	Height	Height%	Area	Area%
1	0.621	708	0.160	449	0.107
2	0.753	1381	0.312	695	0.166
3	0.778	440113	99.527	417208	99.727
Total		442202	100.000	418351	100.000

Retention time: 0.778

Spectrum Mode:Single 0.778(99) Base Peak:397.70(316268)
BG Mode:Averaged 0.678-1.094(87-137) Segment 1 - Event 1



Compound 36 (¹H-NMR, LCMS)

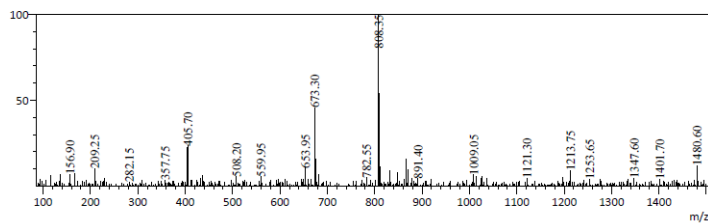


Peak Table

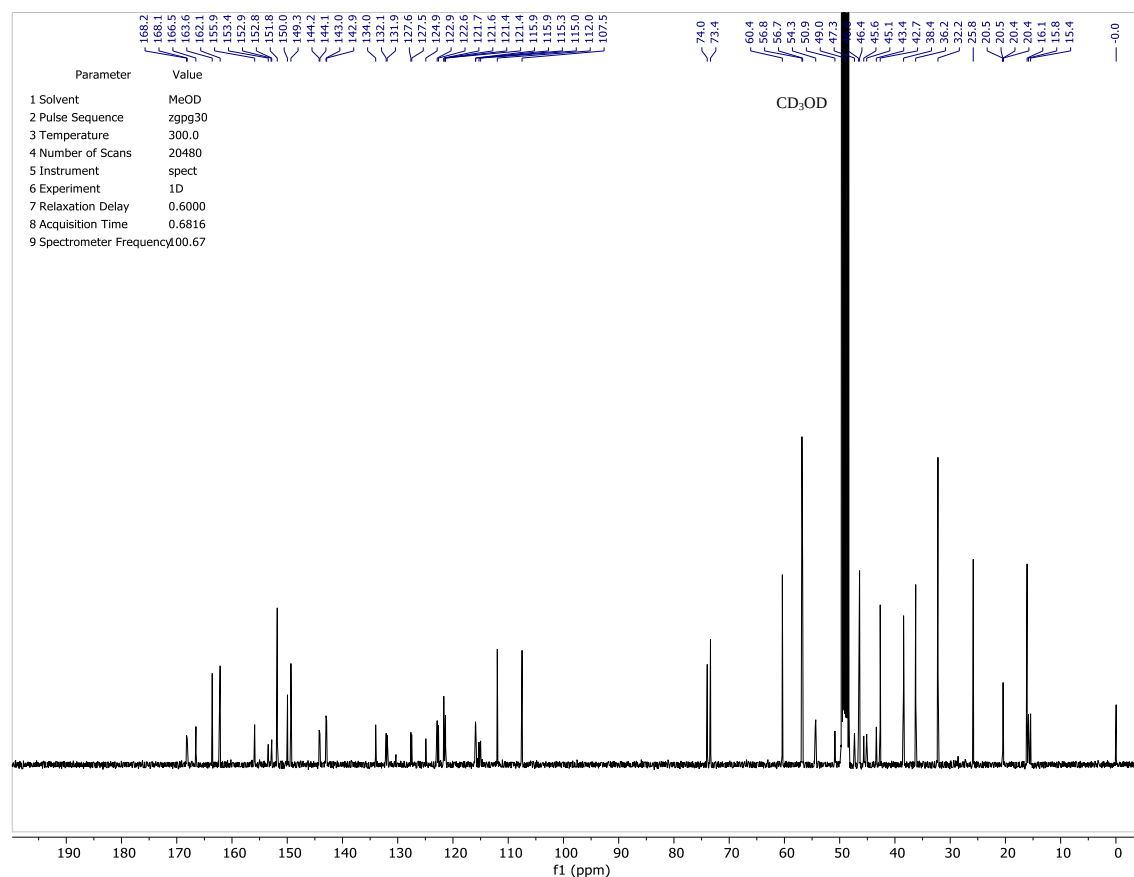
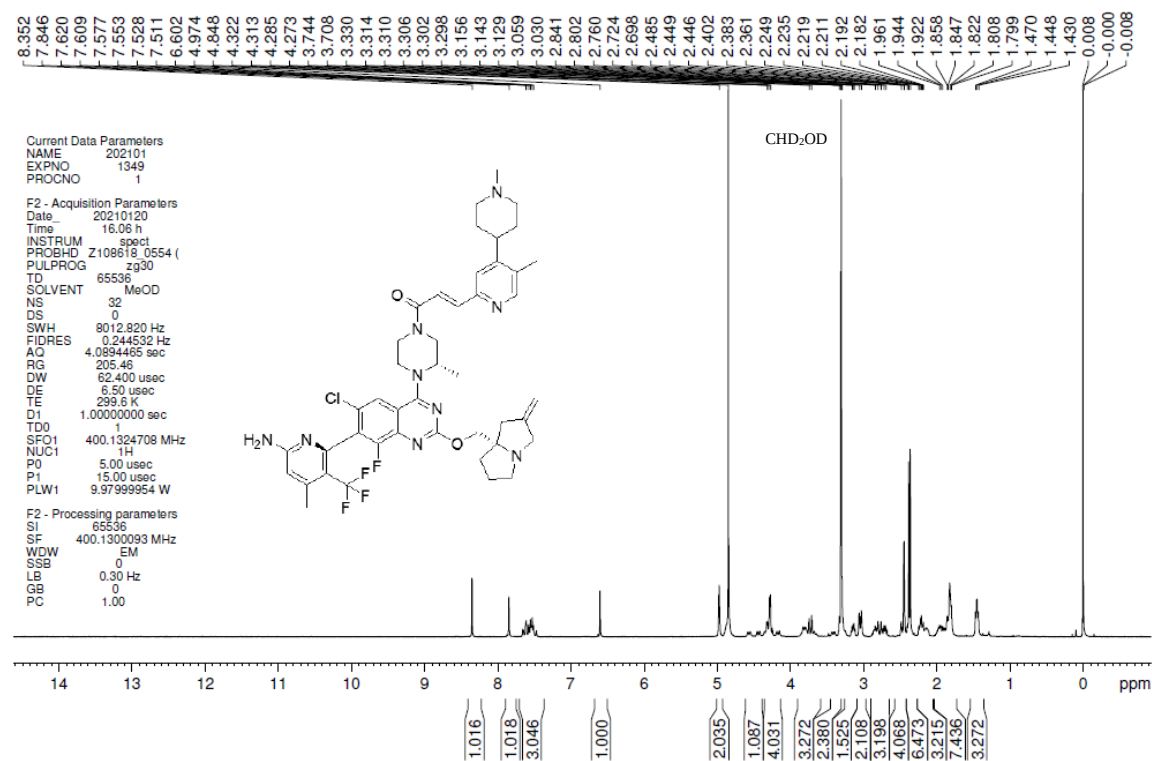
Peak#	Ret. Time	Height	Height%	Area	Area%
1	0.661	2031	0.224	1353	0.180
2	0.685	893	0.099	453	0.060
3	0.809	902780	99.677	748859	99.759
Total		905704	100.000	750665	100.000

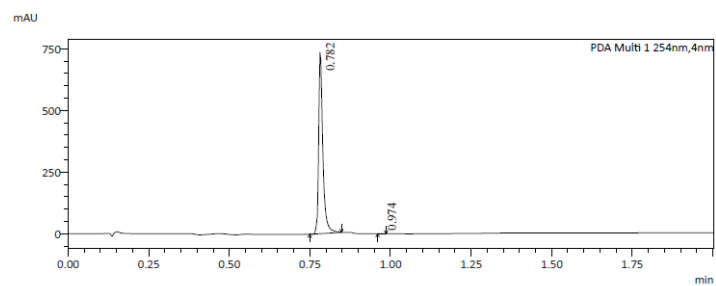
Retention time: 0.819

Spectrum Mode: Single 0.819(103) Base Peak: 808.35(8102)
 BG Mode: Averaged 0.802-0.827(101-104) Segment 1 - Event 1



Compound 37 (¹H-NMR, ¹³C-NMR, LCMS)



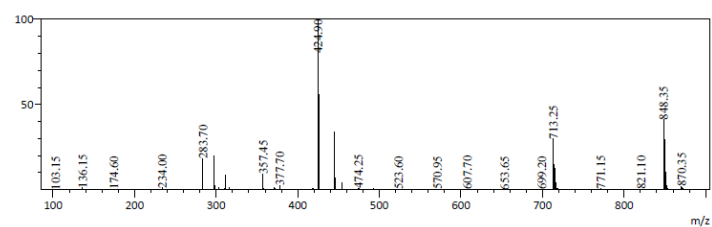


Peak Table
PDA Ch1 254nm

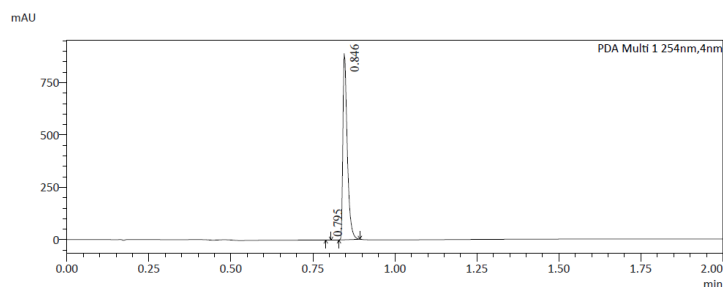
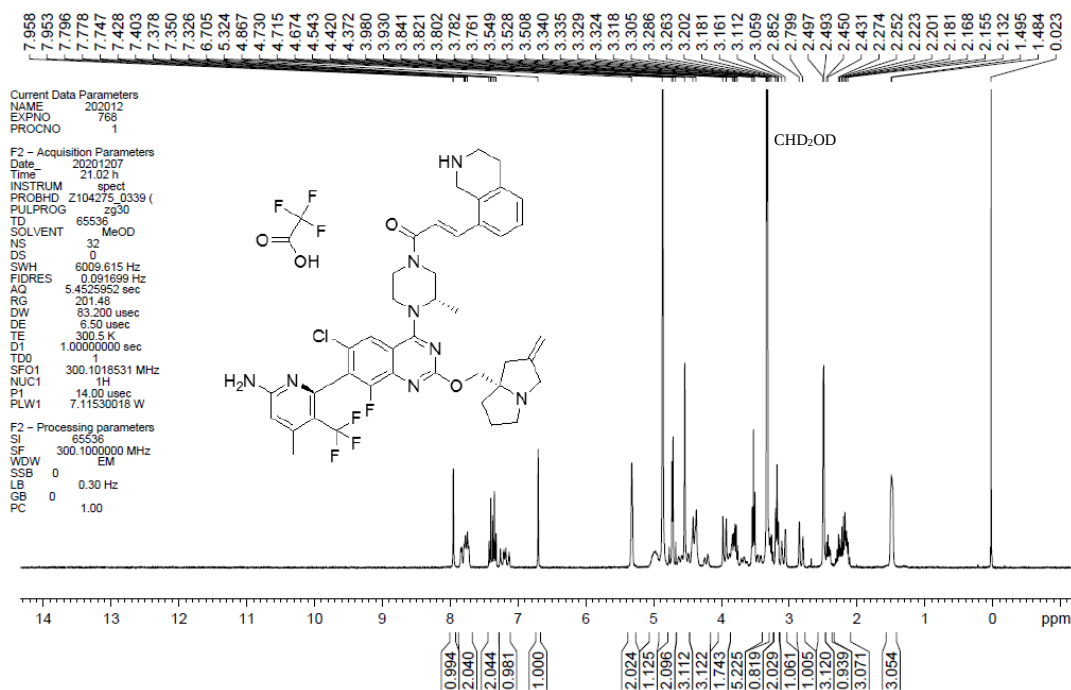
Peak#	Ret. Time	Height	Height%	Area	Area%
1	0.782	736095	99.902	663992	99.941
2	0.974	723	0.098	393	0.059
Total		736818	100.000	664386	100.000

Retention time: 0.778

Spectrum Mode:Single 0.778(99) Base Peak:424.90(293151)
BG Mode:Averaged 0.736-0.836(94-106) Segment 1 - Event 1



Compound 38 (¹H-NMR, LCMS)

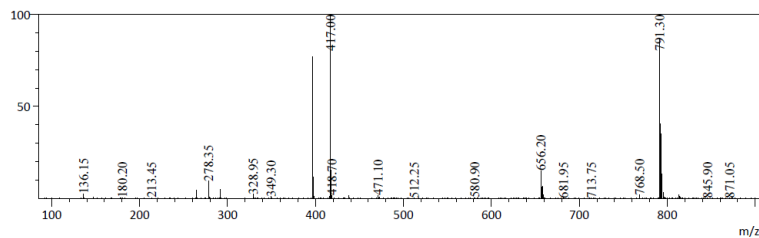


Peak Table

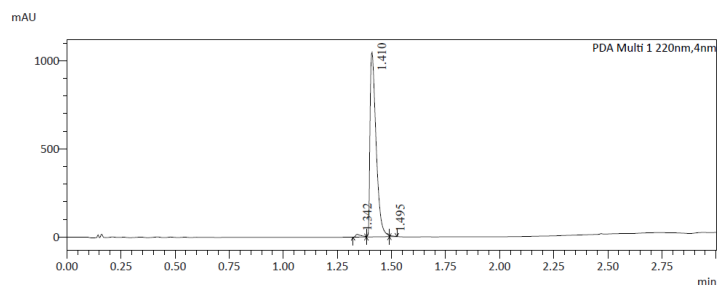
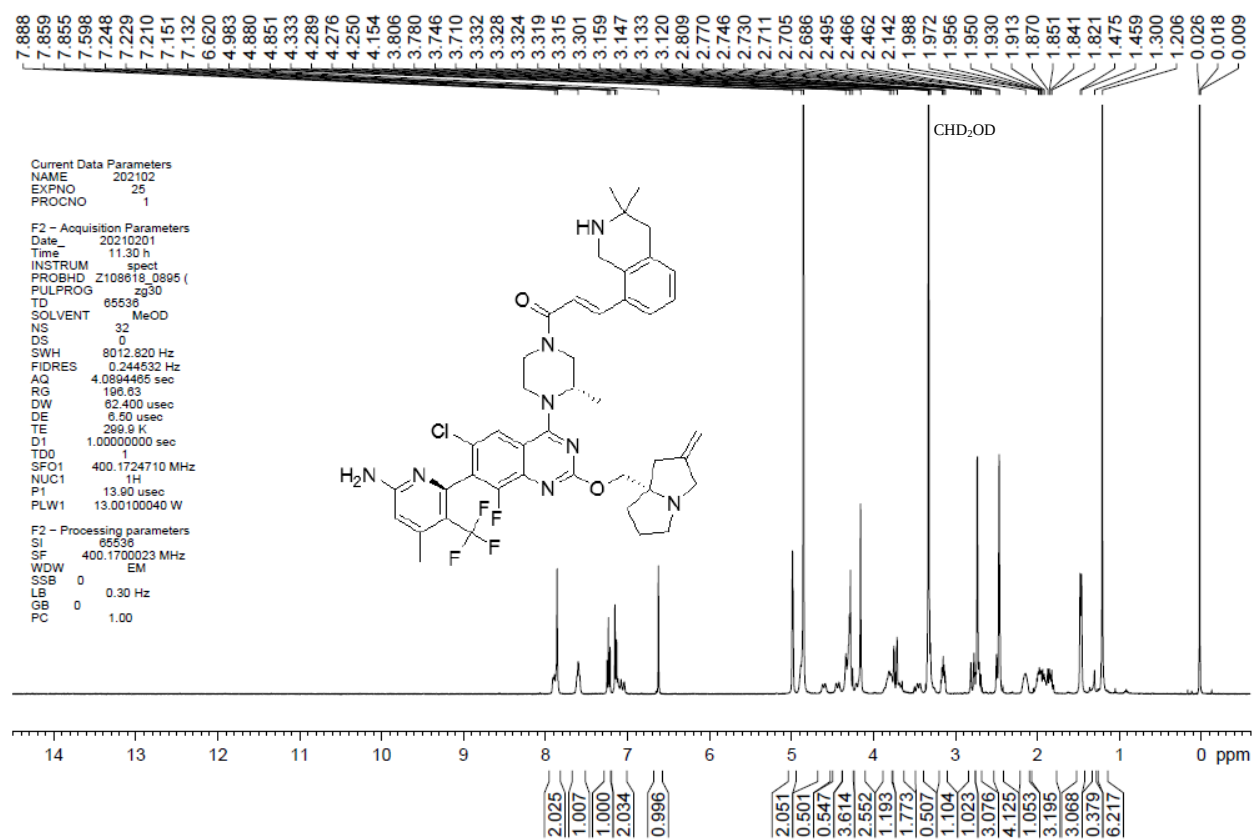
Peak#	Ret. Time	Height	Height%	Area	Area%
1	0.795	377	0.042	177	0.021
2	0.846	890902	99.958	828819	99.979
Total		891279	100.000	828997	100.000

Retention time: 0.844

Spectrum Mode:Single 0.844(107) Base Peak:417.00(630242)
BG Mode:Averaged 0.686-1.244(88-155) Segment 1 - Event 1



Compound 39 (¹H-NMR, LCMS)

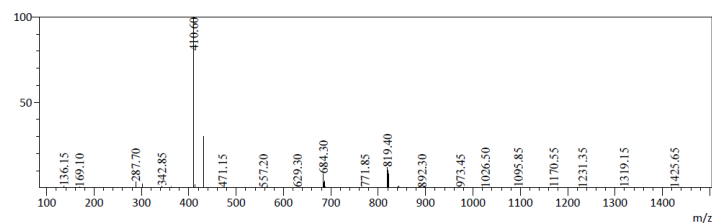


Peak Table

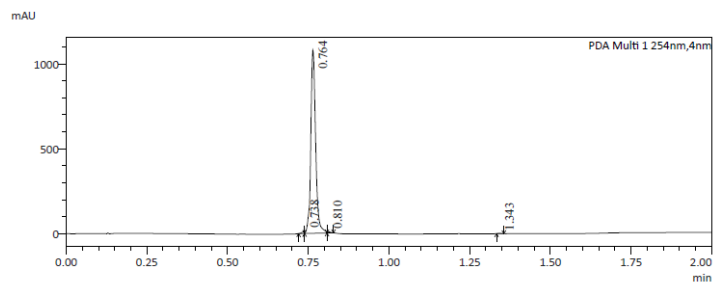
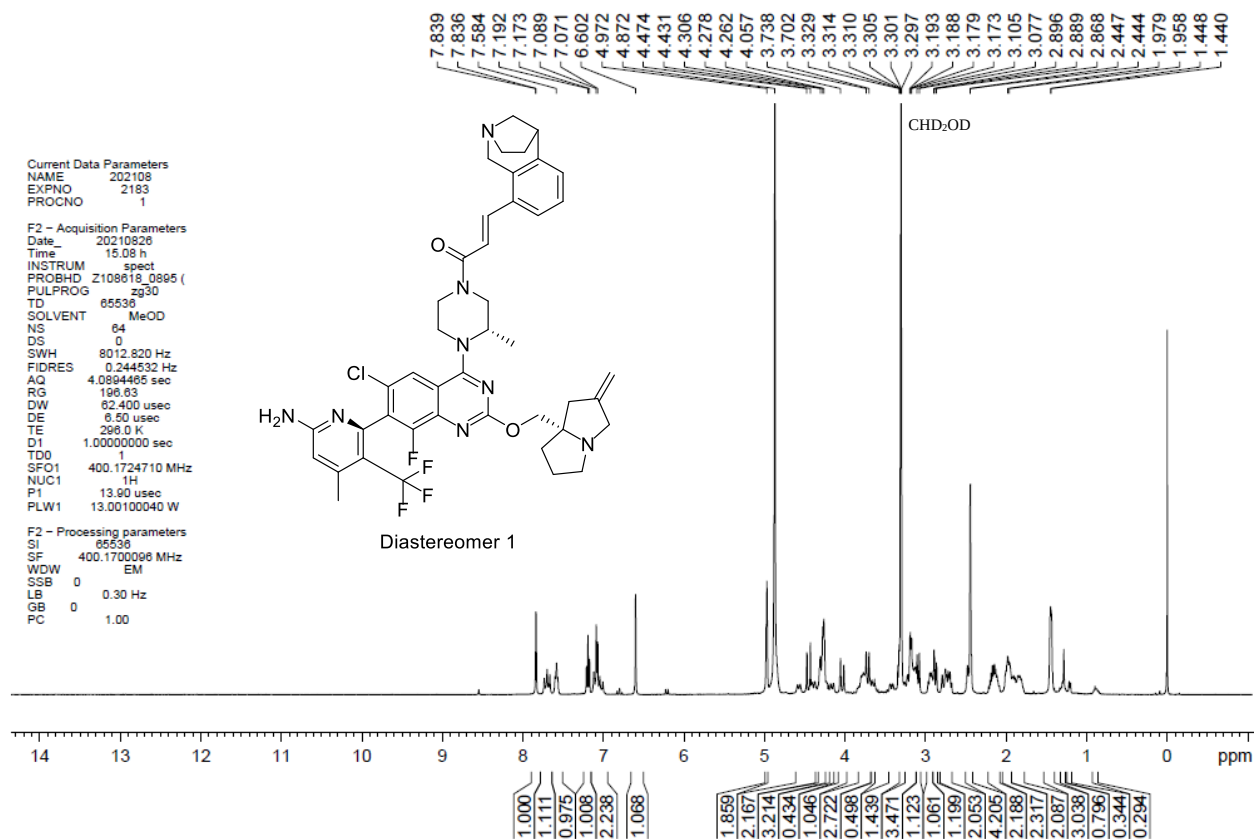
Peak#	Ret. Time	Height	Height%	Area	Area%
1	1.342	15717	1.471	28517	1.381
2	1.410	1046810	97.961	2030914	98.318
3	1.495	6068	0.568	6236	0.302
Total		1068594	100.000	2065667	100.000

Retention time: 1.410

Spectrum Mode:Single 1.410(175) Base Peak:410.60(728648)
BG Mode:Averaged 1.385-1.460(172-181) Segment 1 - Event 1



Compound 40 (¹H-NMR, LCMS)

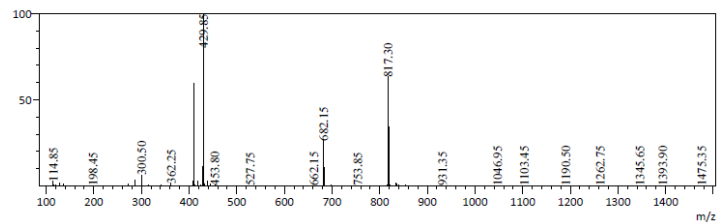


Peak Table

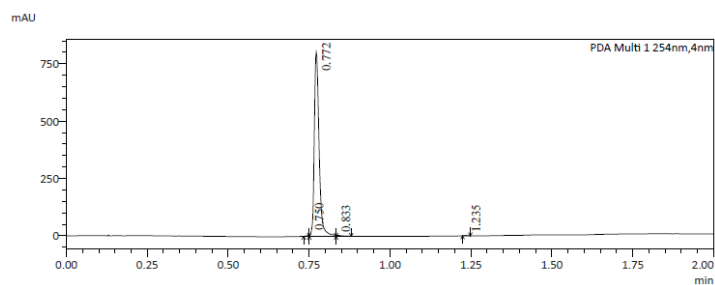
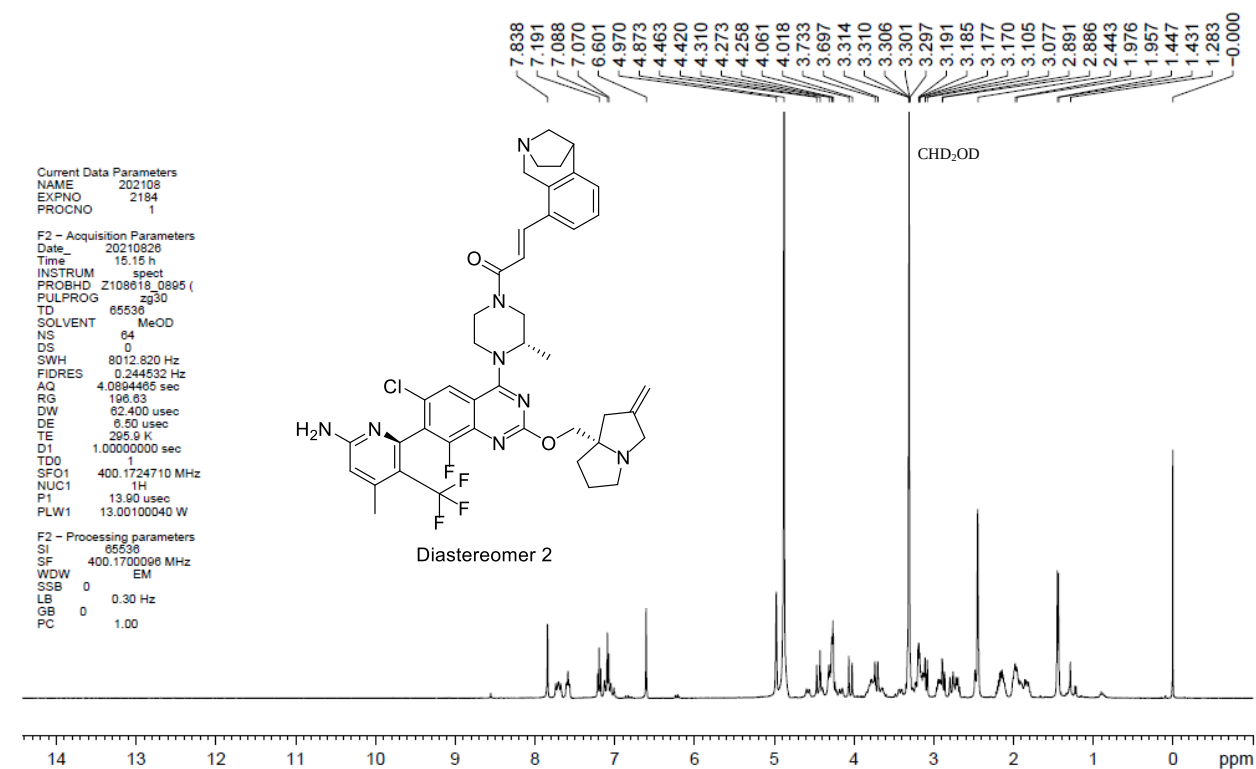
Peak#	Ret. Time	Height	Height%	Area	Area%
11	0.738	10889	0.991	6589	0.579
12	0.764	1079147	98.180	1125842	98.926
13	0.810	7190	0.654	4437	0.390
17	1.343	1930	0.176	1194	0.105
Total		1099156	100.000	1138061	100.000

Retention time: 0.763

Spectrum Mode:Averaged 0.746-0.780(48-50) Base Peak:429.85(862911)
BG Mode:Averaged 0.496-0.896(33-57) Segment 1 - Event 1



Compound 41 (¹H-NMR, LCMS)

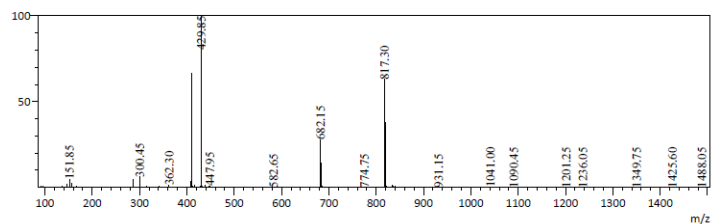


Peak Table
PDA Ch1 254nm

Peak#	Ret. Time	Height	Height%	Area	Area%
10	0.750	4854	0.597	2484	0.299
11	0.772	802692	98.708	823368	99.015
12	0.833	4275	0.526	4697	0.565
16	1.235	1377	0.169	1010	0.121
Total		813198	100.000	831559	100.000

Retention time: 0.763

Spectrum Mode:Single 0.763(49) Base Peak:429.85(931881)
BG Mode:Peak Start 0.863(55) Segment 1 - Event 1



4. X-Ray crystallography

The gene encoding the His6-Tev-KRAS4B(1-169)G13D was codon optimized, synthesized, cloned into a pET24a expression vector and transformed into BL(21)DE3 *E. coli* cells. Cells were grown in LB medium and protein expression was induced by addition of 0.4 mM IPTG for 4 hours at 37°C. Cells were harvested and lysed with a microfluidizer in 50 mM Tris, 500 mM NaCl, 10 mM MgCl₂, pH 8.0, 5 mM mercaptoethanol, 1 mM PMSF, 0.1% Triton X-100 at 4°C. Cell debris was removed by centrifugation. KRAS-G13D was purified from the supernatant using HisTrap HP column equilibrated with 50 mM Tris pH 8.0, 500 mM NaCl, 10 mM MgCl₂, 1 mM TCEP, and eluted using an imidazole gradient. Fractions containing KRAS-G13D were combined, treated with TEV protease, and dialyzed overnight at 4°C. The dialyzed sample was added to a second HisTrap HP column and the flow-through collected. Gel filtration was performed as a final purification step using a Superdex 75 10/300 GL SEC column (Cytiva, Marlborough MA) equilibrated in 50 mM Tris, 100 mM NaCl, 5 mM MgCl₂, 1 mM TCEP, pH 8.0. KRAS-G13D was concentrated to 15 mg/mL, aliquoted, flash frozen, and stored at -80°C for use in crystallization screening. Purity was confirmed by SDS-page analysis. Purified KRAS G13D was thawed and combined with a 4-fold molar excess of each compound. Co-crystals of each KRAS G13D inhibitor complex were grown by vapor diffusion using a 1:1 mixture of protein and reservoir solutions. The crystals of the G13D-Cpd5 complex grew from 0.2M Sodium Chloride, 0.1M Imidazole pH8.0, 0.4M sodium di-hydrogen phosphate, 1.6M Di-potassium hydrogen phosphate, the crystals of the G13D-Cpd36 complex grew from 0.1M MMT pH 5.0, 25 % w/v Polyethylene glycol 1,500 and the crystals of the G13D-Cpd38 grew from 0.1 M Sodium acetate pH 4.5, 30% w/v PEG 5,000MME. Crystals were harvested, cryo-protected with in a solution of reservoir plus 10-15% glycerol, and flash frozen in liquid nitrogen.

Synchrotron X-ray diffraction data was collected on frozen crystals at the Advanced Photon Source (Beamline 23-ID-B) and Shanghai Synchrotron Radiation Facility (Beamline 18U1). Diffraction data was indexed, processed, and scaled using autoPROC, XDS and aimless. An initial KRAS G13D structure was determined by molecular replacement using Phaser and PDB ID 4TQA. The structure was refined using iterative cycles of manual fitting in Coot and Phenix refine. Subsequent crystal structures were solved by molecular replacement using PDB ID 4TQA as a starting model. Coordinates and structure factors are deposited in the PDB with the following accession PDB ID

codes G13D-Cpd5=8UN3 G13D-Cpd36=8UN4 G13D-Cpd38=8UN5. See Table S2 and Figure S1 for more details.

Table S2: X-ray Data collection and refinement statistics on G13D-GDP complexes

	G13D-Cpd5	G13D-Cpd36	G13D-Cpd38^o
Synchrotron Beamline	APS 23-ID-B	APS 23-ID-B	SSRF BL18U1
Wavelength	1.033	0.979	0.979
Resolution range*	48.69 - 2.07 (2.14 - 2.07)	46.66 - 1.57 (1.72 - 1.57)	31.19 - 1.31 (1.45 - 1.31)
Space group	P 32	P 21 2 21	P 1
Unit cell	69.2, 69.2, 166.9 90 90 120	40.7, 55.7, 85.5 90 90 90	38.9, 41.6, 55.6 90.2 91.4 116.3
Total reflections	432438 (34777)	196335 (8258)	120853 (6688)
Unique reflections	54585 (4286)	21157 (1058)	45350 (2267)
Multiplicity	7.9 (8.1)	9.3 (7.8)	2.7 (3.0)
Spherical completeness (%)	99.99 (100.00)	76.4 (16.6)	60.6 (11.5)
Ellipsoidal completeness (%)		93.4 (54.8)	82.5 (38.0)
Mean I/sigma(I)	12.3 (2.1)	16.7 (1.3)	11.5 (0.7)
R-merge	0.12 (1.16)	0.077 (1.873)	0.047 (1.697)
R-meas	0.13 (1.24)	0.081 (2.006)	0.059 (2.065)
CC1/2	0.998 (0.45)	0.999 (0.532)	0.998 (0.366)
Reflections used in refinement	108961 (1096)	21126 (200)	45343 (79)
R/Rfree	0.16/0.19	0.19/0.22	0.18/0.21
Number of non-hydrogen atoms	6119	1649	3384
macromolecules	5496	1409	2799
ligands	313	104	228
solvent	430	182	459
Protein residues	672	170	337
RMS(bonds)	0.004	0.007	0.008
RMS(angles)	0.7	1.06	1.21
Ramachandran f/a/o (%)	96/3.8/0.15	98/2/0	98/2/0
Clashscore	6.62	10.52	9.85
Average B-factor	40.28	30.05	21.64
macromolecules	40.34	29.4	20.17
ligands	33.99	25.94	18.69
solvent	42.56	37.57	32.48

*Statistics for the highest-resolution shell are shown in parentheses

°High resolution cut-off based only on CC1/2 value of 0.3 in highest shell

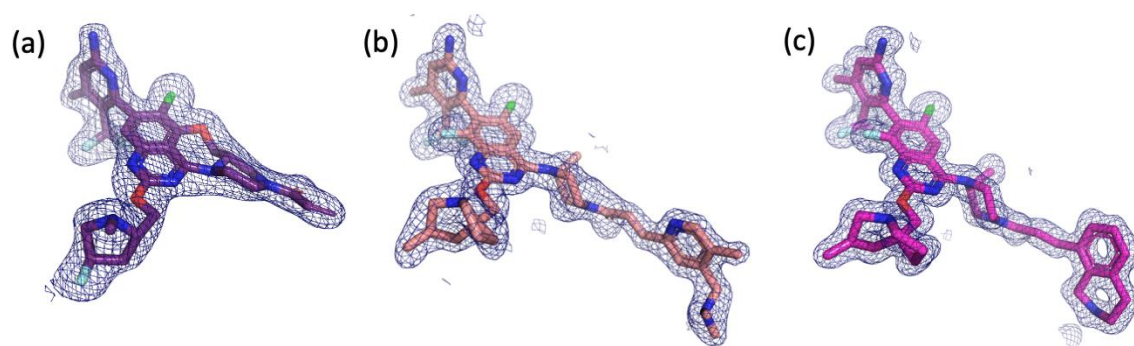


Figure S1: 2FoFc electron density for (a) Cpd5 (b) Cpd36 (c) Cpd38. Electron density map for each compound is represented by blue mesh contoured a 1rmsd.

5. Biochemical and biological assays

HTRF assay. Compound IC₅₀ values were determined in parallel in TR-FRET assays that evaluate the exchange of BODIPY-labeled GDP for the bound nucleotide in the GDP states of KRAS G13D or WT KRAS. In the WT KRAS assay, the nucleotide exchange is mediated by SOS1, but in the KRASG13D assay, the intrinsic nucleotide exchange is rapid and was monitored in absence of SOS1. Compound titrations in DMSO were dispensed into the assay plates using an Echo acoustic dispenser (Beckman). Biotinylated-KRAS G13D (M1-M188, 0.08 nM final) or biotinylated-WT KRAS (M1-M188, 0.5 nM final) preincubated with terbium-labeled streptavidin (0.5 nM final, Cisbio) was added to the assay plates and incubated for 10 minutes. The nucleotide exchange reaction was initiated by the addition of 100 nM BODIPY-GDP (ThermoFisher), and for WT KRAS, also 100 nM SOS1. The TR-FRET signal (excitation at 337 nm and emission at 520 nm and 495 nm) was monitored after 10 min for KRAS G13D and after 15 min for WT KRAS. The TR-FRET ratios (fluorescence at 520 nm / fluorescence at 495) were normalized to controls and plotted against compound concentration. The data were fit to a 4-parameter Hill equation to determine the IC₅₀ values.

Cell viability assay. HCT-116 cells were obtained from American Type Culture Collection (ATCC). They were maintained in RPMI 1640 supplemented with 10% FBS (Sigma-Aldrich Corporation) and 2 mM L-glutamine (Gibco Life Sciences) in a humidified incubator maintained at 37°C with 5% CO₂. Cell line authentication was conducted for Short Tandem Repeat (STR) Profiling using the Promega PowerPlex 16 System (Promega Corporation). This was performed once when receiving new cell lines and compared to external STR profiles of cell lines (when available) to determine cell line ancestry. Cell line authentication was routinely conducted by SNP-based genotyping using Fluidigm multiplexed assays at the Genentech cell line core facility. For the 7 day viability assay, cells were seeded at 3000 cells per U bottom ultra-low adhesion 96-well plate and were treated with a 9-point titration (1:3) of the desired chemical compounds using the HP D300 drug dispenser 24 hours after seeding. Cell growth was assessed using CellTiter-Glo Luminescent Cell Viability Assays (Promega Corporation), and luminescence was read by a 2104 EnVision Multilabel Plate Reader (PerkinElmer). All cell viability data was collected and calculated for at least 3 replicates per condition. IC₅₀s for the compounds were determined by fitting the nonlinear regression curves generated by GraphPad Prism.

6. In vivo pharmacokinetics

In vivo study

The study performed was approved by the Institutional Animal Care and Use Committee at Genentech, Inc. (South San Francisco, CA) and WuXi AppTec Co., Ltd (Suzhou, Jiangsu Province 215104, P. R. China)

Pharmacokinetic study in mouse

Three female CD-1 mice (Lingchang/Vital River Laboratory Animal Co., Ltd., Shanghai/Beijing, P.R. China) were given a 1 mg/kg intravenous (IV) cassette dose of **29** and **38** in 10%DMSO, 35% PEG400, 55% Water.

At the initiation of the study, the mice weighed from 29 to 31 g. 15 μ L blood was be collected accurately for each blood sample on wet ice. Once collected, the blood sample will be immediately placed into labeled polypropylene micro-centrifuge tubes which containing 60 uL of 1.7mg/ml Potassium (K2) EDTA in water on wet ice, mixed, then frozen and stored in a freezer set to maintain -70°C or lower until bio-analysis. The frozen blood samples will be thawed on wet ice and analyzed on the day after dosing day. Blood samples were taken at 0.033, 0.083, 0.25, 0.5, 1, 3, 6, and 9 h after the IV administration.

The concentration of **29** and **38** in each blood sample was determined by LC–MS/MS analysis.

Pharmacokinetic analysis was performed using Phoenix® WinNonlin® version 8.2 (Certara USA, Inc., Princeton, NJ). Nominal doses and sampling times were used. Descriptive statistics were calculated using WinNonlin.